EPR pre-generation on ten benchmarks

ODG, CGP, ACGP and compiler-driven Dynamic, memories per core $K = 1$ to ∞

BTP, MARS Lab, IIT Guwahati

Started 4 November 2026

Contents

1 Setup **4**

1.1 Architecture and workloads 4

1.2 Strategies and policy 4

1.3 Memories per core and the split 5

1.4 Metrics 5

2 Results **6**

2.1 QFT, XY routing 6

2.1.1 Headline: 8 memories per core, 6 for pre-generation 6

2.1.2 Headline: 4 memories per core, all 4 for pre-generation 6

2.1.3 Memories per core K 7

2.2 QFT, history routing 10

2.2.1 Headline: 8 memories per core, 6 for pre-generation 10

2.2.2 Headline: 4 memories per core, all 4 for pre-generation 10

2.2.3 Memories per core K 11

2.3 BV, XY routing 14

2.3.1 Headline: 8 memories per core, 6 for pre-generation 14

2.3.2 Headline: 4 memories per core, all 4 for pre-generation 14

2.3.3 Memories per core K 15

2.4 BV, history routing 18

2.4.1 Headline: 8 memories per core, 6 for pre-generation 18

2.4.2 Headline: 4 memories per core, all 4 for pre-generation 18

2.4.3 Memories per core K 19

2.5 GHZ, XY routing 22

2.5.1 Headline: 8 memories per core, 6 for pre-generation 22

2.5.2 Headline: 4 memories per core, all 4 for pre-generation 22

2.5.3 Memories per core K 23

2.6 GHZ, history routing 26

2.6.1 Headline: 8 memories per core, 6 for pre-generation 26

2.6.2 Headline: 4 memories per core, all 4 for pre-generation 26

2.6.3 Memories per core K 27

2.7 DJ, XY routing . . . 30
2.7.1 Headline: 8 memories per core, 6 for pre-generation . . . 30
2.7.2 Headline: 4 memories per core, all 4 for pre-generation . . . 30
2.7.3 Memories per core K . . . 31
2.8 DJ, history routing . . . 34
2.8.1 Headline: 8 memories per core, 6 for pre-generation . . . 34
2.8.2 Headline: 4 memories per core, all 4 for pre-generation . . . 34
2.8.3 Memories per core K . . . 35
2.9 QPE (exact), XY routing . . . 38
2.9.1 Headline: 8 memories per core, 6 for pre-generation . . . 38
2.9.2 Headline: 4 memories per core, all 4 for pre-generation . . . 38
2.9.3 Memories per core K . . . 39
2.10 QPE (exact), history routing . . . 42
2.10.1 Headline: 8 memories per core, 6 for pre-generation . . . 42
2.10.2 Headline: 4 memories per core, all 4 for pre-generation . . . 42
2.10.3 Memories per core K . . . 43
2.11 QAOA, XY routing . . . 46
2.11.1 Memories per core K . . . 46
2.12 QAOA, history routing . . . 49
2.12.1 Headline: 8 memories per core, 6 for pre-generation . . . 49
2.12.2 Memories per core K . . . 49
2.13 Draper adder, XY routing . . . 53
2.13.1 Headline: 8 memories per core, 6 for pre-generation . . . 53
2.13.2 Headline: 4 memories per core, all 4 for pre-generation . . . 53
2.13.3 Memories per core K . . . 54
2.14 Draper adder, history routing . . . 57
2.14.1 Headline: 8 memories per core, 6 for pre-generation . . . 57
2.14.2 Headline: 4 memories per core, all 4 for pre-generation . . . 57
2.14.3 Memories per core K . . . 58
2.15 Graph state, XY routing . . . 61
2.15.1 Headline: 8 memories per core, 6 for pre-generation . . . 61
2.15.2 Headline: 4 memories per core, all 4 for pre-generation . . . 61
2.15.3 Memories per core K . . . 62
2.16 Graph state, history routing . . . 65
2.16.1 Headline: 8 memories per core, 6 for pre-generation . . . 65
2.16.2 Headline: 4 memories per core, all 4 for pre-generation . . . 65
2.16.3 Memories per core K . . . 66
2.17 BMW cardinality, XY routing . . . 69
2.17.1 Headline: 8 memories per core, 6 for pre-generation . . . 69

2.17.2 Headline: 4 memories per core, all 4 for pre-generation 69
2.17.3 Memories per core K 70
2.18 BMW cardinality, history routing 73
2.18.1 Headline: 8 memories per core, 6 for pre-generation 73
2.18.2 Headline: 4 memories per core, all 4 for pre-generation 73
2.18.3 Memories per core K 74
2.19 BMW copula, XY routing 77
2.19.1 Headline: 8 memories per core, 6 for pre-generation 77
2.19.2 Headline: 4 memories per core, all 4 for pre-generation 77
2.19.3 Memories per core K 78
2.20 BMW copula, history routing 81
2.20.1 Headline: 8 memories per core, 6 for pre-generation 81
2.20.2 Headline: 4 memories per core, all 4 for pre-generation 81
2.20.3 Memories per core K 82

Chapter 1

Setup

1.1 Architecture and workloads

- 4×4 mesh of 16 cores; each core has 16 data slots, 14 occupied at the start and 2 kept free for migration, so circuits have $16 \times 14 = 224$ logical qubits.
- Benchmarks: ten MQT Bench 2.3.0 circuits at 224 qubits (`benchmarks/`, verified by `benchmarks/verify.py`). Each is mapped once (T-HQA placer, OEE initial placement, mapping seed 42) and routed with XY and with history routing (history counts the route visits of all preceding layers, i.e. cumulative history, routing capacity 8); every strategy and every K sees the same frozen trace.
- Physics (`config/final_config/grid_4x4_ace_3.json`): memory efficiency 0.8, memory fidelity 0.95, coherence 1 s, detector efficiency 0.95. The classical delay is set by the topology code to about 0.1 ms between neighbours.

1.2 Strategies and policy

ODG generates every pair on demand. CGP and ACGP generate pairs continuously into a cache. Dynamic is compiler driven: the planner chooses, before the run, the layer in which each pair is started. Everything except K and the memory split is the default policy of the code (branch `history-algo-update`, `ace/compiler/settings.py`):

Setting	Value
Layer basis, lead, lookahead, age limit	active layers; 1, 2, 10 layers
Generation budget	8 launches per core per active layer
Preparation priority	trace order (`trace`)
Demand guard / refill on release	on / on
Admission	per-qubit, event driven, ready pairs first, longer chains first
Empty-layer gap	10 ns
Compiler reservation, pair use	50 ms, handoff
Physics seeds	60 and 61

Table 1.1: Default policy used in every run.

1.3 Memories per core and the split

K (memories per core)	1	2	3	4	6	8	∞ (64)
Dynamic may use	1	2	3	4	6	6	64
CGP / ACGP cache ($\lfloor 0.75K \rfloor$)	0	1	2	3	4	6	48

Table 1.2: Pre-generation share of the K memories. At $K = 8$ six memories are reserved for pre-generation. CGP/ACGP use $\lfloor 0.75K \rfloor$ because caching in all K memories starves requests (smoke test: 150–1060 ms per hop at $K = 1$–4); at $K = 1$ they therefore equal ODG.

1.4 Metrics

A *hop* is one request between neighbouring cores; a *chain* is all hops of one qubit in one layer.

Latency mean time per hop from eligibility to EPR delivery, including any wait for a free memory.

Fidelity mean fidelity of the delivered pairs (not a circuit fidelity).

Coverage share of hops that found a pair already made: generation phase under 0.35 ms, i.e. only the reservation handshake. Same definition for every strategy. Chain full coverage: every hop of the chain ready.

Utilisation pairs used divided by pairs generated. **Wastage** = 100− utilisation: pairs that expired or were never used. ODG makes no pairs in advance, so both are not applicable to it.

Bars are the mean of the two seeds; whiskers span the two seeds.

Chapter 2

Results

2.1 QFT, XY routing

11076 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.1.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.782	4.60	0.9498	0.0	0.0	–	–
CGP	0.970	3.25	0.9424	77.9	60.1	4.24	95.8
ACGP	0.976	3.30	0.9421	77.6	60.0	4.39	95.6
Dynamic (8)	0.747	2.25	0.9439	76.8	64.5	1.23	98.8
Dynamic (∞)	0.616	1.65	0.9442	84.0	71.9	1.18	98.8

Table 2.1: QFT, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

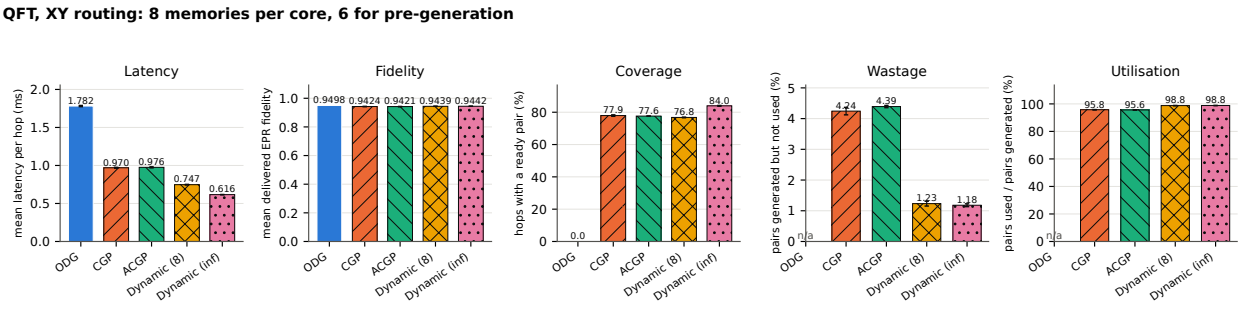


Figure 2.1: QFT, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.1.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.887	4.80	0.9498	0.0	0.0	–	–
CGP	2.331	8.95	0.9401	63.0	41.7	5.87	94.1
ACGP	2.325	8.95	0.9402	63.4	43.9	5.76	94.2
Dynamic (4)	1.295	4.70	0.9425	63.7	54.1	3.89	96.1
Dynamic (∞)	0.616	1.65	0.9442	84.0	71.9	1.18	98.8

Table 2.2: QFT, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

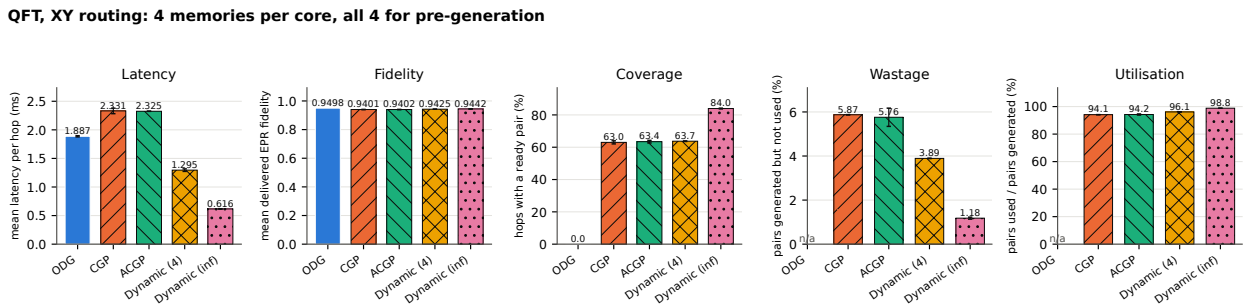


Figure 2.2: QFT, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.1.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	6.275	2.816	2.109	1.887	1.788	1.782	1.782
CGP	6.275	4.177	2.885	2.331	1.237	0.970	0.457
ACGP	6.275	4.118	2.974	2.325	1.247	0.976	0.458
Dynamic	9.041	3.528	1.877	1.295	0.866	0.747	0.616

Table 2.3: QFT, XY routing: latency (mean latency per hop (ms)) for every K.

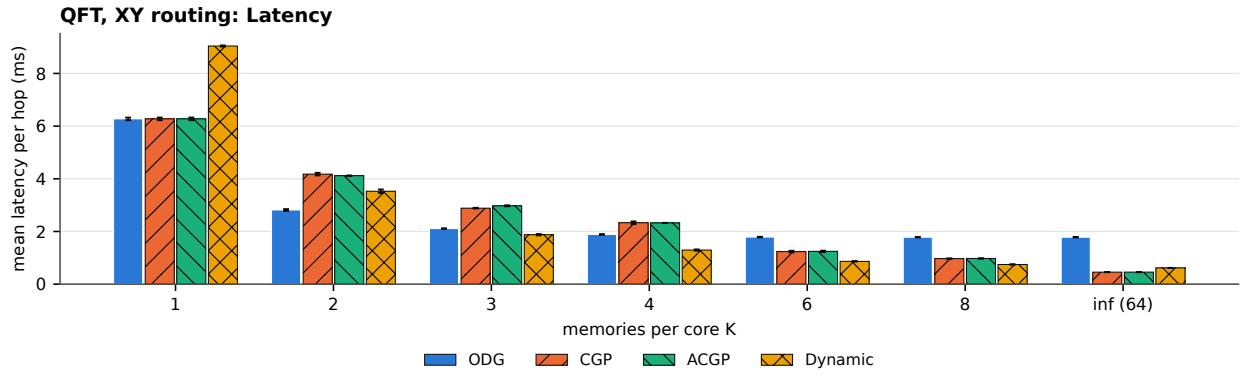


Figure 2.3: QFT, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9441	0.9415	0.9401	0.9427	0.9424	0.9422
ACGP	0.9498	0.9441	0.9414	0.9402	0.9425	0.9421	0.9422
Dynamic	0.9486	0.9440	0.9429	0.9425	0.9432	0.9439	0.9442

Table 2.4: QFT, XY routing: fidelity (mean delivered EPR fidelity) for every K.

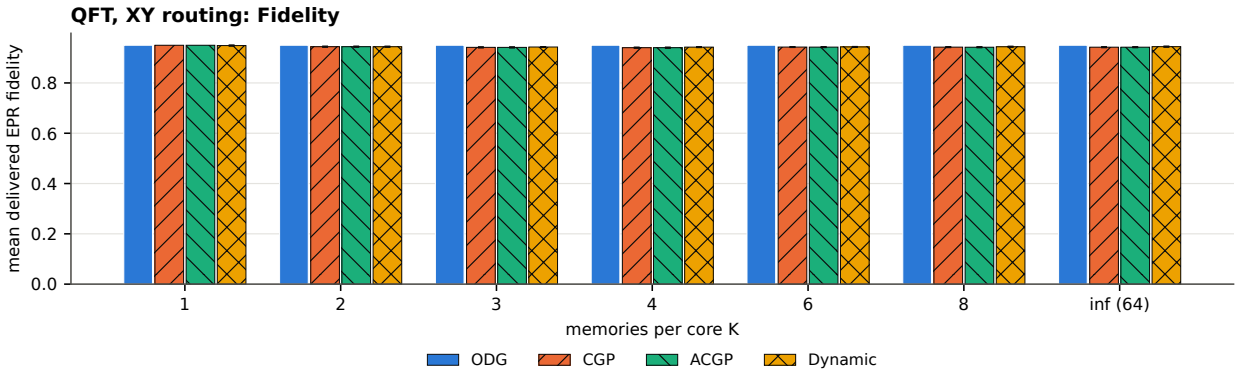


Figure 2.4: QFT, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	26.8	50.5	63.0	64.2	77.9	99.3
ACGP	0.0	27.0	49.0	63.4	63.9	77.6	99.1
Dynamic	8.9	36.3	52.4	63.7	76.8	76.8	84.0

Table 2.5: QFT, XY routing: coverage (hops with a ready pair (%)) for every K.

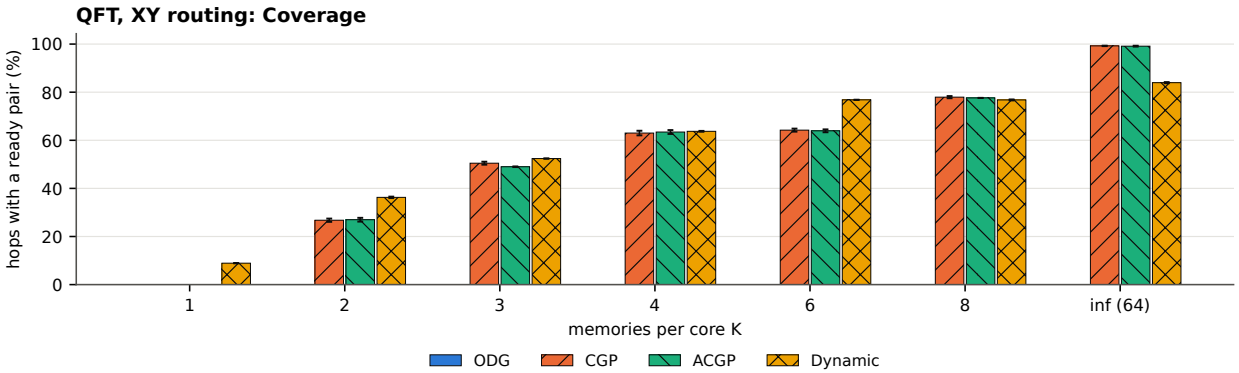


Figure 2.5: QFT, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	7.08	5.66	5.87	4.34	4.24	13.73
ACGP	–	7.04	6.13	5.76	4.49	4.39	13.73
Dynamic	60.80	14.59	6.08	3.89	2.05	1.23	1.18

Table 2.6: QFT, XY routing: wastage (pairs generated but not used (%)) for every K.

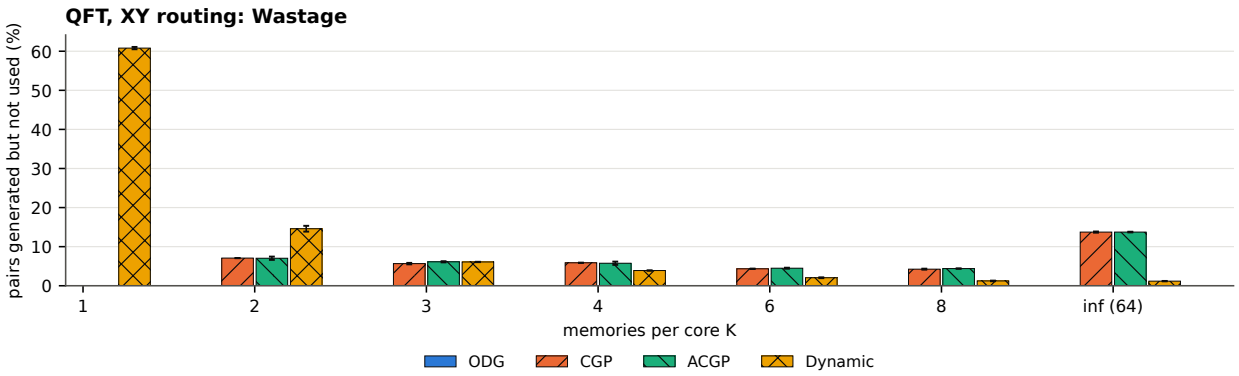


Figure 2.6: QFT, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	92.9	94.3	94.1	95.7	95.8	86.3
ACGP	–	93.0	93.9	94.2	95.5	95.6	86.3
Dynamic	39.2	85.4	93.9	96.1	98.0	98.8	98.8

Table 2.7: QFT, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

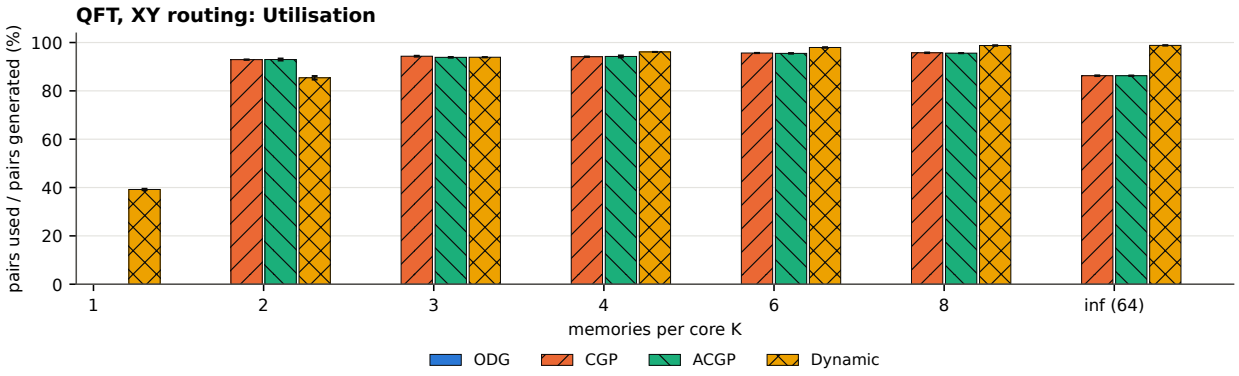


Figure 2.7: QFT, XY routing: utilisation for every K.

2.2 QFT, history routing

11076 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.2.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.773	4.40	0.9498	0.0	0.0	–	–
CGP	0.966	3.10	0.9422	73.8	52.6	4.97	95.0
ACGP	0.917	2.95	0.9425	76.6	57.4	4.34	95.7
Dynamic (8)	0.693	2.00	0.9442	79.6	65.7	1.40	98.6
Dynamic (∞)	0.584	1.45	0.9450	85.6	72.4	1.46	98.5

Table 2.8: QFT, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

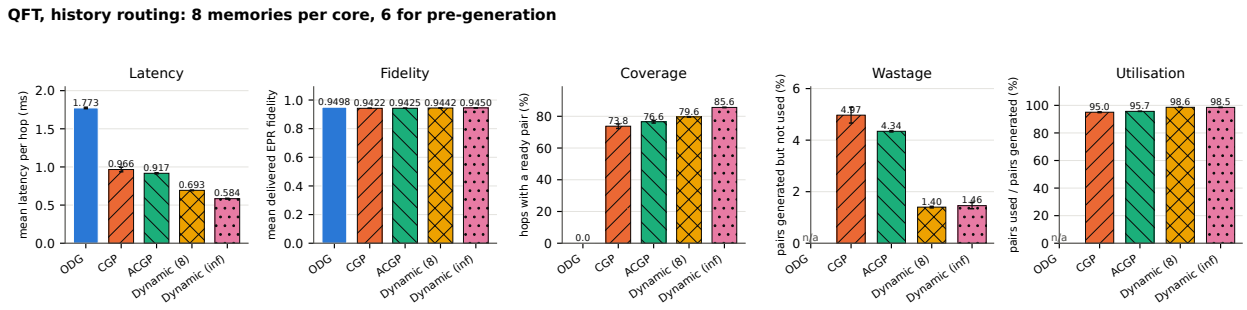


Figure 2.8: QFT, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.2.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.824	4.60	0.9498	0.0	0.0	–	–
CGP	2.192	8.10	0.9413	63.5	40.3	5.37	94.6
ACGP	2.194	8.14	0.9412	62.3	38.3	5.44	94.6
Dynamic (4)	1.199	4.20	0.9423	65.8	55.5	4.45	95.6
Dynamic (∞)	0.584	1.45	0.9450	85.6	72.4	1.46	98.5

Table 2.9: QFT, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

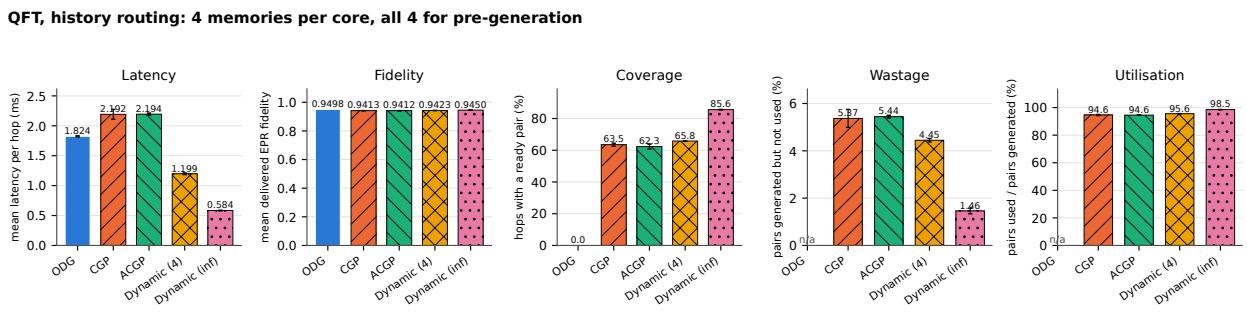


Figure 2.9: QFT, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.2.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	5.820	2.554	1.963	1.824	1.776	1.773	1.773
CGP	5.820	3.739	2.767	2.192	1.112	0.966	0.457
ACGP	5.820	3.748	2.696	2.194	1.113	0.917	0.455
Dynamic	8.812	3.153	1.749	1.199	0.797	0.693	0.584

Table 2.10: QFT, history routing: latency (mean latency per hop (ms)) for every K.

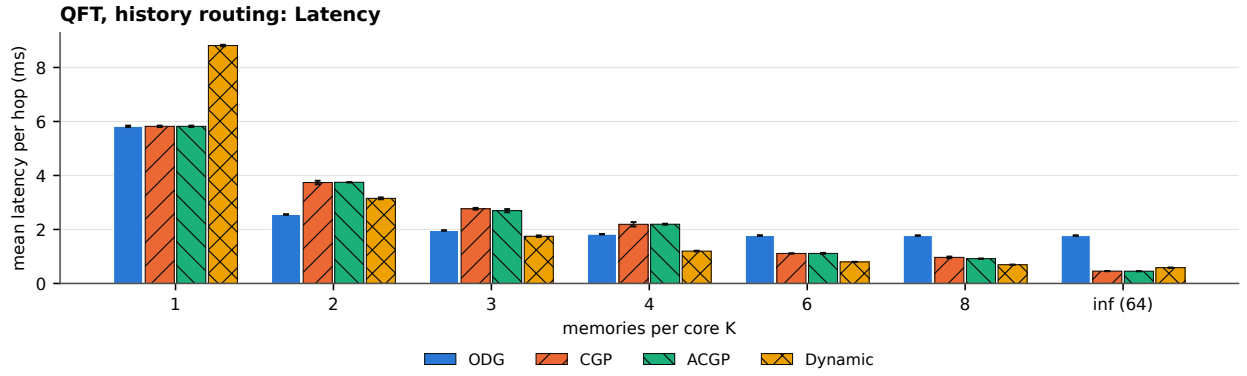


Figure 2.10: QFT, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9452	0.9427	0.9413	0.9433	0.9422	0.9417
ACGP	0.9498	0.9449	0.9426	0.9412	0.9433	0.9425	0.9418
Dynamic	0.9487	0.9442	0.9427	0.9423	0.9439	0.9442	0.9450

Table 2.11: QFT, history routing: fidelity (mean delivered EPR fidelity) for every K.

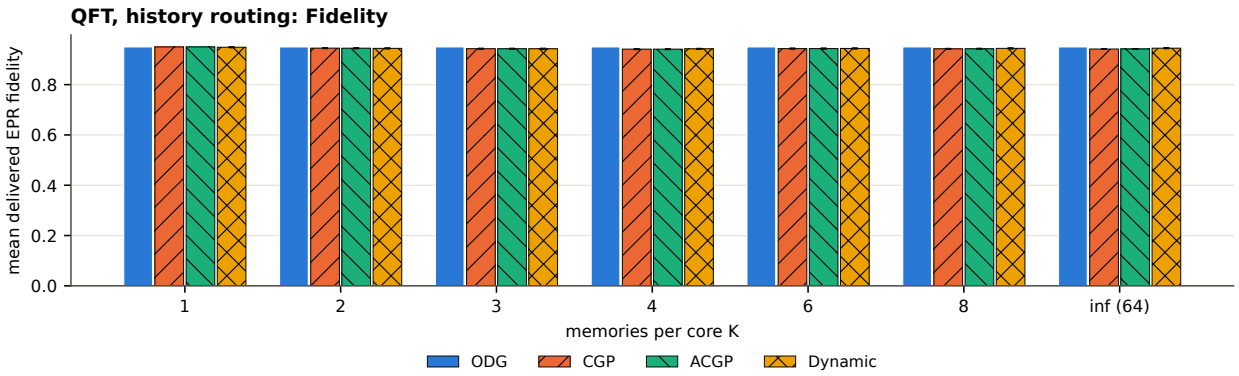


Figure 2.11: QFT, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	27.2	47.8	63.5	64.2	73.8	99.2
ACGP	0.0	28.2	49.7	62.3	65.3	76.6	99.5
Dynamic	8.4	36.4	53.8	65.8	79.2	79.6	85.6

Table 2.12: QFT, history routing: coverage (hops with a ready pair (%)) for every K.

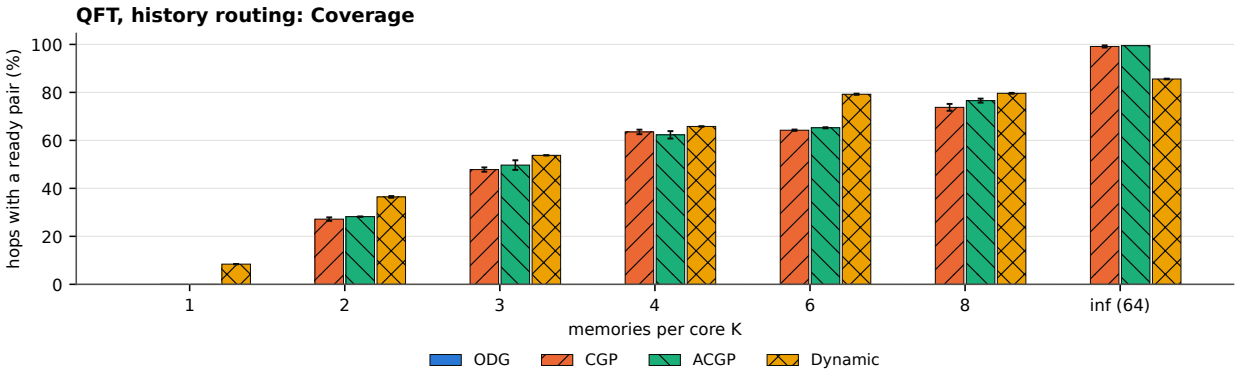


Figure 2.12: QFT, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	6.23	5.85	5.37	3.96	4.97	14.14
ACGP	–	6.07	5.58	5.44	3.74	4.34	13.66
Dynamic	63.26	14.98	6.07	4.45	2.25	1.40	1.46

Table 2.13: QFT, history routing: wastage (pairs generated but not used (%)) for every K.

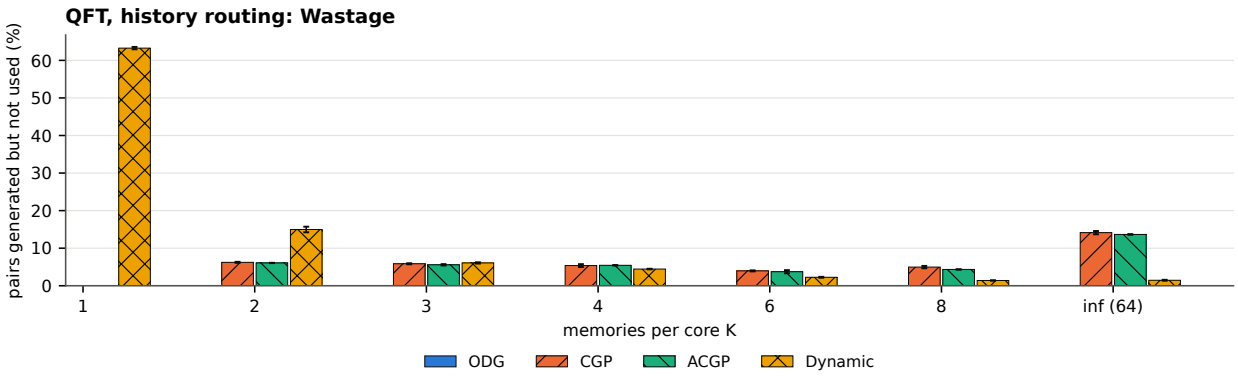


Figure 2.13: QFT, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	93.8	94.1	94.6	96.0	95.0	85.9
ACGP	–	93.9	94.4	94.6	96.3	95.7	86.3
Dynamic	36.7	85.0	93.9	95.6	97.7	98.6	98.5

Table 2.14: QFT, history routing: utilisation (pairs used / pairs generated (%)) for every K.

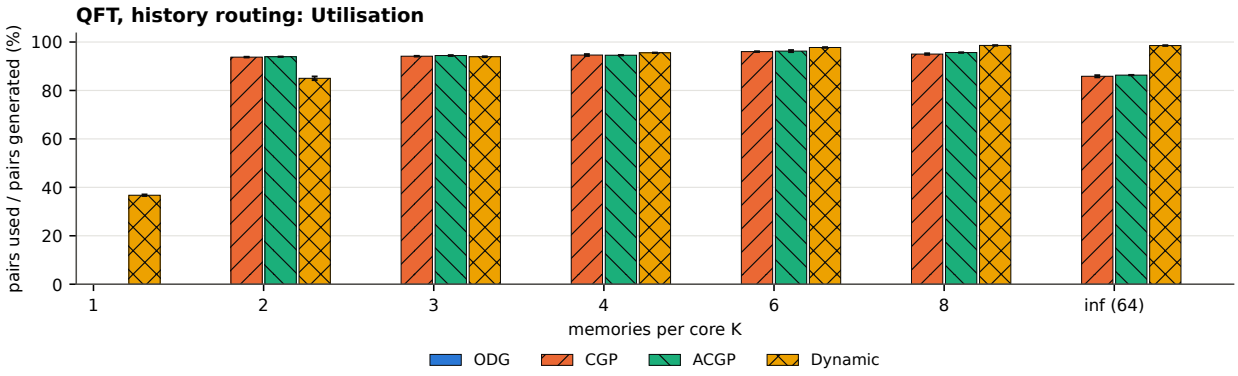


Figure 2.14: QFT, history routing: utilisation for every K.

2.3 BV, XY routing

23 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.3.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.550	3.40	0.9498	0.0	0.0	–	–
CGP	0.648	1.60	0.9447	73.9	60.7	72.36	27.6
ACGP	0.648	1.60	0.9447	73.9	60.7	72.36	27.6
Dynamic (8)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4
Dynamic (∞)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4

Table 2.15: BV, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

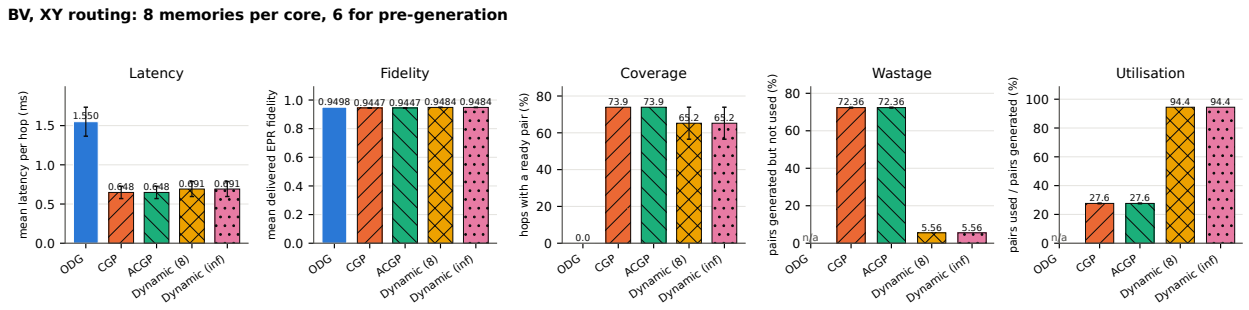


Figure 2.15: BV, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.3.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.550	3.40	0.9498	0.0	0.0	–	–
CGP	0.811	2.40	0.9410	67.4	46.4	56.67	43.3
ACGP	0.811	2.40	0.9410	67.4	46.4	56.67	43.3
Dynamic (4)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4
Dynamic (∞)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4

Table 2.16: BV, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

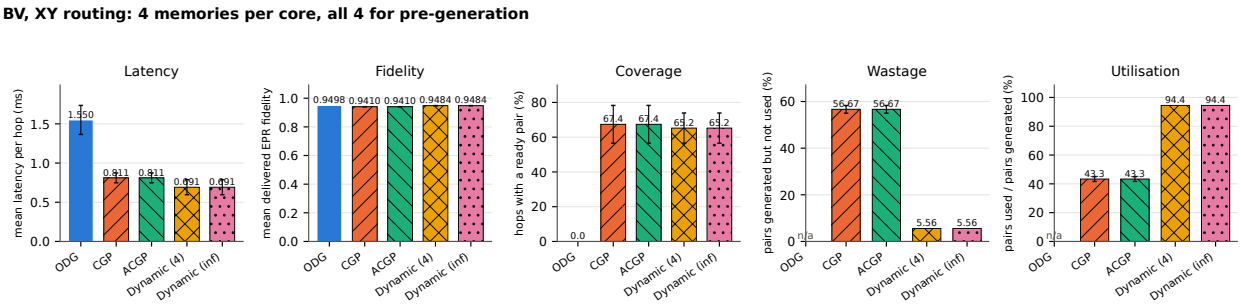


Figure 2.16: BV, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.3.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.550	1.550	1.550	1.550	1.550	1.550	1.550
CGP	1.550	1.228	0.948	0.811	0.526	0.648	0.504
ACGP	1.550	1.228	0.948	0.811	0.526	0.648	0.504
Dynamic	2.065	1.065	0.691	0.691	0.691	0.691	0.691

Table 2.17: BV, XY routing: latency (mean latency per hop (ms)) for every K.

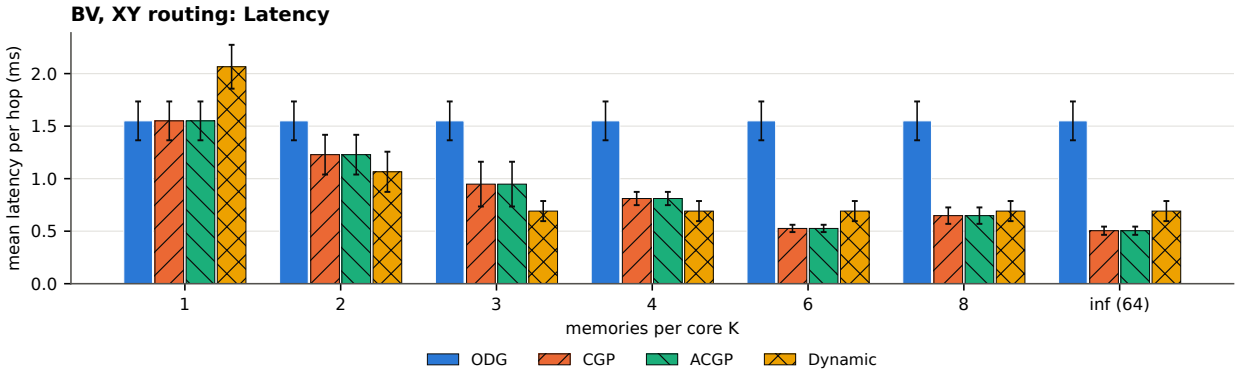


Figure 2.17: BV, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9460	0.9432	0.9410	0.9434	0.9447	0.9473
ACGP	0.9498	0.9460	0.9432	0.9410	0.9434	0.9447	0.9473
Dynamic	0.9493	0.9477	0.9484	0.9484	0.9484	0.9484	0.9484

Table 2.18: BV, XY routing: fidelity (mean delivered EPR fidelity) for every K.

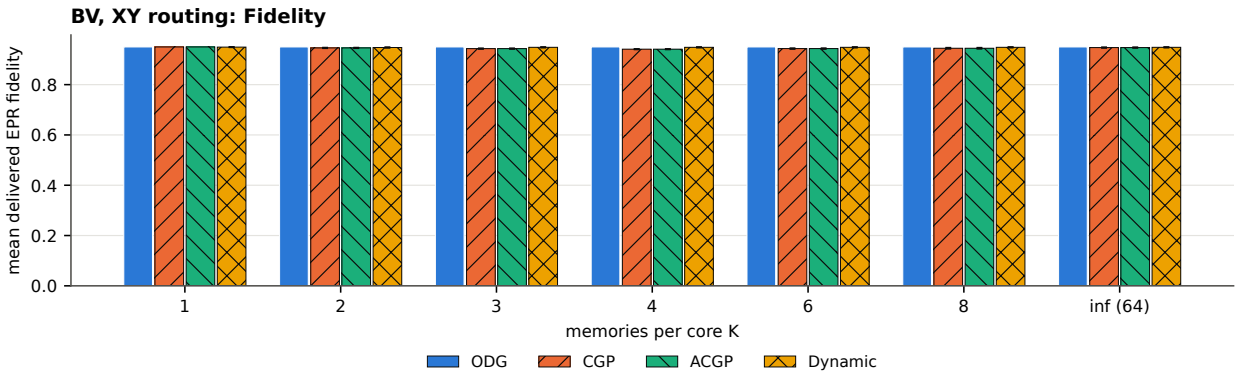


Figure 2.18: BV, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	23.9	45.7	67.4	84.8	73.9	93.5
ACGP	0.0	23.9	45.7	67.4	84.8	73.9	93.5
Dynamic	43.5	63.0	65.2	65.2	65.2	65.2	65.2

Table 2.19: BV, XY routing: coverage (hops with a ready pair (%)) for every K.

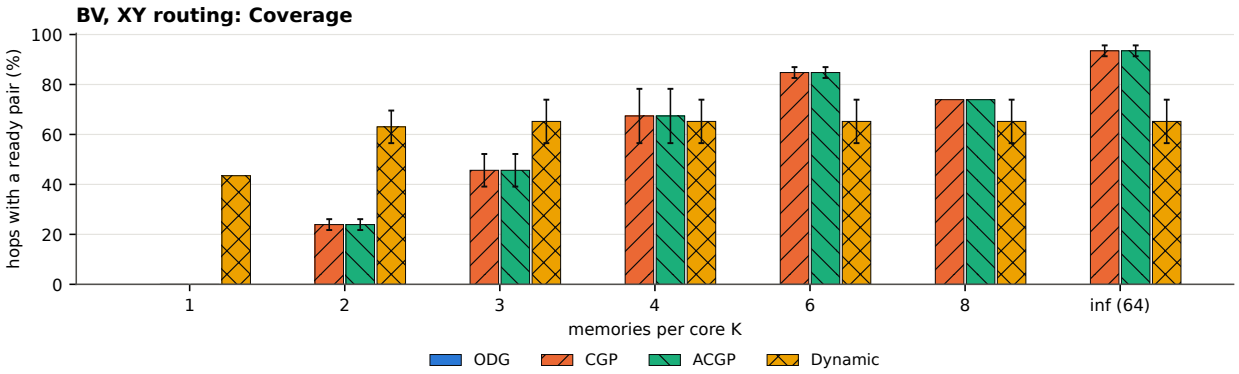


Figure 2.19: BV, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	46.43	54.19	56.67	57.98	72.36	90.97
ACGP	–	46.43	54.19	56.67	57.98	72.36	90.97
Dynamic	37.50	15.00	5.56	5.56	5.56	5.56	5.56

Table 2.20: BV, XY routing: wastage (pairs generated but not used (%)) for every K.

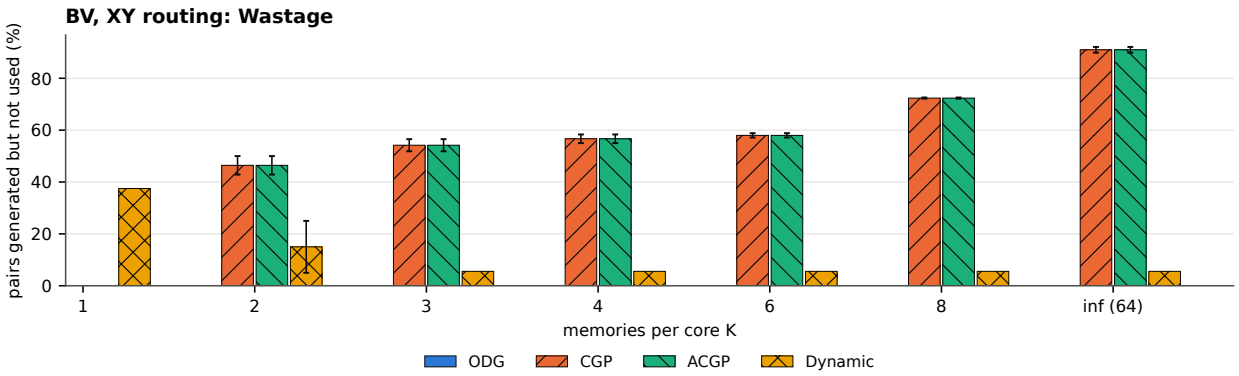


Figure 2.20: BV, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	53.6	45.8	43.3	42.0	27.6	9.0
ACGP	–	53.6	45.8	43.3	42.0	27.6	9.0
Dynamic	62.5	85.0	94.4	94.4	94.4	94.4	94.4

Table 2.21: BV, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

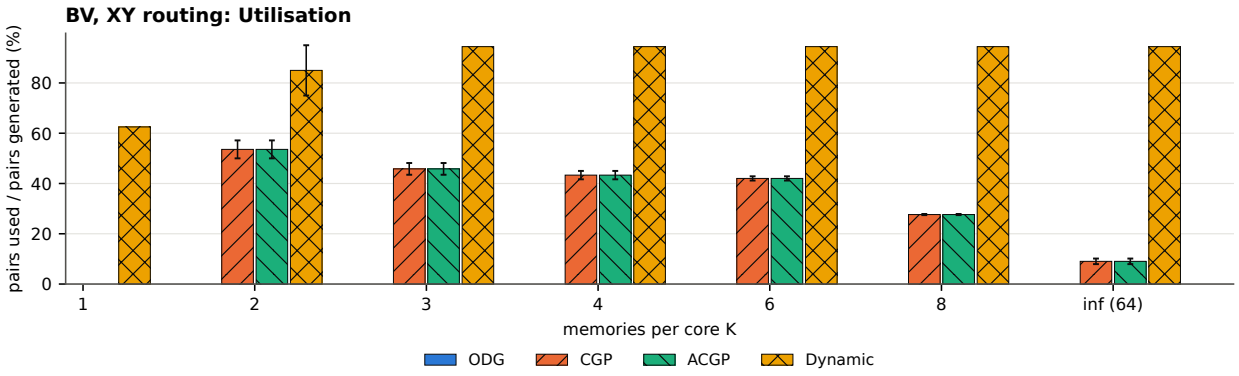


Figure 2.21: BV, XY routing: utilisation for every K.

2.4 BV, history routing

23 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.4.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.550	3.40	0.9498	0.0	0.0	–	–
CGP	0.648	1.60	0.9447	73.9	60.7	72.36	27.6
ACGP	0.648	1.60	0.9447	73.9	60.7	72.36	27.6
Dynamic (8)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4
Dynamic (∞)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4

Table 2.22: BV, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

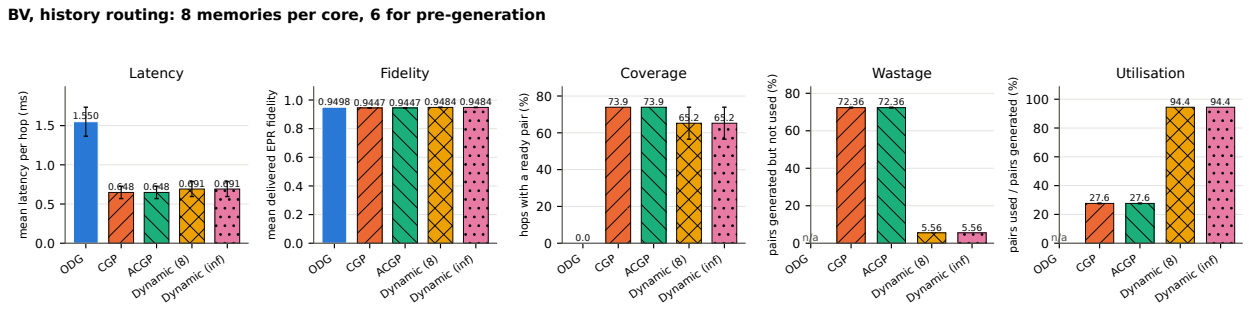


Figure 2.22: BV, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.4.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.550	3.40	0.9498	0.0	0.0	–	–
CGP	0.811	2.40	0.9410	67.4	46.4	56.67	43.3
ACGP	0.811	2.40	0.9410	67.4	46.4	56.67	43.3
Dynamic (4)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4
Dynamic (∞)	0.691	1.70	0.9484	65.2	53.6	5.56	94.4

Table 2.23: BV, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

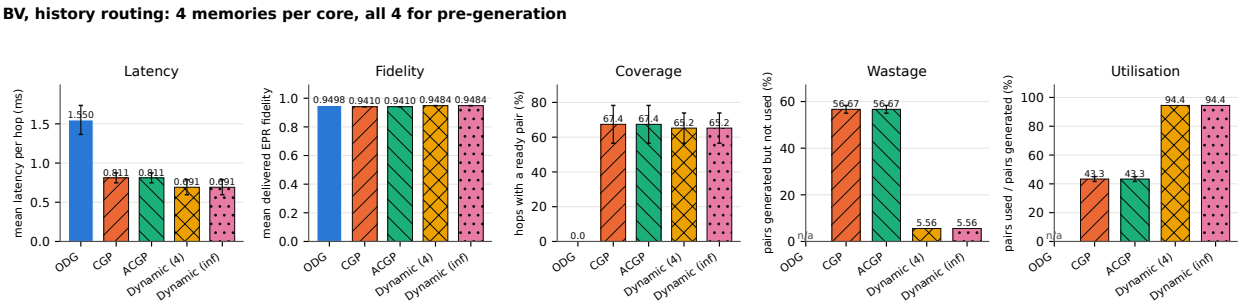


Figure 2.23: BV, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.4.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.550	1.550	1.550	1.550	1.550	1.550	1.550
CGP	1.550	1.228	0.948	0.811	0.526	0.648	0.504
ACGP	1.550	1.228	0.948	0.811	0.526	0.648	0.504
Dynamic	2.065	1.065	0.691	0.691	0.691	0.691	0.691

Table 2.24: BV, history routing: latency (mean latency per hop (ms)) for every K.

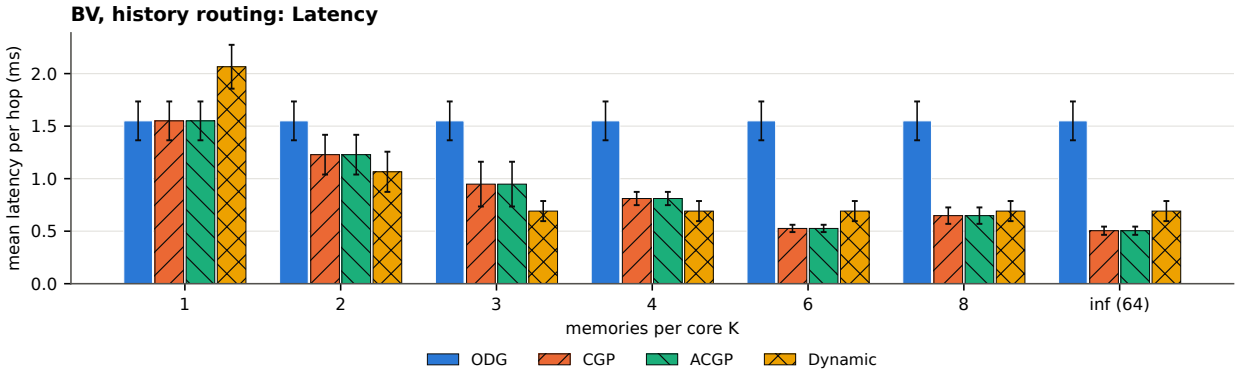


Figure 2.24: BV, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9460	0.9432	0.9410	0.9434	0.9447	0.9473
ACGP	0.9498	0.9460	0.9432	0.9410	0.9434	0.9447	0.9473
Dynamic	0.9493	0.9477	0.9484	0.9484	0.9484	0.9484	0.9484

Table 2.25: BV, history routing: fidelity (mean delivered EPR fidelity) for every K.

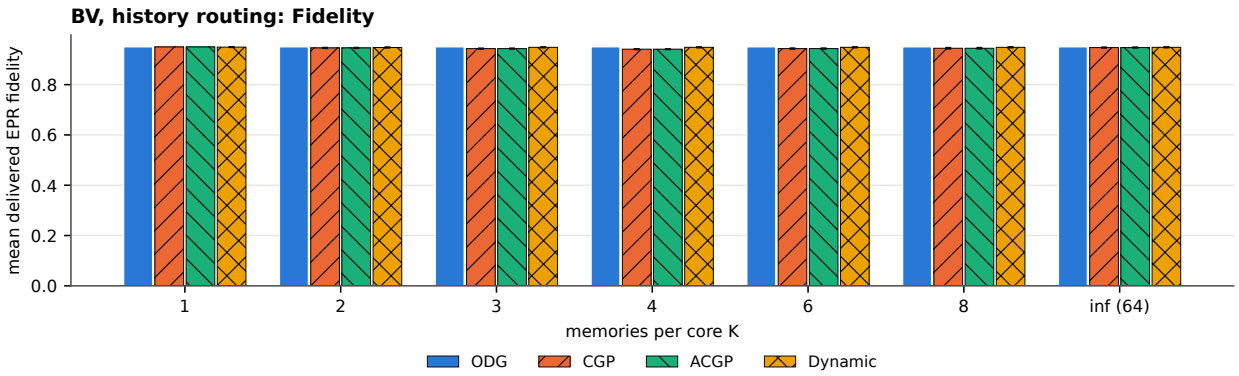


Figure 2.25: BV, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	23.9	45.7	67.4	84.8	73.9	93.5
ACGP	0.0	23.9	45.7	67.4	84.8	73.9	93.5
Dynamic	43.5	63.0	65.2	65.2	65.2	65.2	65.2

Table 2.26: BV, history routing: coverage (hops with a ready pair (%)) for every K.

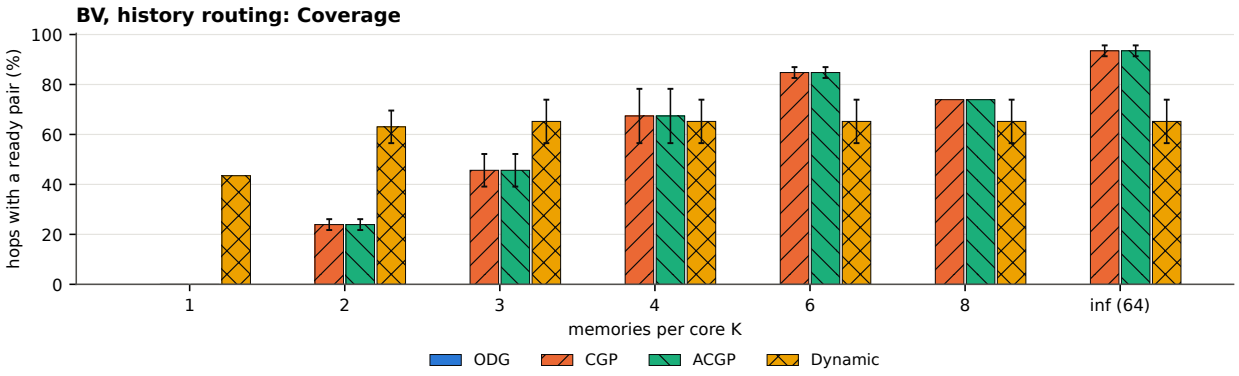


Figure 2.26: BV, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	46.43	54.19	56.67	57.98	72.36	90.97
ACGP	–	46.43	54.19	56.67	57.98	72.36	90.97
Dynamic	37.50	15.00	5.56	5.56	5.56	5.56	5.56

Table 2.27: BV, history routing: wastage (pairs generated but not used (%)) for every K.

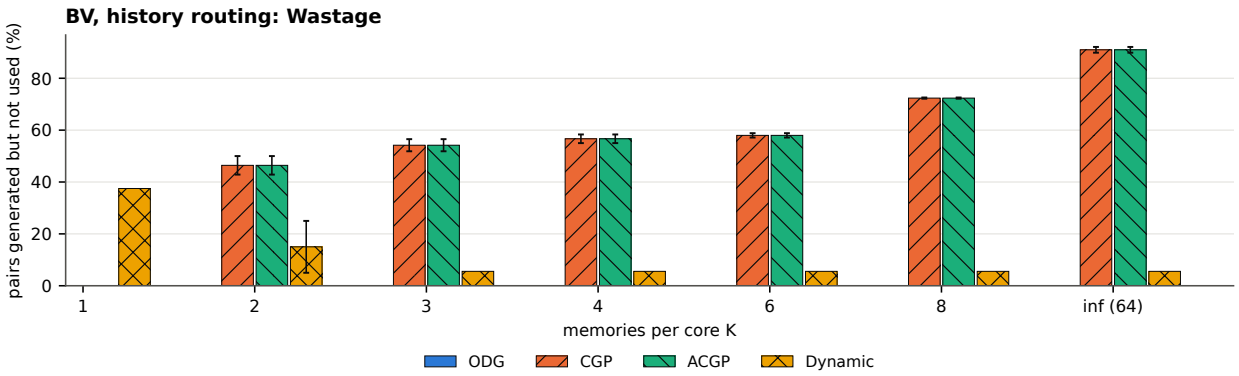


Figure 2.27: BV, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	53.6	45.8	43.3	42.0	27.6	9.0
ACGP	–	53.6	45.8	43.3	42.0	27.6	9.0
Dynamic	62.5	85.0	94.4	94.4	94.4	94.4	94.4

Table 2.28: BV, history routing: utilisation (pairs used / pairs generated (%)) for every K.

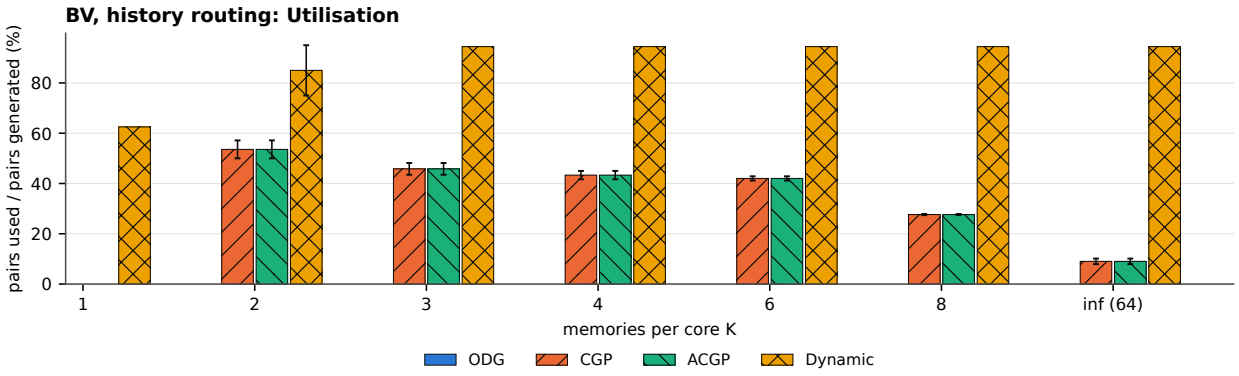


Figure 2.28: BV, history routing: utilisation for every K.

2.5 GHZ, XY routing

24 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.5.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.923	4.20	0.9498	0.0	0.0	–	–
CGP	0.633	1.45	0.9432	79.2	70.0	69.81	30.2
ACGP	0.633	1.45	0.9432	79.2	70.0	69.81	30.2
Dynamic (8)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5
Dynamic (∞)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5

Table 2.29: GHZ, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

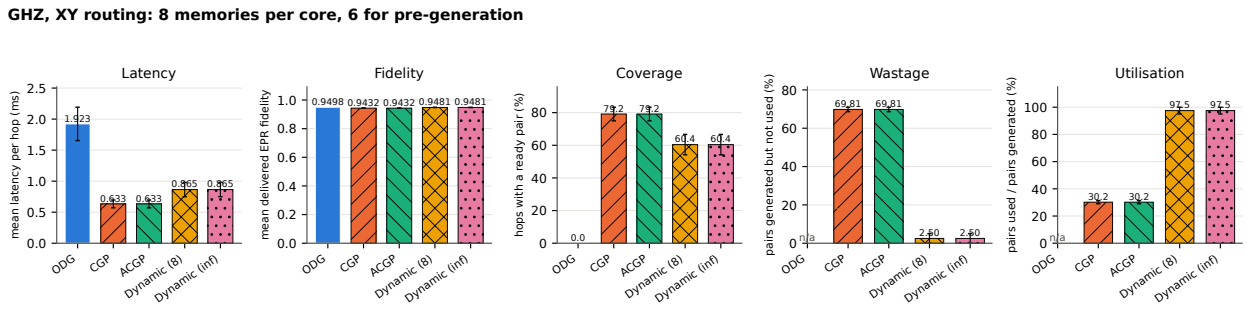


Figure 2.29: GHZ, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.5.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.923	4.20	0.9498	0.0	0.0	–	–
CGP	0.940	3.15	0.9412	64.6	46.7	55.71	44.3
ACGP	0.940	3.15	0.9412	64.6	46.7	55.71	44.3
Dynamic (4)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5
Dynamic (∞)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5

Table 2.30: GHZ, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

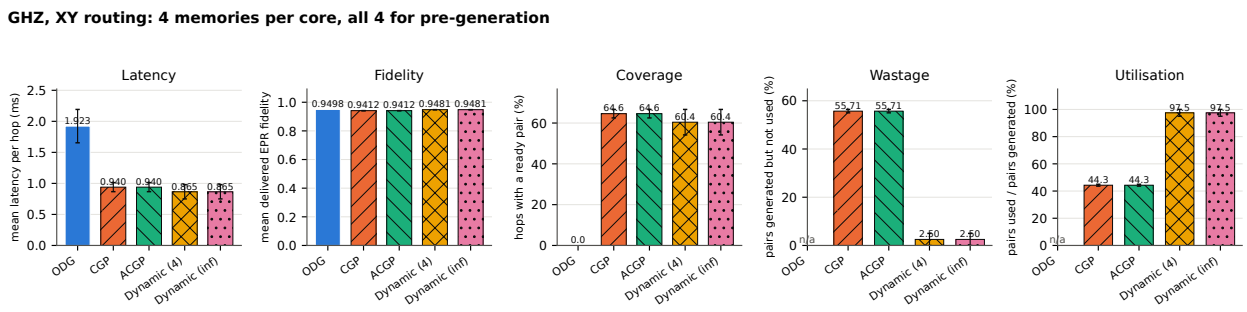


Figure 2.30: GHZ, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.5.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.923	1.923	1.923	1.923	1.923	1.923	1.923
CGP	1.923	1.623	1.225	0.940	0.671	0.633	0.558
ACGP	1.923	1.623	1.225	0.940	0.671	0.633	0.558
Dynamic	2.406	1.260	0.865	0.865	0.865	0.865	0.865

Table 2.31: GHZ, XY routing: latency (mean latency per hop (ms)) for every K.

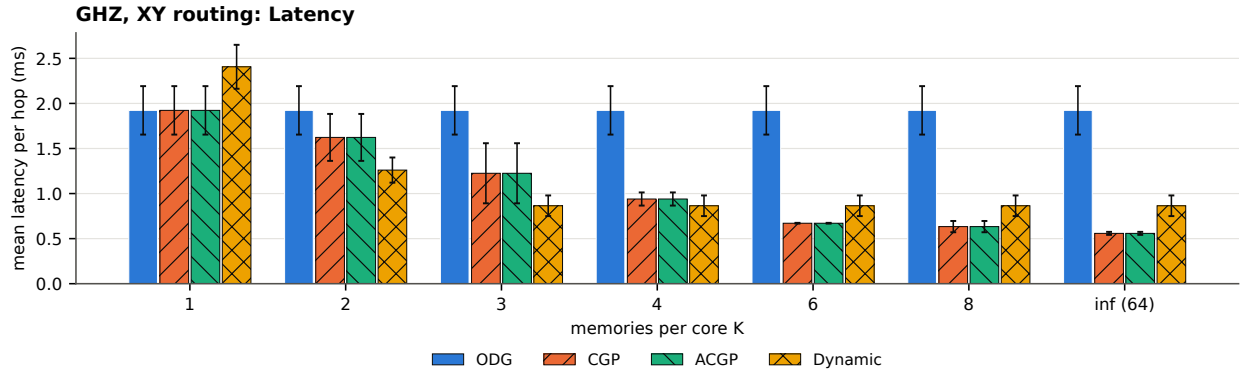


Figure 2.31: GHZ, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9442	0.9436	0.9412	0.9437	0.9432	0.9471
ACGP	0.9498	0.9442	0.9436	0.9412	0.9437	0.9432	0.9471
Dynamic	0.9489	0.9479	0.9481	0.9481	0.9481	0.9481	0.9481

Table 2.32: GHZ, XY routing: fidelity (mean delivered EPR fidelity) for every K.

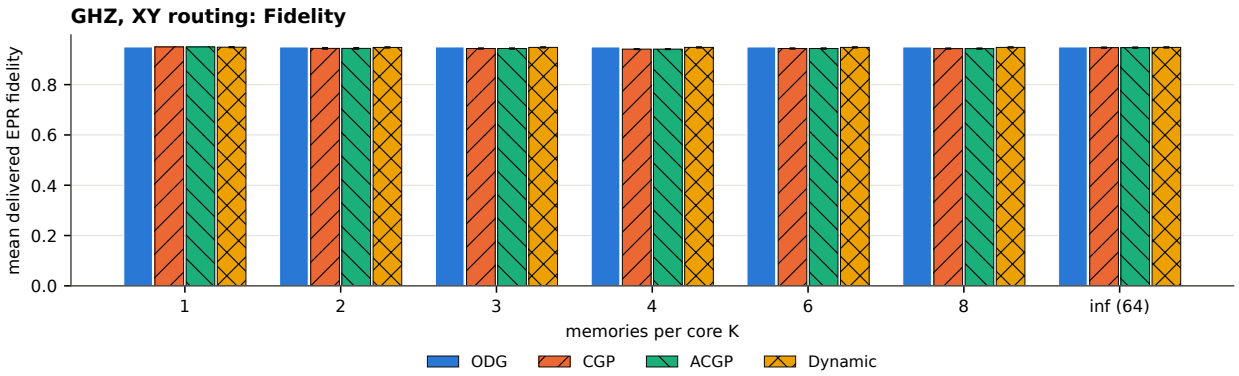


Figure 2.32: GHZ, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	25.0	47.9	64.6	79.2	79.2	89.6
ACGP	0.0	25.0	47.9	64.6	79.2	79.2	89.6
Dynamic	45.8	56.2	60.4	60.4	60.4	60.4	60.4

Table 2.33: GHZ, XY routing: coverage (hops with a ready pair (%)) for every K.

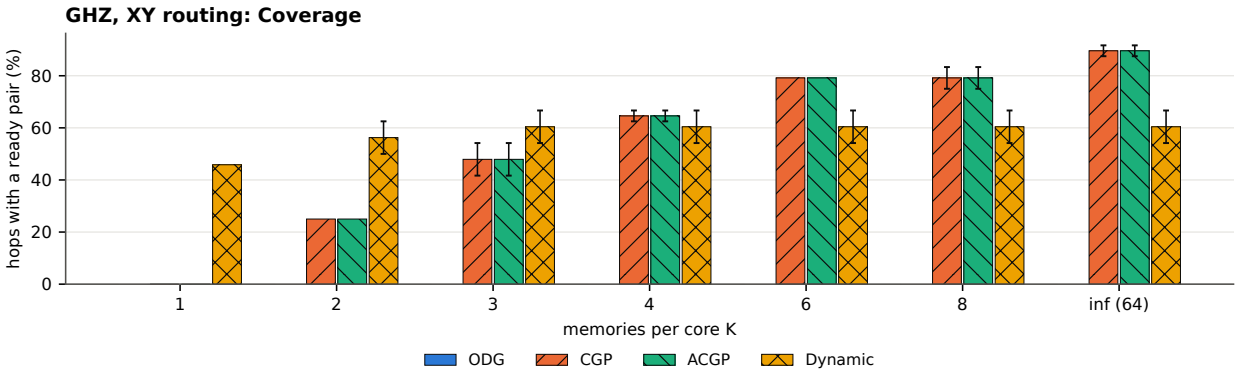


Figure 2.33: GHZ, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	46.41	53.00	55.71	58.82	69.81	91.75
ACGP	–	46.41	53.00	55.71	58.82	69.81	91.75
Dynamic	35.29	22.04	2.50	2.50	2.50	2.50	2.50

Table 2.34: GHZ, XY routing: wastage (pairs generated but not used (%)) for every K.

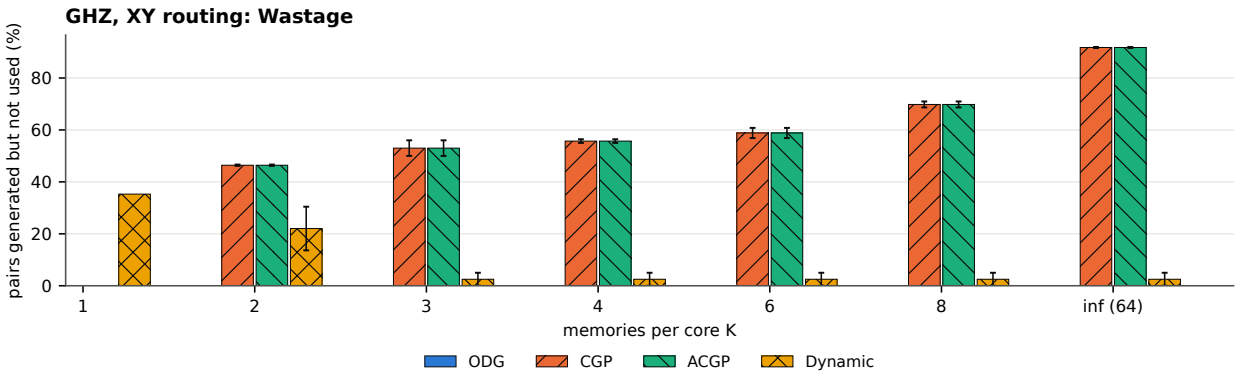


Figure 2.34: GHZ, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	53.6	47.0	44.3	41.2	30.2	8.3
ACGP	–	53.6	47.0	44.3	41.2	30.2	8.3
Dynamic	64.7	78.0	97.5	97.5	97.5	97.5	97.5

Table 2.35: GHZ, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

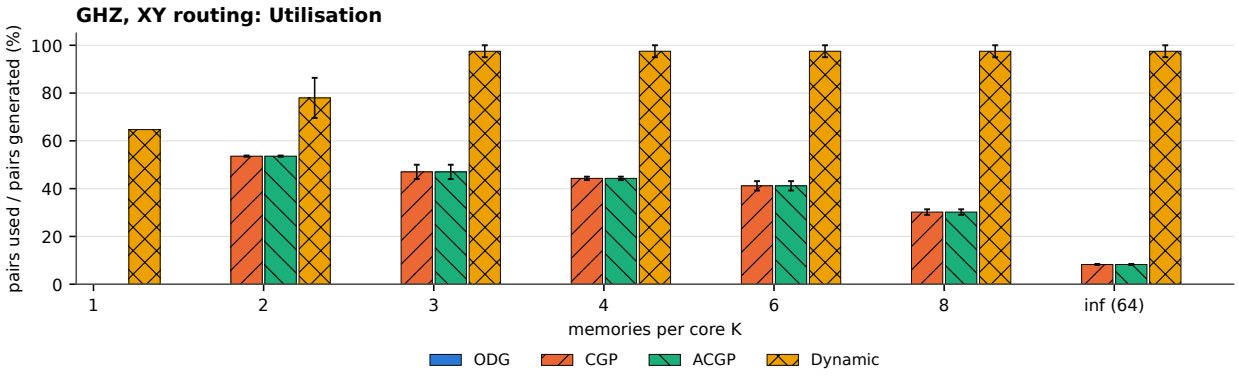


Figure 2.35: GHZ, XY routing: utilisation for every K.

2.6 GHZ, history routing

24 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.6.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.923	4.20	0.9498	0.0	0.0	–	–
CGP	0.633	1.45	0.9432	79.2	70.0	69.81	30.2
ACGP	0.633	1.45	0.9432	79.2	70.0	69.81	30.2
Dynamic (8)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5
Dynamic (∞)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5

Table 2.36: GHZ, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

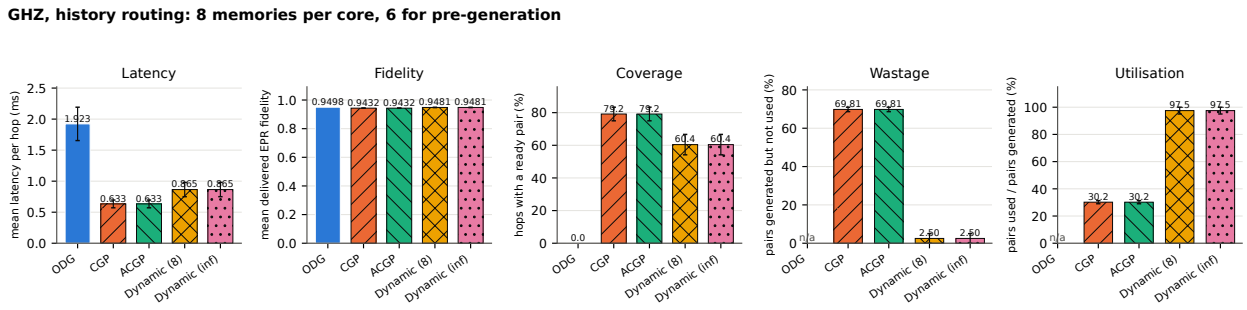


Figure 2.36: GHZ, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.6.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.923	4.20	0.9498	0.0	0.0	–	–
CGP	0.940	3.15	0.9412	64.6	46.7	55.71	44.3
ACGP	0.940	3.15	0.9412	64.6	46.7	55.71	44.3
Dynamic (4)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5
Dynamic (∞)	0.865	2.00	0.9481	60.4	40.0	2.50	97.5

Table 2.37: GHZ, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

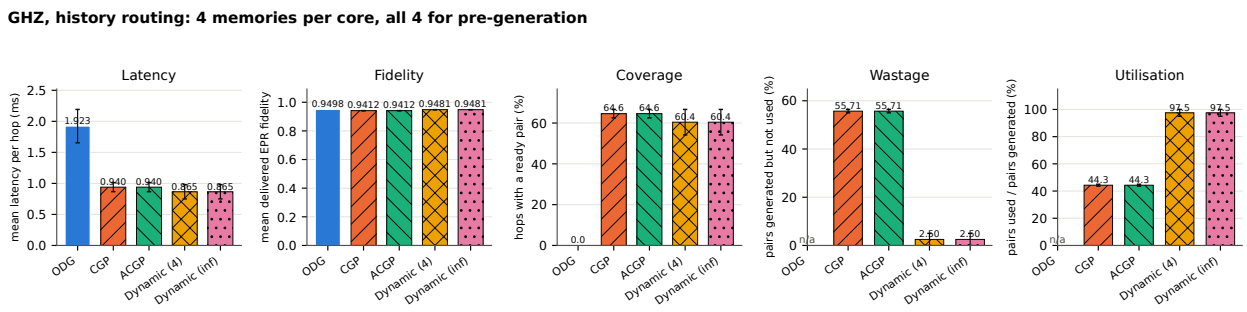


Figure 2.37: GHZ, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.6.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.923	1.923	1.923	1.923	1.923	1.923	1.923
CGP	1.923	1.623	1.225	0.940	0.671	0.633	0.558
ACGP	1.923	1.623	1.225	0.940	0.671	0.633	0.558
Dynamic	2.406	1.260	0.865	0.865	0.865	0.865	0.865

Table 2.38: GHZ, history routing: latency (mean latency per hop (ms)) for every K.

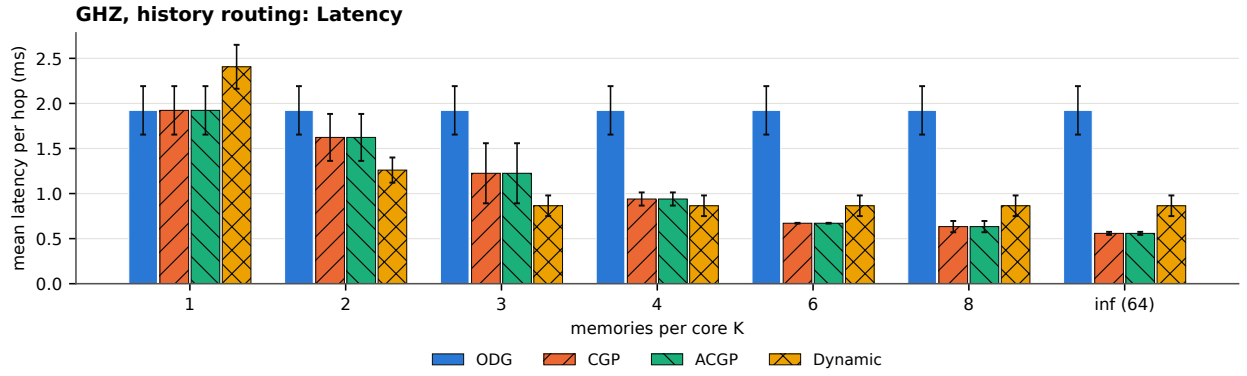


Figure 2.38: GHZ, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9442	0.9436	0.9412	0.9437	0.9432	0.9471
ACGP	0.9498	0.9442	0.9436	0.9412	0.9437	0.9432	0.9471
Dynamic	0.9489	0.9479	0.9481	0.9481	0.9481	0.9481	0.9481

Table 2.39: GHZ, history routing: fidelity (mean delivered EPR fidelity) for every K.

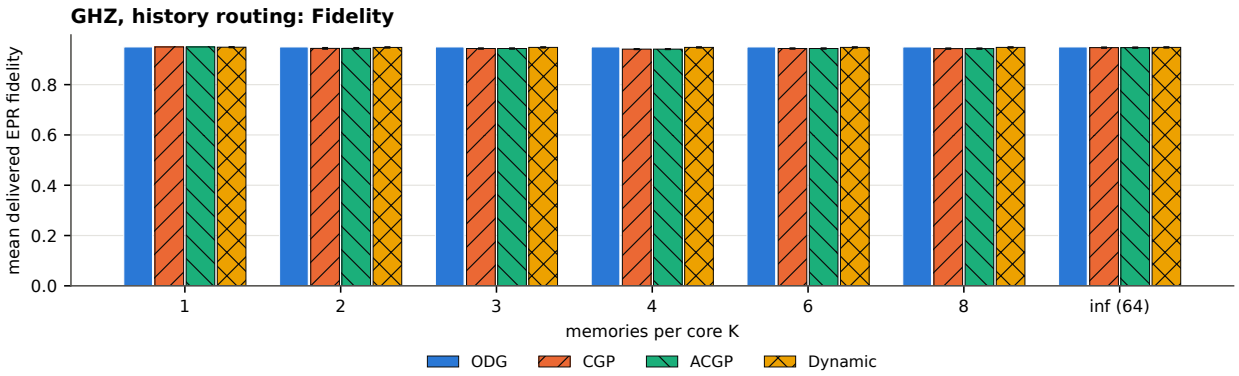


Figure 2.39: GHZ, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	25.0	47.9	64.6	79.2	79.2	89.6
ACGP	0.0	25.0	47.9	64.6	79.2	79.2	89.6
Dynamic	45.8	56.2	60.4	60.4	60.4	60.4	60.4

Table 2.40: GHZ, history routing: coverage (hops with a ready pair (%)) for every K.

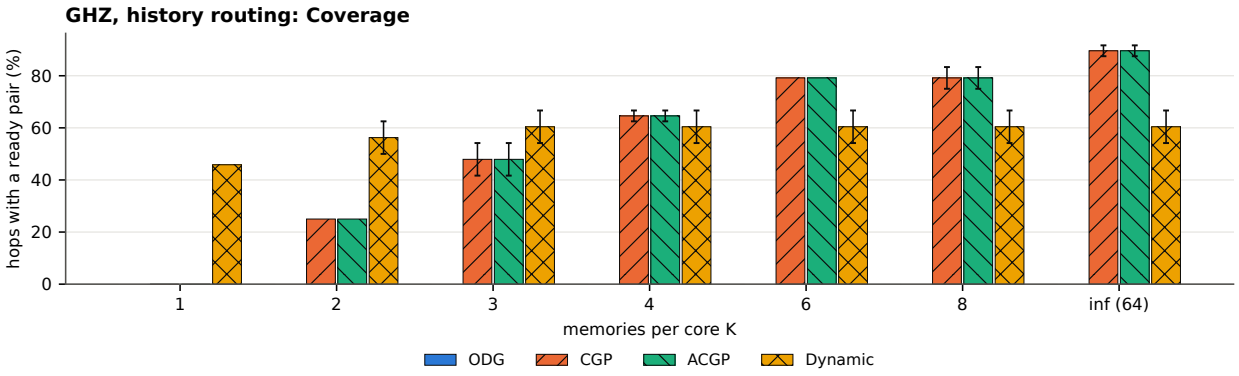


Figure 2.40: GHZ, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	46.41	53.00	55.71	58.82	69.81	91.75
ACGP	–	46.41	53.00	55.71	58.82	69.81	91.75
Dynamic	35.29	22.04	2.50	2.50	2.50	2.50	2.50

Table 2.41: GHZ, history routing: wastage (pairs generated but not used (%)) for every K.

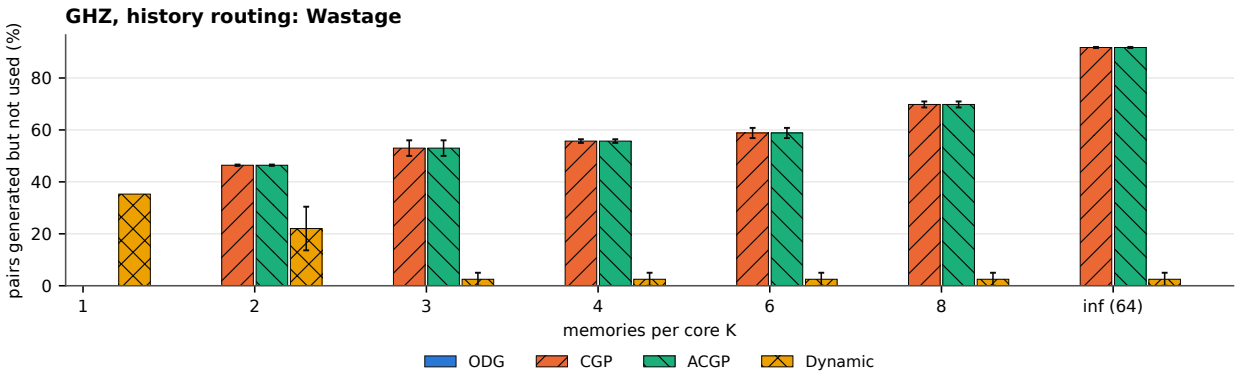


Figure 2.41: GHZ, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	53.6	47.0	44.3	41.2	30.2	8.3
ACGP	–	53.6	47.0	44.3	41.2	30.2	8.3
Dynamic	64.7	78.0	97.5	97.5	97.5	97.5	97.5

Table 2.42: GHZ, history routing: utilisation (pairs used / pairs generated (%)) for every K.

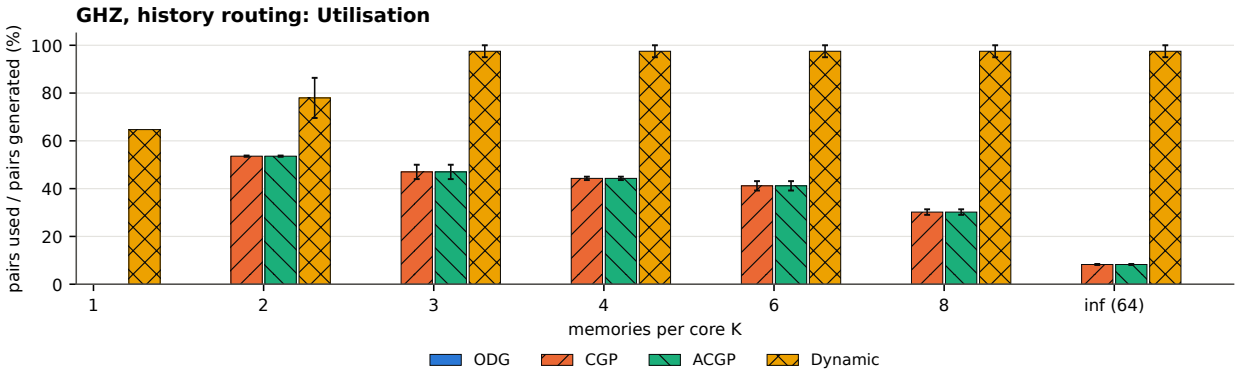


Figure 2.42: GHZ, history routing: utilisation for every K.

2.7 DJ, XY routing

30 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.7.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.777	3.95	0.9498	0.0	0.0	–	–
CGP	0.643	1.45	0.9427	71.7	62.5	65.73	34.3
ACGP	0.643	1.45	0.9427	71.7	62.5	65.73	34.3
Dynamic (8)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5
Dynamic (∞)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5

Table 2.43: DJ, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

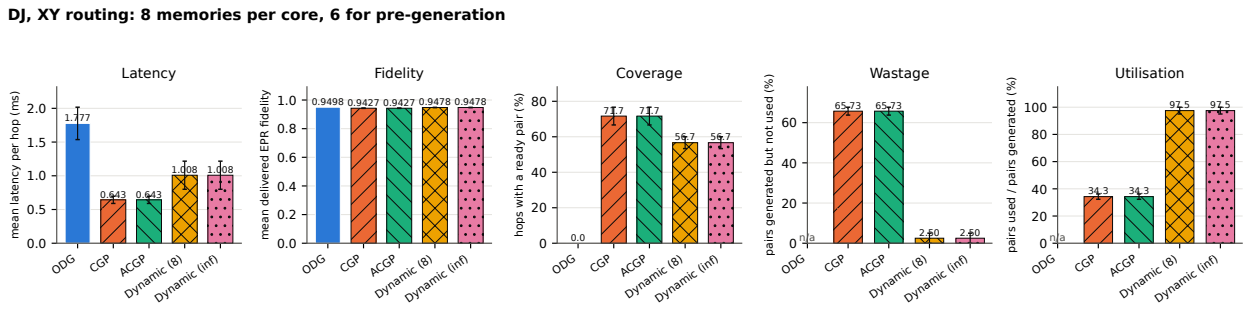


Figure 2.43: DJ, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.7.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.777	3.95	0.9498	0.0	0.0	–	–
CGP	0.858	3.10	0.9377	68.3	50.0	49.44	50.6
ACGP	0.858	3.10	0.9377	68.3	50.0	49.44	50.6
Dynamic (4)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5
Dynamic (∞)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5

Table 2.44: DJ, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

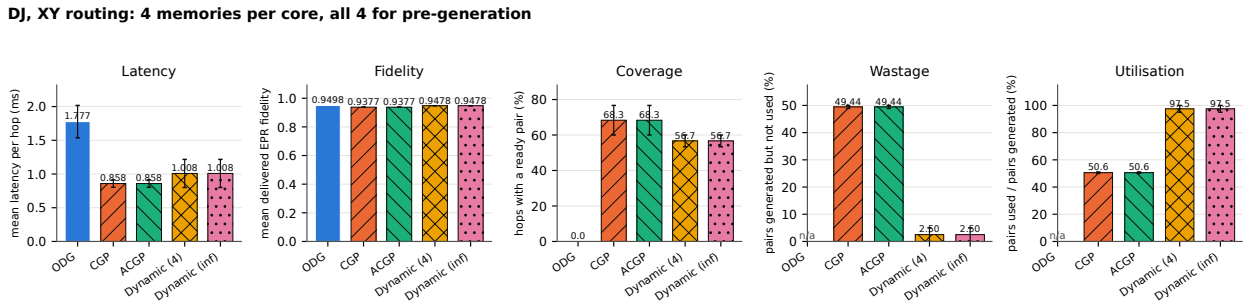


Figure 2.44: DJ, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.7.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.777	1.777	1.777	1.777	1.777	1.777	1.777
CGP	1.777	1.457	1.152	0.858	0.612	0.643	0.508
ACGP	1.777	1.457	1.152	0.858	0.612	0.643	0.508
Dynamic	2.167	1.375	1.008	1.008	1.008	1.008	1.008

Table 2.45: DJ, XY routing: latency (mean latency per hop (ms)) for every K.

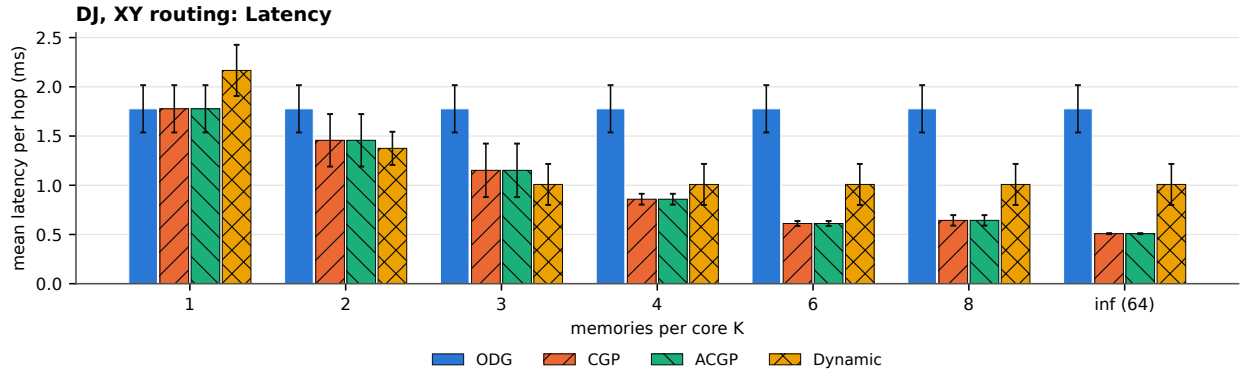


Figure 2.45: DJ, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9433	0.9386	0.9377	0.9392	0.9427	0.9473
ACGP	0.9498	0.9433	0.9386	0.9377	0.9392	0.9427	0.9473
Dynamic	0.9493	0.9470	0.9478	0.9478	0.9478	0.9478	0.9478

Table 2.46: DJ, XY routing: fidelity (mean delivered EPR fidelity) for every K.

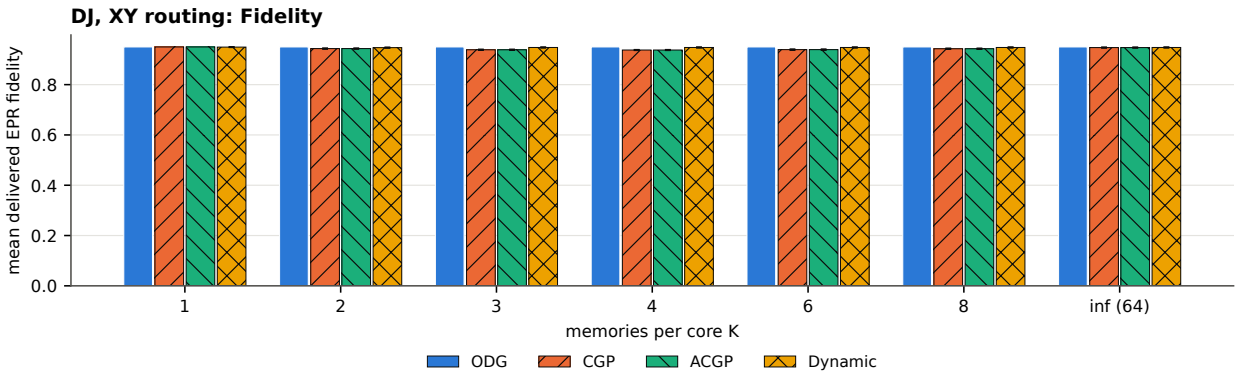


Figure 2.46: DJ, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	26.7	45.0	68.3	80.0	71.7	90.0
ACGP	0.0	26.7	45.0	68.3	80.0	71.7	90.0
Dynamic	36.7	46.7	56.7	56.7	56.7	56.7	56.7

Table 2.47: DJ, XY routing: coverage (hops with a ready pair (%)) for every K.

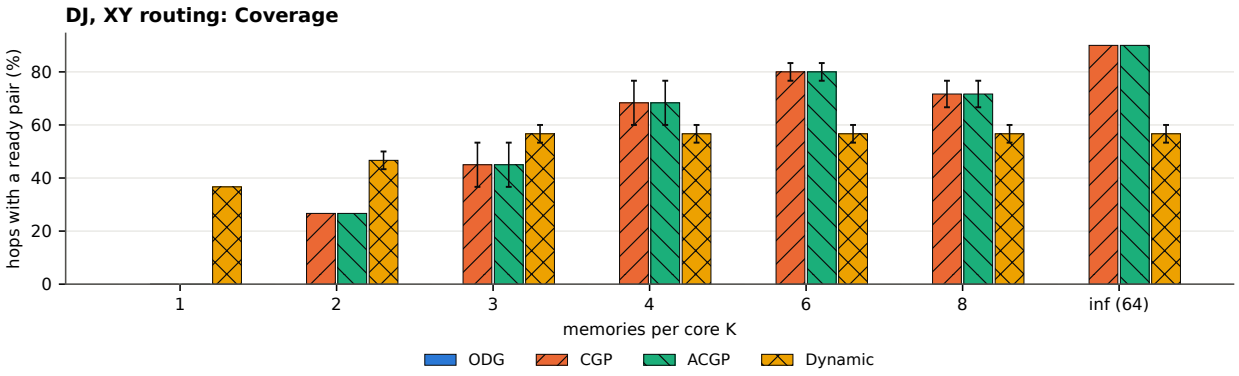


Figure 2.47: DJ, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	40.18	47.88	49.44	52.33	65.73	91.10
ACGP	–	40.18	47.88	49.44	52.33	65.73	91.10
Dynamic	38.89	14.64	2.50	2.50	2.50	2.50	2.50

Table 2.48: DJ, XY routing: wastage (pairs generated but not used (%)) for every K.

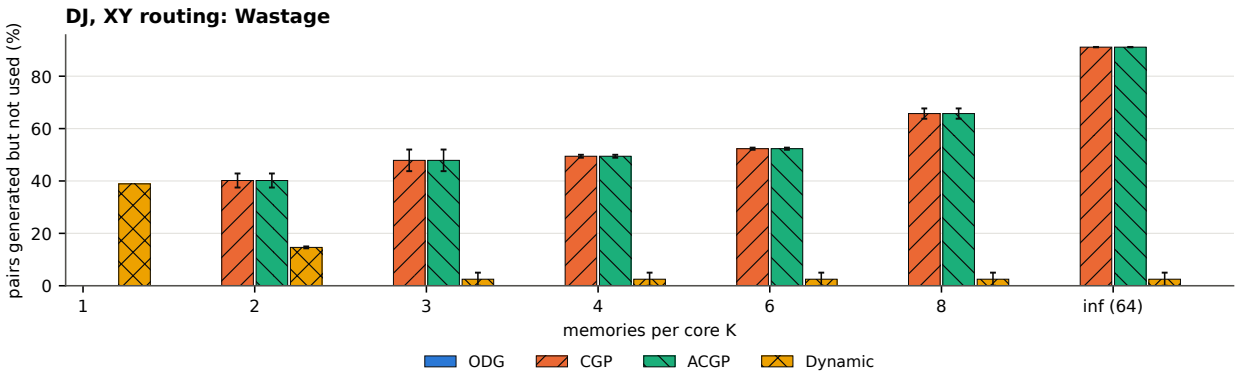


Figure 2.48: DJ, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	59.8	52.1	50.6	47.7	34.3	8.9
ACGP	–	59.8	52.1	50.6	47.7	34.3	8.9
Dynamic	61.1	85.4	97.5	97.5	97.5	97.5	97.5

Table 2.49: DJ, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

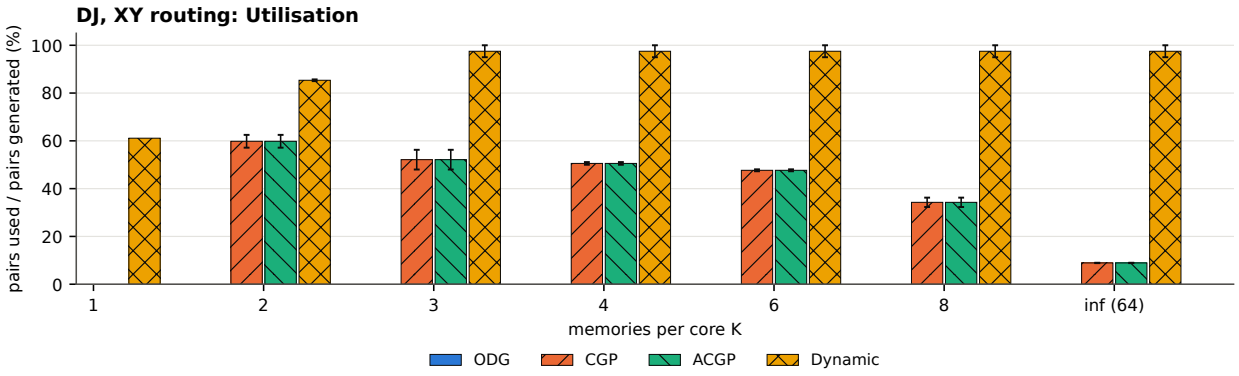


Figure 2.49: DJ, XY routing: utilisation for every K.

2.8 DJ, history routing

30 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.8.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.777	3.95	0.9498	0.0	0.0	–	–
CGP	0.643	1.45	0.9427	71.7	62.5	65.73	34.3
ACGP	0.643	1.45	0.9427	71.7	62.5	65.73	34.3
Dynamic (8)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5
Dynamic (∞)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5

Table 2.50: DJ, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

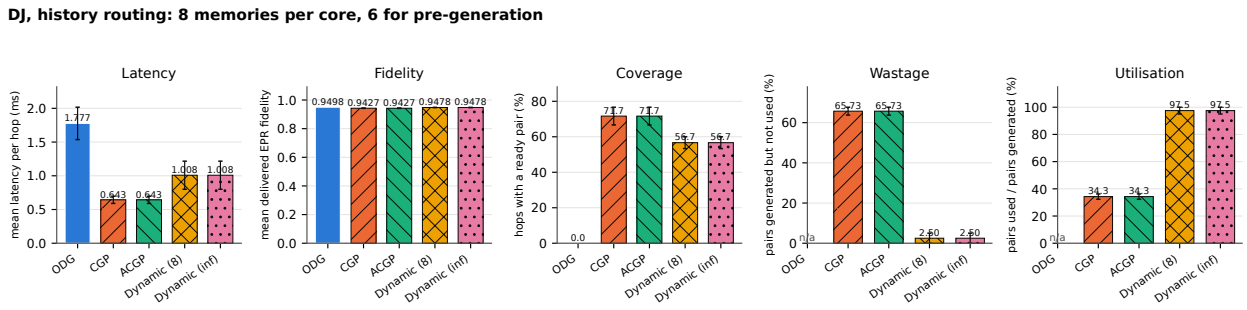


Figure 2.50: DJ, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.8.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.777	3.95	0.9498	0.0	0.0	–	–
CGP	0.858	3.10	0.9377	68.3	50.0	49.44	50.6
ACGP	0.858	3.10	0.9377	68.3	50.0	49.44	50.6
Dynamic (4)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5
Dynamic (∞)	1.008	2.85	0.9478	56.7	53.1	2.50	97.5

Table 2.51: DJ, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

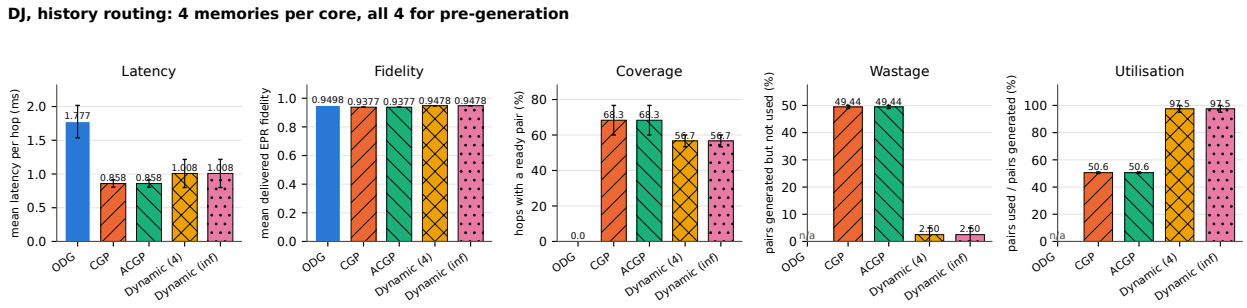


Figure 2.51: DJ, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.8.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.777	1.777	1.777	1.777	1.777	1.777	1.777
CGP	1.777	1.457	1.152	0.858	0.612	0.643	0.508
ACGP	1.777	1.457	1.152	0.858	0.612	0.643	0.508
Dynamic	2.167	1.375	1.008	1.008	1.008	1.008	1.008

Table 2.52: DJ, history routing: latency (mean latency per hop (ms)) for every K.

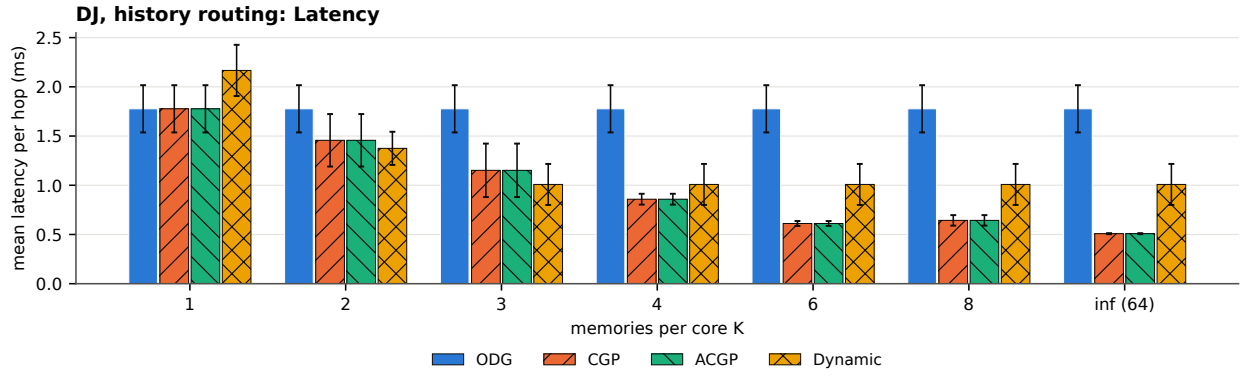


Figure 2.52: DJ, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9433	0.9386	0.9377	0.9392	0.9427	0.9473
ACGP	0.9498	0.9433	0.9386	0.9377	0.9392	0.9427	0.9473
Dynamic	0.9493	0.9470	0.9478	0.9478	0.9478	0.9478	0.9478

Table 2.53: DJ, history routing: fidelity (mean delivered EPR fidelity) for every K.

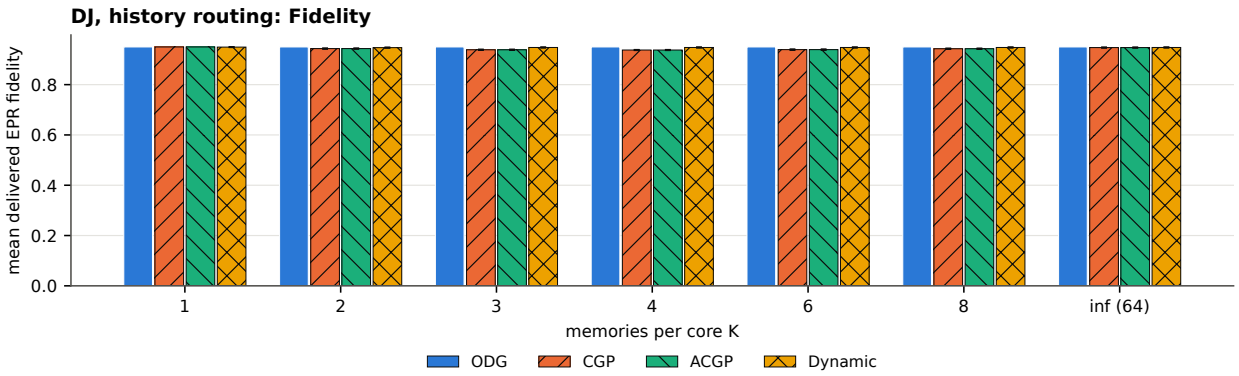


Figure 2.53: DJ, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	26.7	45.0	68.3	80.0	71.7	90.0
ACGP	0.0	26.7	45.0	68.3	80.0	71.7	90.0
Dynamic	36.7	46.7	56.7	56.7	56.7	56.7	56.7

Table 2.54: DJ, history routing: coverage (hops with a ready pair (%)) for every K.

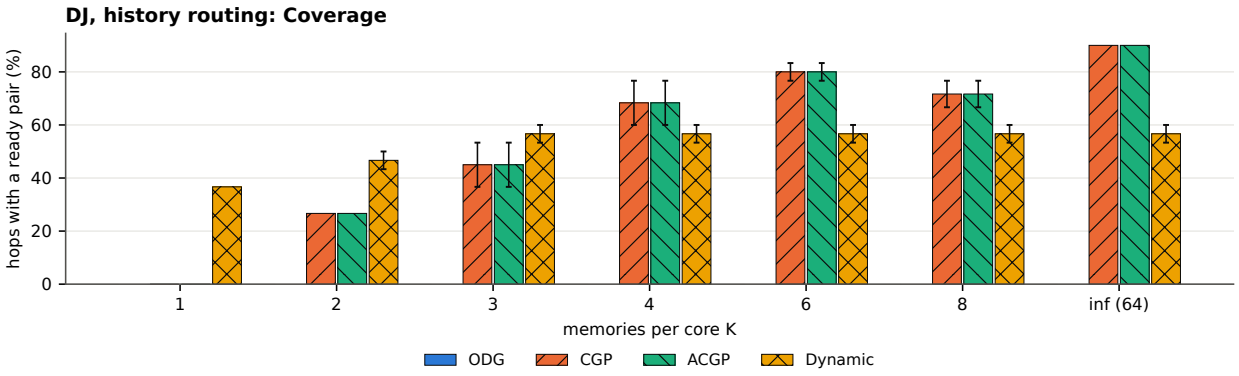


Figure 2.54: DJ, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	40.18	47.88	49.44	52.33	65.73	91.10
ACGP	–	40.18	47.88	49.44	52.33	65.73	91.10
Dynamic	38.89	14.64	2.50	2.50	2.50	2.50	2.50

Table 2.55: DJ, history routing: wastage (pairs generated but not used (%)) for every K.

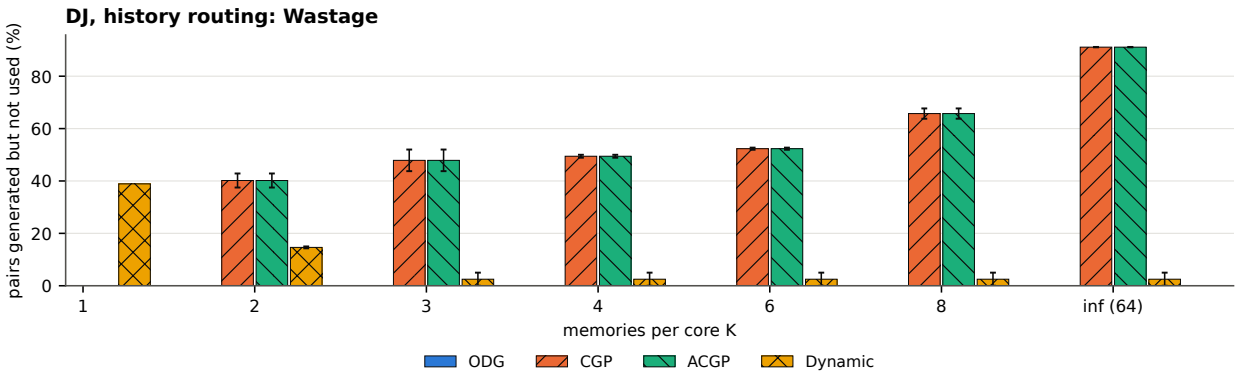


Figure 2.55: DJ, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	59.8	52.1	50.6	47.7	34.3	8.9
ACGP	–	59.8	52.1	50.6	47.7	34.3	8.9
Dynamic	61.1	85.4	97.5	97.5	97.5	97.5	97.5

Table 2.56: DJ, history routing: utilisation (pairs used / pairs generated (%)) for every K.

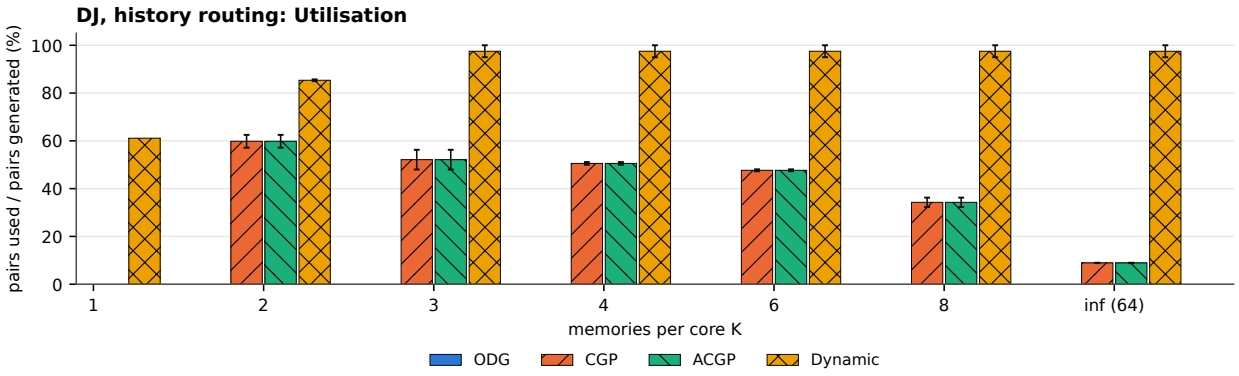


Figure 2.56: DJ, history routing: utilisation for every K.

2.9 QPE (exact), XY routing

11921 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.9.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.784	4.60	0.9498	0.0	0.0	–	–
CGP	1.020	3.40	0.9420	75.4	54.8	4.42	95.6
ACGP	1.016	3.35	0.9422	76.1	56.3	4.32	95.7
Dynamic (8)	0.758	2.40	0.9435	77.0	64.6	0.99	99.0
Dynamic (∞)	0.615	1.65	0.9440	84.9	72.9	1.07	98.9

Table 2.57: QPE (exact), XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

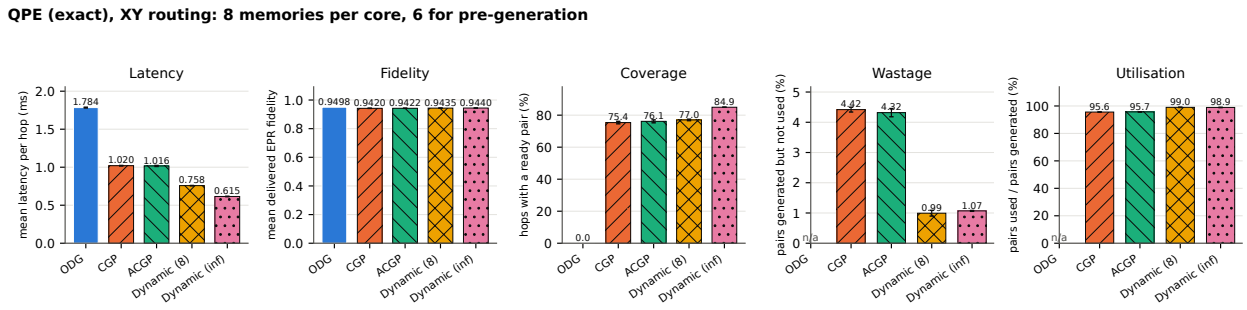


Figure 2.57: QPE (exact), XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.9.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.894	4.70	0.9498	0.0	0.0	–	–
CGP	2.527	9.72	0.9400	61.5	36.4	5.96	94.0
ACGP	2.408	9.05	0.9405	63.2	39.6	5.48	94.5
Dynamic (4)	1.358	4.80	0.9424	60.6	49.4	4.00	96.0
Dynamic (∞)	0.615	1.65	0.9440	84.9	72.9	1.07	98.9

Table 2.58: QPE (exact), XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

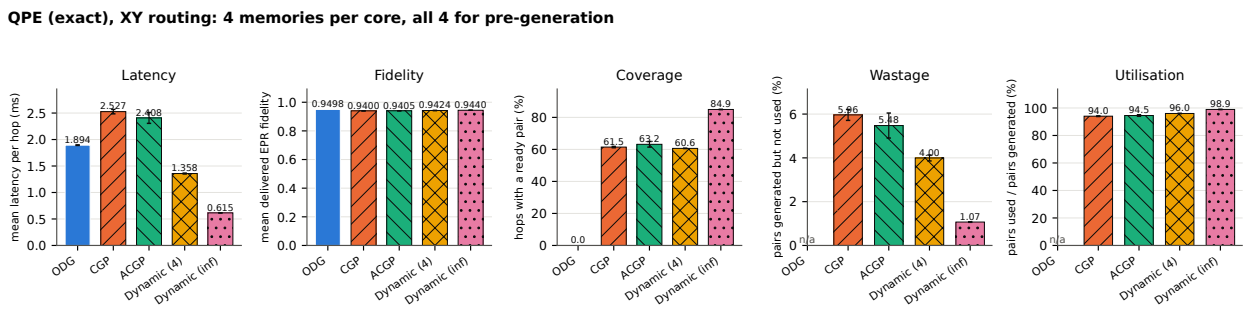


Figure 2.58: QPE (exact), XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.9.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	6.455	2.860	2.122	1.894	1.792	1.784	1.784
CGP	6.455	4.218	3.079	2.527	1.205	1.020	0.458
ACGP	6.455	4.345	3.096	2.408	1.223	1.016	0.457
Dynamic	8.929	3.513	1.897	1.358	0.897	0.758	0.615

Table 2.59: QPE (exact), XY routing: latency (mean latency per hop (ms)) for every K.

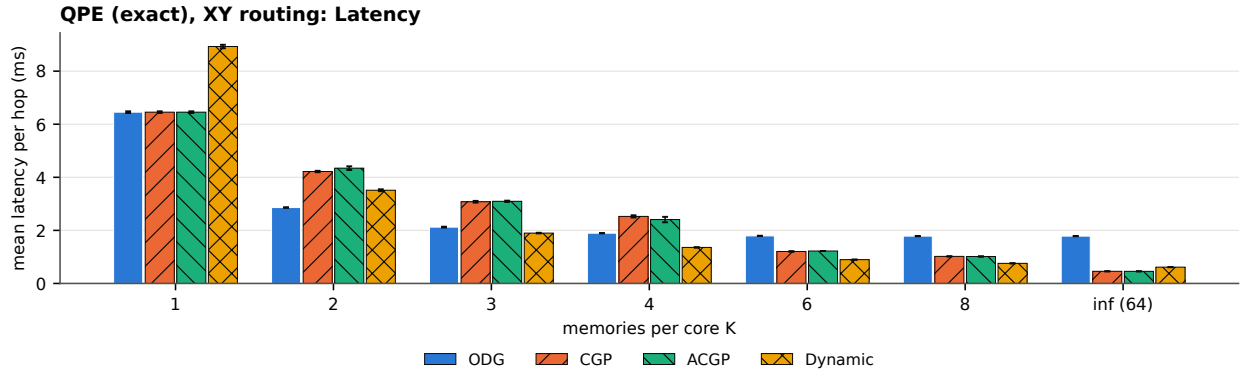


Figure 2.59: QPE (exact), XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9445	0.9419	0.9400	0.9435	0.9420	0.9421
ACGP	0.9498	0.9445	0.9419	0.9405	0.9431	0.9422	0.9423
Dynamic	0.9488	0.9444	0.9432	0.9424	0.9429	0.9435	0.9440

Table 2.60: QPE (exact), XY routing: fidelity (mean delivered EPR fidelity) for every K.

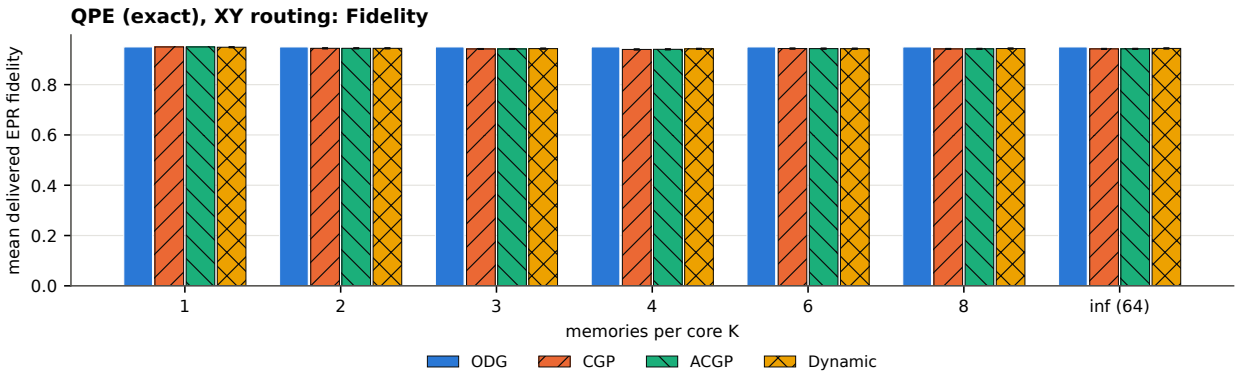


Figure 2.60: QPE (exact), XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	26.4	47.6	61.5	64.1	75.4	99.2
ACGP	0.0	26.1	47.1	63.2	64.0	76.1	99.1
Dynamic	10.0	32.8	48.4	60.6	76.7	77.0	84.9

Table 2.61: QPE (exact), XY routing: coverage (hops with a ready pair (%)) for every K.

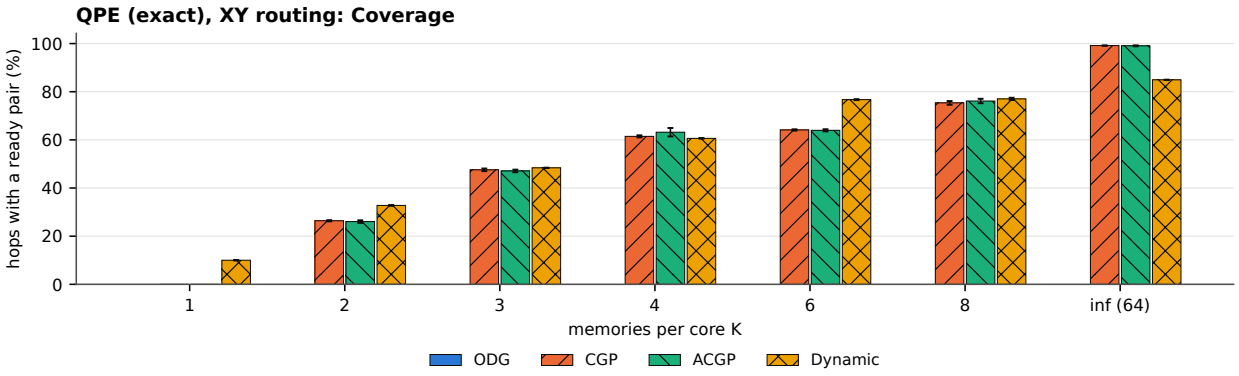


Figure 2.61: QPE (exact), XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	6.46	6.04	5.96	3.60	4.42	12.61
ACGP	–	6.71	5.66	5.48	3.78	4.32	12.50
Dynamic	49.72	13.88	6.38	4.00	2.09	0.99	1.07

Table 2.62: QPE (exact), XY routing: wastage (pairs generated but not used (%)) for every K.

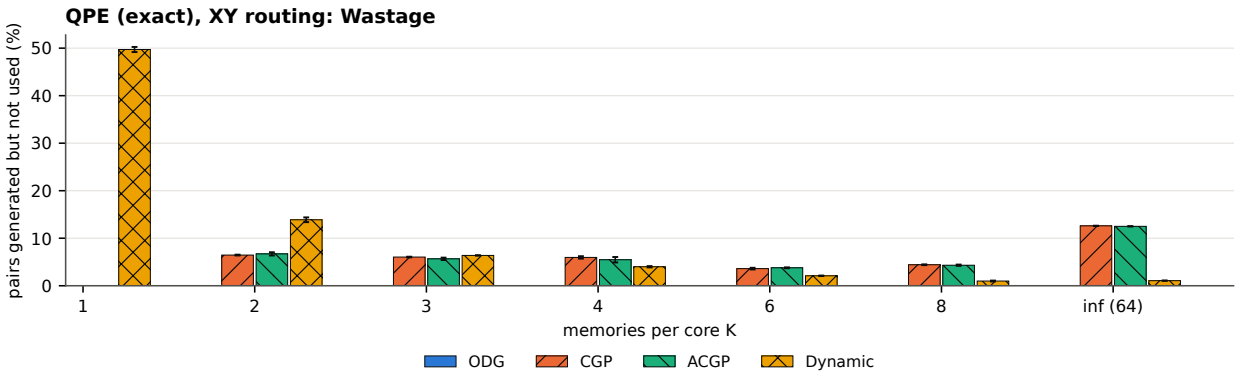


Figure 2.62: QPE (exact), XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	93.5	94.0	94.0	96.4	95.6	87.4
ACGP	–	93.3	94.3	94.5	96.2	95.7	87.5
Dynamic	50.3	86.1	93.6	96.0	97.9	99.0	98.9

Table 2.63: QPE (exact), XY routing: utilisation (pairs used / pairs generated (%)) for every K.

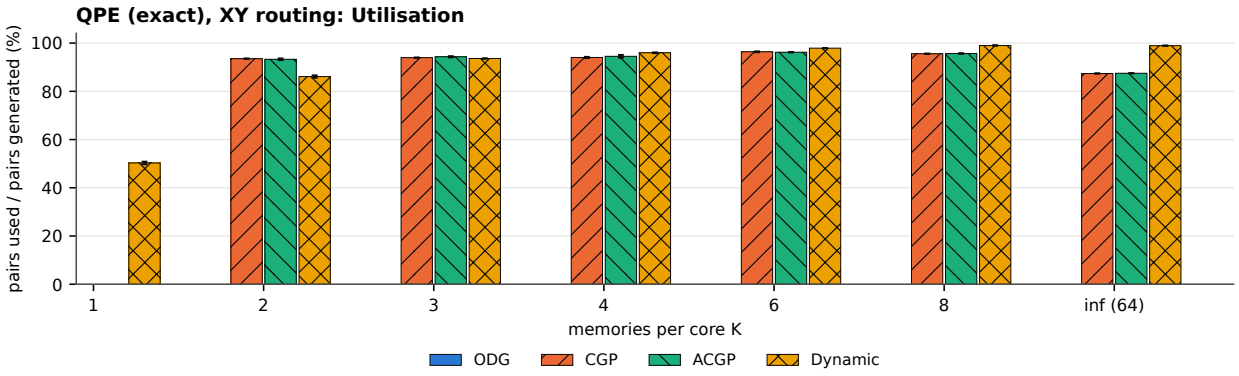


Figure 2.63: QPE (exact), XY routing: utilisation for every K.

2.10 QPE (exact), history routing

11921 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.10.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.774	4.40	0.9498	0.0	0.0	–	–
CGP	0.947	2.95	0.9428	76.3	55.6	4.16	95.8
ACGP	0.960	3.10	0.9425	76.2	55.0	3.97	96.0
Dynamic (8)	0.714	2.20	0.9434	79.5	67.0	1.10	98.9
Dynamic (∞)	0.578	1.40	0.9446	86.7	74.4	1.19	98.8

Table 2.64: QPE (exact), history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

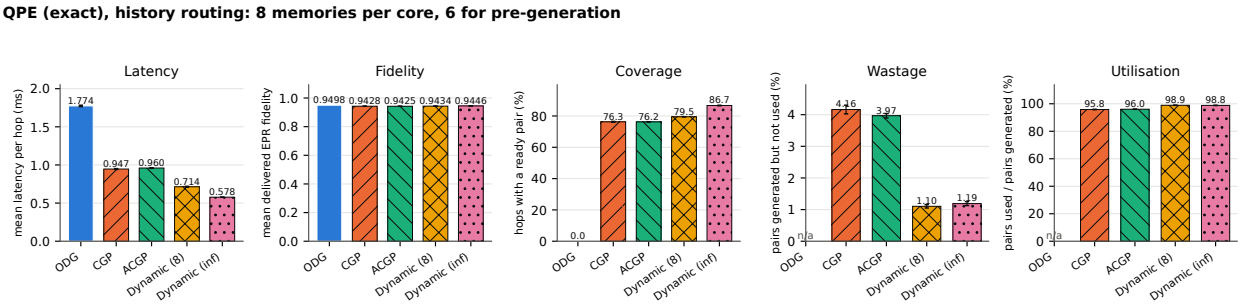


Figure 2.64: QPE (exact), history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.10.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.830	4.65	0.9498	0.0	0.0	–	–
CGP	2.256	8.20	0.9416	61.6	36.2	5.21	94.8
ACGP	2.248	8.31	0.9413	63.4	38.6	5.33	94.7
Dynamic (4)	1.305	4.60	0.9417	62.6	51.4	4.57	95.4
Dynamic (∞)	0.578	1.40	0.9446	86.7	74.4	1.19	98.8

Table 2.65: QPE (exact), history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

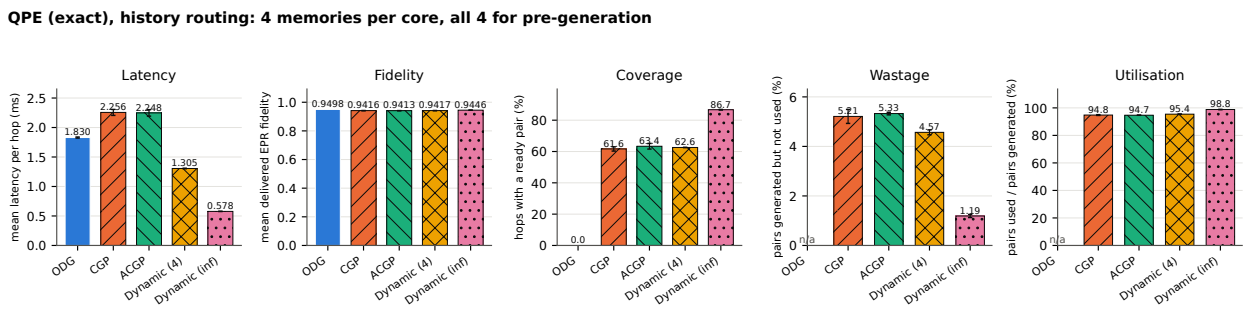


Figure 2.65: QPE (exact), history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.10.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	6.043	2.654	2.011	1.830	1.776	1.774	1.775
CGP	6.043	3.831	2.846	2.256	1.162	0.947	0.456
ACGP	6.043	3.843	2.809	2.248	1.127	0.960	0.456
Dynamic	8.359	3.260	1.832	1.305	0.856	0.714	0.578

Table 2.66: QPE (exact), history routing: latency (mean latency per hop (ms)) for every K.

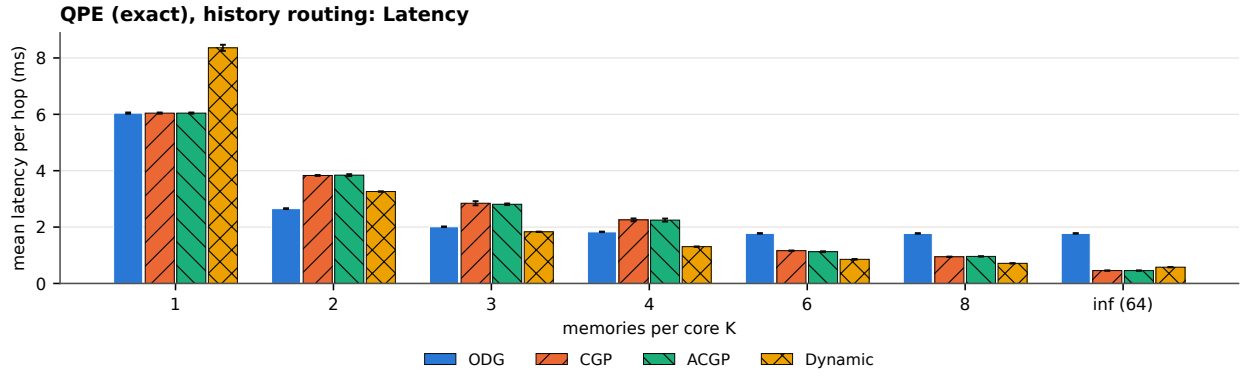


Figure 2.66: QPE (exact), history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9451	0.9426	0.9416	0.9434	0.9428	0.9421
ACGP	0.9498	0.9450	0.9426	0.9413	0.9435	0.9425	0.9419
Dynamic	0.9485	0.9446	0.9426	0.9417	0.9428	0.9434	0.9446

Table 2.67: QPE (exact), history routing: fidelity (mean delivered EPR fidelity) for every K.

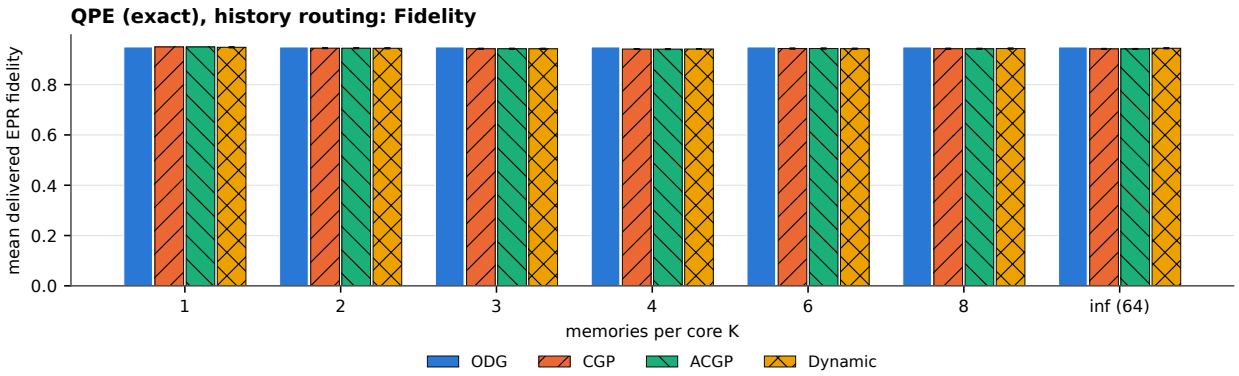


Figure 2.67: QPE (exact), history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	26.7	47.6	61.6	63.6	76.3	99.3
ACGP	0.0	27.9	48.4	63.4	65.1	76.2	99.3
Dynamic	11.3	33.1	49.1	62.6	78.8	79.5	86.7

Table 2.68: QPE (exact), history routing: coverage (hops with a ready pair (%)) for every K.

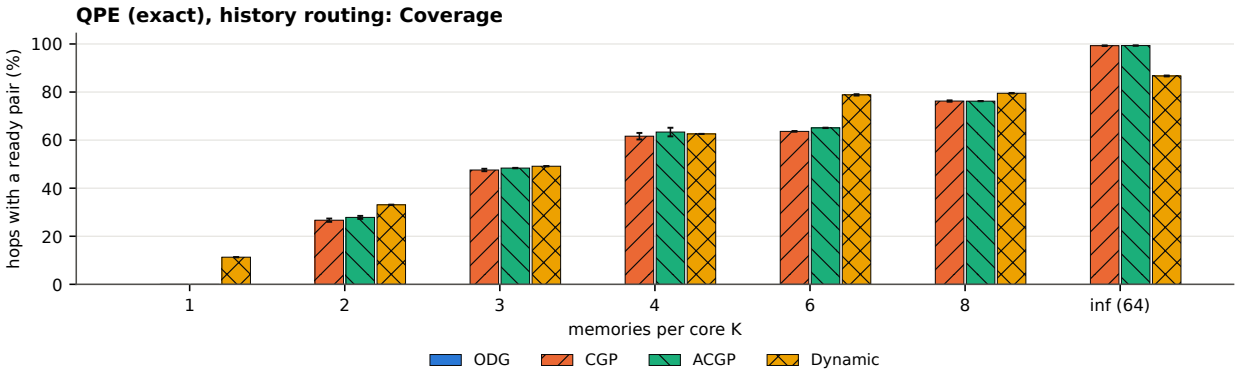


Figure 2.68: QPE (exact), history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	5.93	5.44	5.21	3.71	4.16	12.64
ACGP	–	5.61	5.27	5.33	3.54	3.97	12.64
Dynamic	45.07	15.64	7.46	4.57	2.19	1.10	1.19

Table 2.69: QPE (exact), history routing: wastage (pairs generated but not used (%)) for every K.

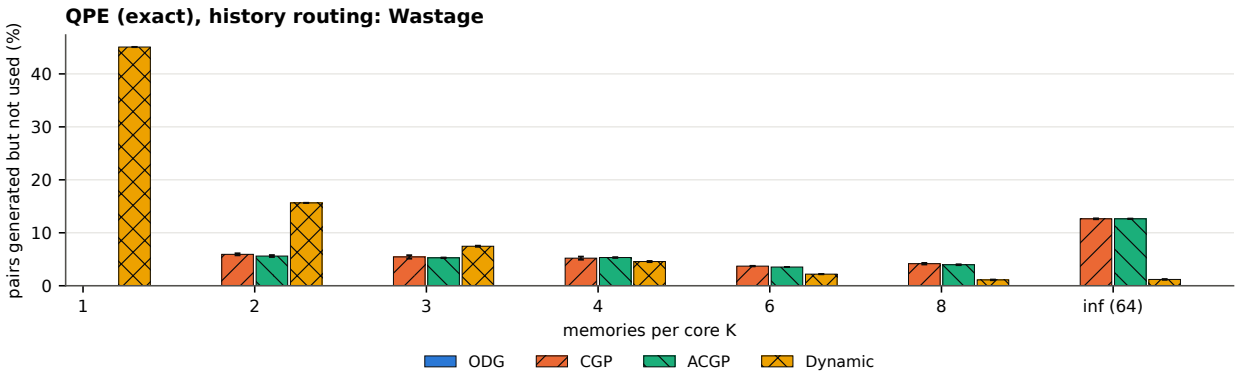


Figure 2.69: QPE (exact), history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	94.1	94.6	94.8	96.3	95.8	87.4
ACGP	–	94.4	94.7	94.7	96.5	96.0	87.4
Dynamic	54.9	84.4	92.5	95.4	97.8	98.9	98.8

Table 2.70: QPE (exact), history routing: utilisation (pairs used / pairs generated (%)) for every K.

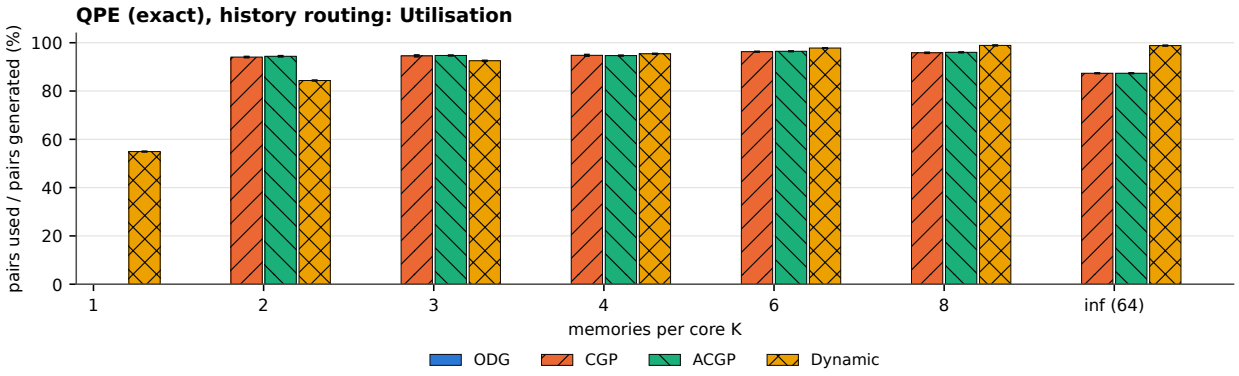


Figure 2.70: QPE (exact), history routing: utilisation for every K.

2.11 QAOA, XY routing

29605 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.11.1 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	8.860	3.673	2.493	2.076	–	–	–
CGP	8.860	5.593	3.951	2.984	–	–	–
ACGP	8.860	5.613	3.924	2.998	–	–	–
Dynamic	11.684	3.773	2.129	–	–	–	–

Table 2.71: QAOA, XY routing: latency (mean latency per hop (ms)) for every K.

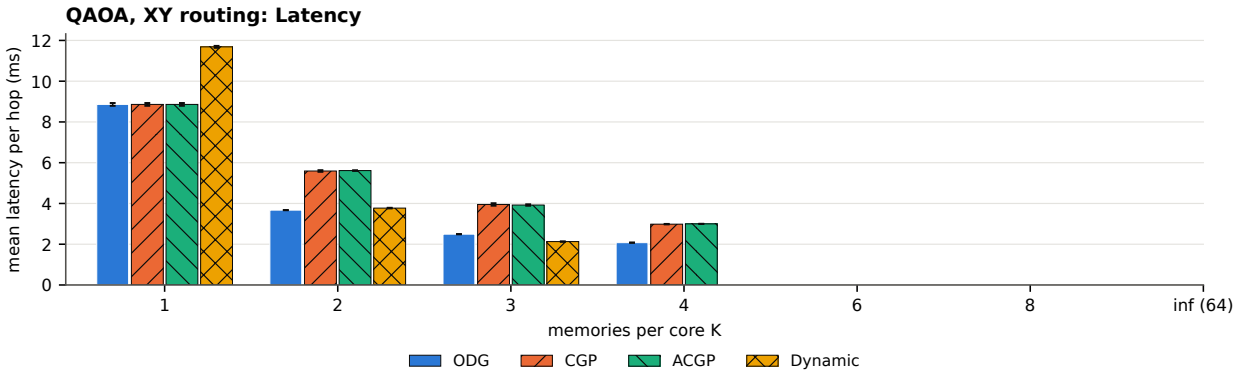


Figure 2.71: QAOA, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	–	–	–
CGP	0.9498	0.9447	0.9416	0.9400	–	–	–
ACGP	0.9498	0.9447	0.9416	0.9398	–	–	–
Dynamic	0.9487	0.9462	0.9449	–	–	–	–

Table 2.72: QAOA, XY routing: fidelity (mean delivered EPR fidelity) for every K.

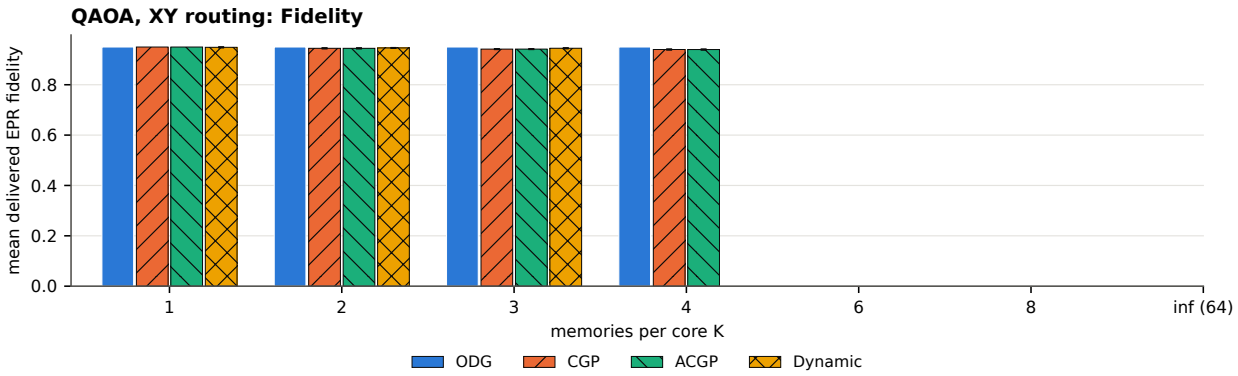


Figure 2.72: QAOA, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	–	–	–
CGP	0.0	26.9	48.8	63.5	–	–	–
ACGP	0.0	27.5	48.4	63.3	–	–	–
Dynamic	9.8	29.3	43.0	–	–	–	–

Table 2.73: QAOA, XY routing: coverage (hops with a ready pair (%)) for every K.

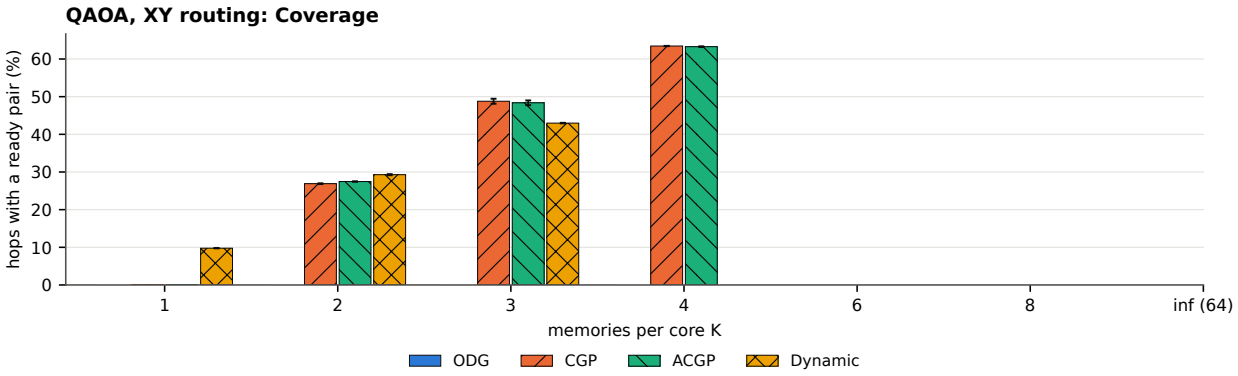


Figure 2.73: QAOA, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	5.39	4.93	4.45	–	–	–
ACGP	–	5.51	4.71	4.61	–	–	–
Dynamic	37.78	8.38	4.63	–	–	–	–

Table 2.74: QAOA, XY routing: wastage (pairs generated but not used (%)) for every K.

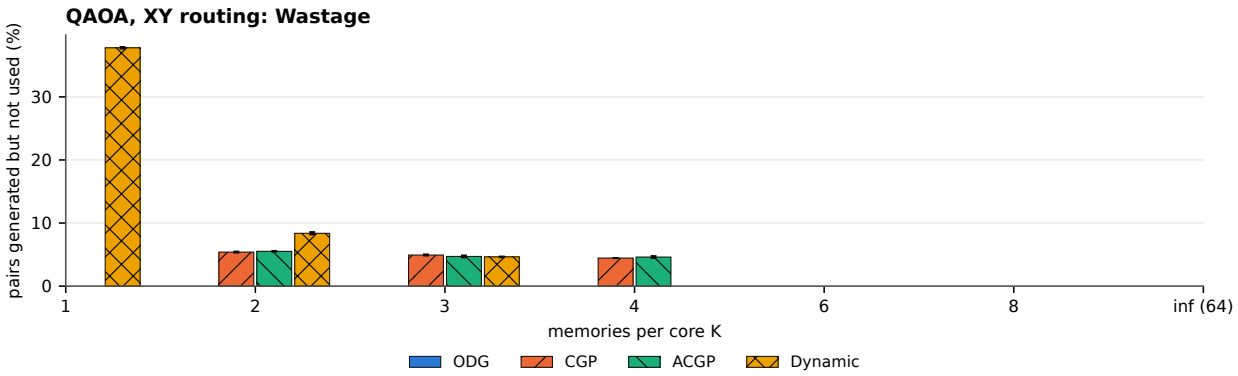


Figure 2.74: QAOA, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	94.6	95.1	95.5	–	–	–
ACGP	–	94.5	95.3	95.4	–	–	–
Dynamic	62.2	91.6	95.4	–	–	–	–

Table 2.75: QAOA, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

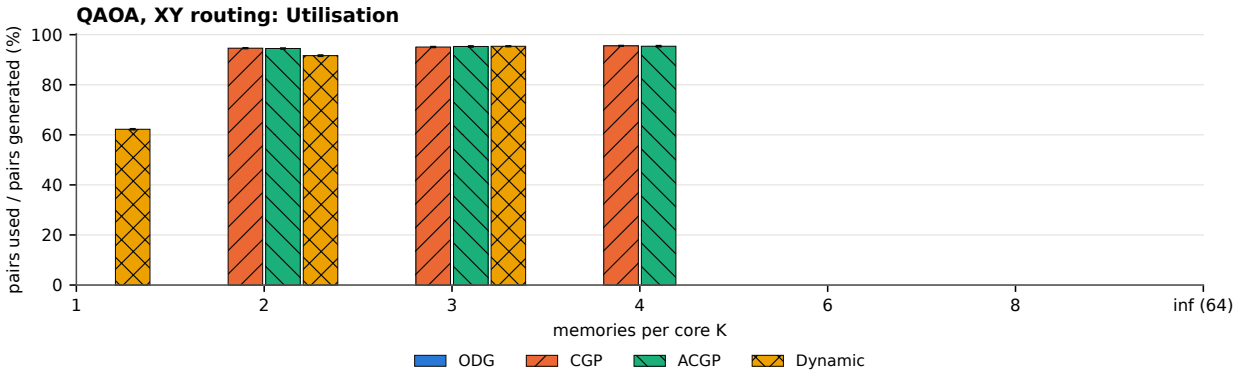


Figure 2.75: QAOA, XY routing: utilisation for every K.

2.12 QAOA, history routing

29605 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.12.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.819	4.65	0.9498	0.0	0.0	–	–
CGP	1.170	3.65	0.9422	74.1	57.1	3.06	96.9
ACGP	1.170	3.65	0.9421	74.5	57.7	3.07	96.9
Dynamic (8)	0.846	2.70	0.9444	71.1	60.8	1.50	98.5
Dynamic (∞)	0.609	1.60	0.9452	83.3	72.7	1.63	98.4

Table 2.76: QAOA, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

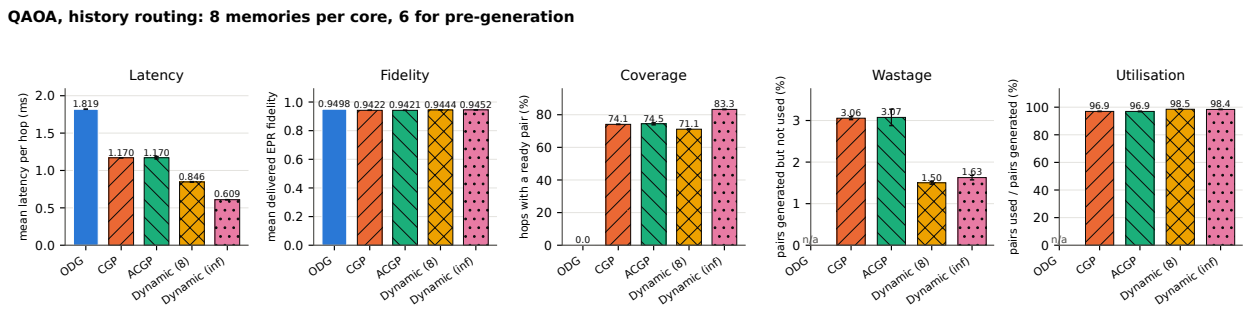


Figure 2.76: QAOA, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.12.2 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	8.382	3.463	2.378	2.019	1.841	1.819	1.818
CGP	8.382	5.103	3.671	2.787	1.438	1.170	0.458
ACGP	8.382	5.163	3.595	2.802	1.402	1.170	0.457
Dynamic	11.608	3.533	1.969	1.455	0.977	0.846	0.609

Table 2.77: QAOA, history routing: latency (mean latency per hop (ms)) for every K.

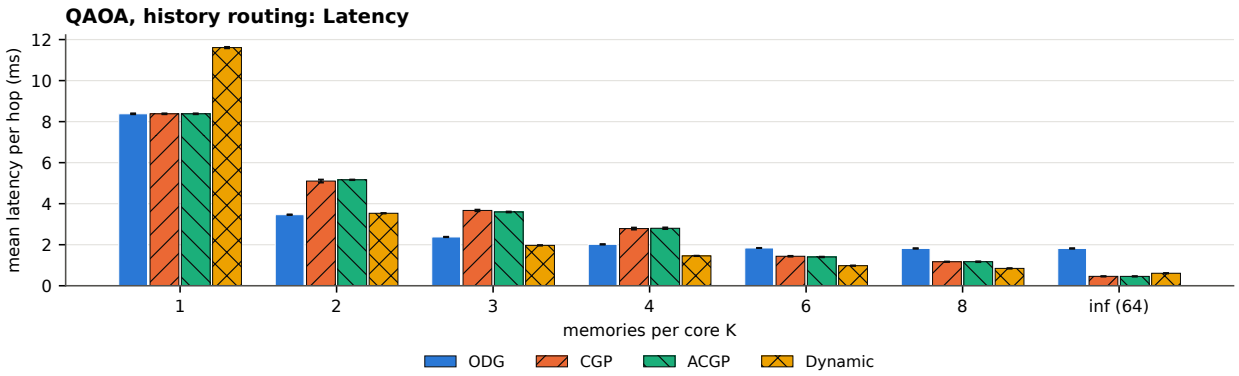


Figure 2.77: QAOA, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9451	0.9424	0.9409	0.9434	0.9422	0.9418
ACGP	0.9498	0.9450	0.9425	0.9409	0.9436	0.9421	0.9417
Dynamic	0.9486	0.9460	0.9448	0.9438	0.9439	0.9444	0.9452

Table 2.78: QAOA, history routing: fidelity (mean delivered EPR fidelity) for every K.

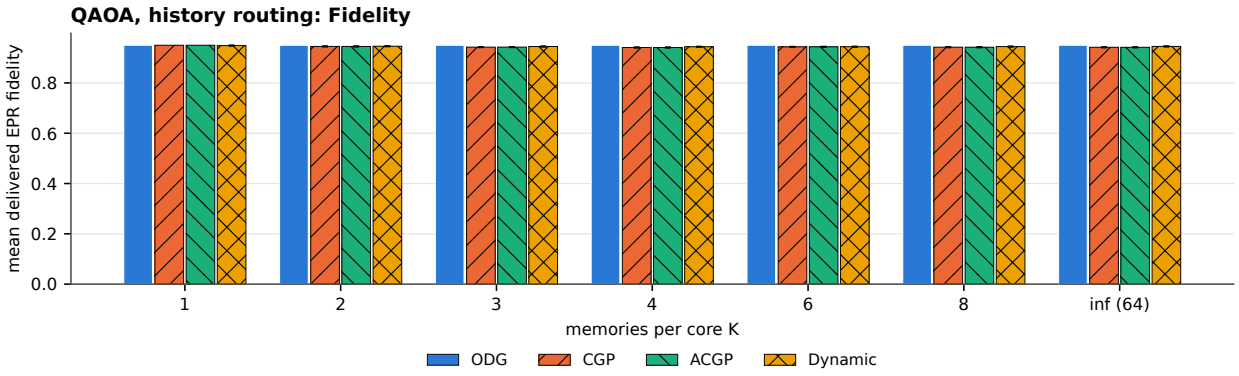


Figure 2.78: QAOA, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	28.0	48.8	63.4	60.6	74.1	99.1
ACGP	0.0	28.5	49.5	63.4	61.9	74.5	99.2
Dynamic	10.1	30.2	44.3	56.3	72.0	71.1	83.3

Table 2.79: QAOA, history routing: coverage (hops with a ready pair (%)) for every K.

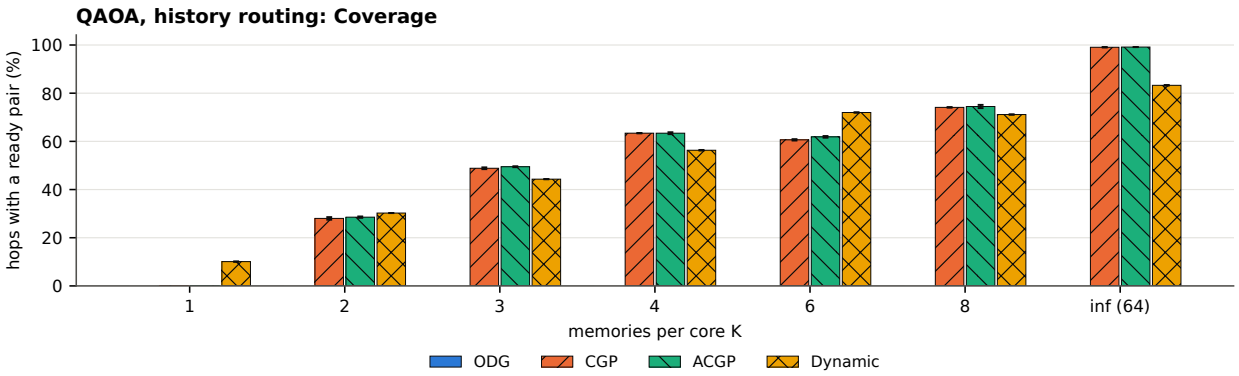


Figure 2.79: QAOA, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	4.78	4.31	4.02	2.85	3.06	8.36
ACGP	–	4.73	4.21	3.85	2.67	3.07	8.36
Dynamic	37.35	8.46	4.74	3.27	1.98	1.50	1.63

Table 2.80: QAOA, history routing: wastage (pairs generated but not used (%)) for every K.

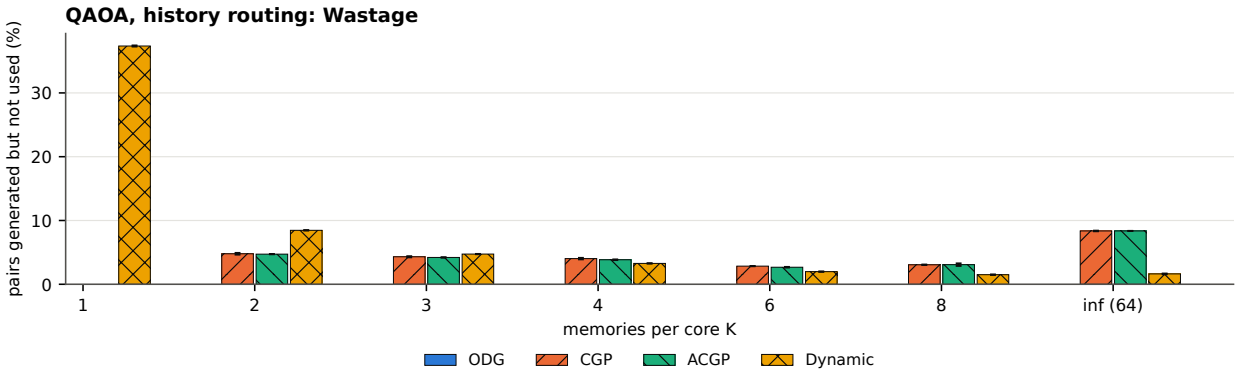


Figure 2.80: QAOA, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	95.2	95.7	96.0	97.2	96.9	91.6
ACGP	–	95.3	95.8	96.2	97.3	96.9	91.6
Dynamic	62.6	91.5	95.3	96.7	98.0	98.5	98.4

Table 2.81: QAOA, history routing: utilisation (pairs used / pairs generated (%)) for every K.

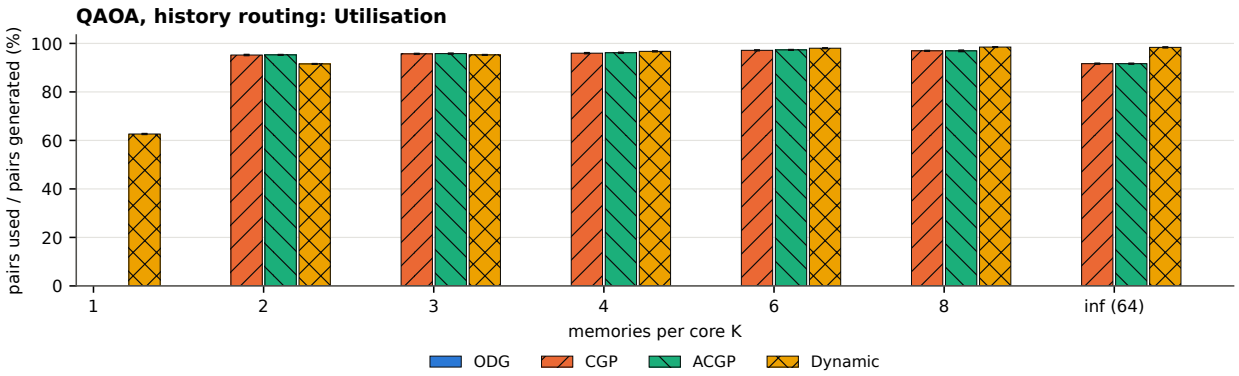


Figure 2.81: QAOA, history routing: utilisation for every K.

2.13 Draper adder, XY routing

6510 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.13.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.761	4.20	0.9498	0.0	0.0	–	–
CGP	0.687	1.90	0.9404	81.5	68.7	6.00	94.0
ACGP	0.671	1.80	0.9405	81.7	69.6	6.03	94.0
Dynamic (8)	0.623	1.60	0.9467	79.2	65.8	2.43	97.6
Dynamic (∞)	0.611	1.50	0.9467	79.9	66.6	2.74	97.3

Table 2.82: Draper adder, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

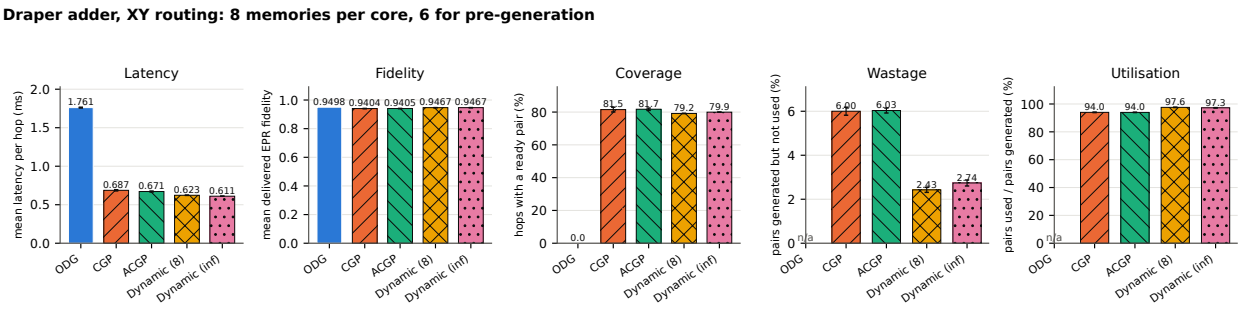


Figure 2.82: Draper adder, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.13.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.765	4.20	0.9498	0.0	0.0	–	–
CGP	1.351	4.50	0.9400	65.5	46.6	6.00	94.0
ACGP	1.347	4.50	0.9395	65.8	47.7	6.14	93.9
Dynamic (4)	0.783	2.35	0.9465	77.6	64.0	3.00	97.0
Dynamic (∞)	0.611	1.50	0.9467	79.9	66.6	2.74	97.3

Table 2.83: Draper adder, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

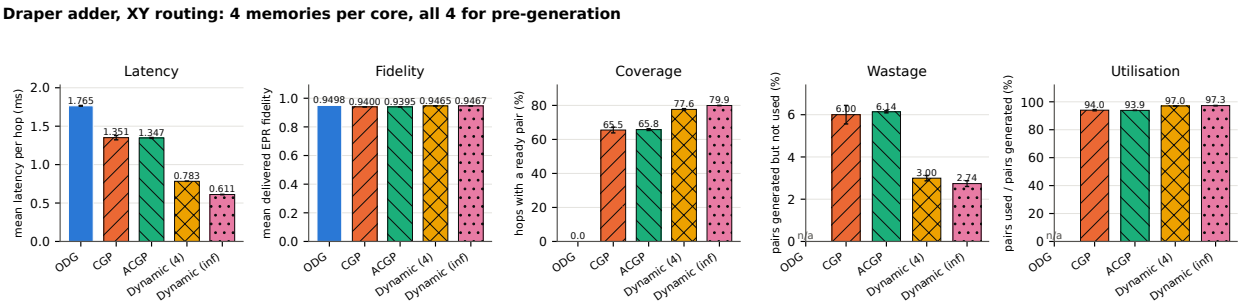


Figure 2.83: Draper adder, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.13.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	3.506	1.918	1.783	1.765	1.761	1.761	1.761
CGP	3.506	2.263	1.699	1.351	0.826	0.687	0.456
ACGP	3.506	2.271	1.668	1.347	0.784	0.671	0.455
Dynamic	3.975	1.546	1.014	0.783	0.645	0.623	0.611

Table 2.84: Draper adder, XY routing: latency (mean latency per hop (ms)) for every K.

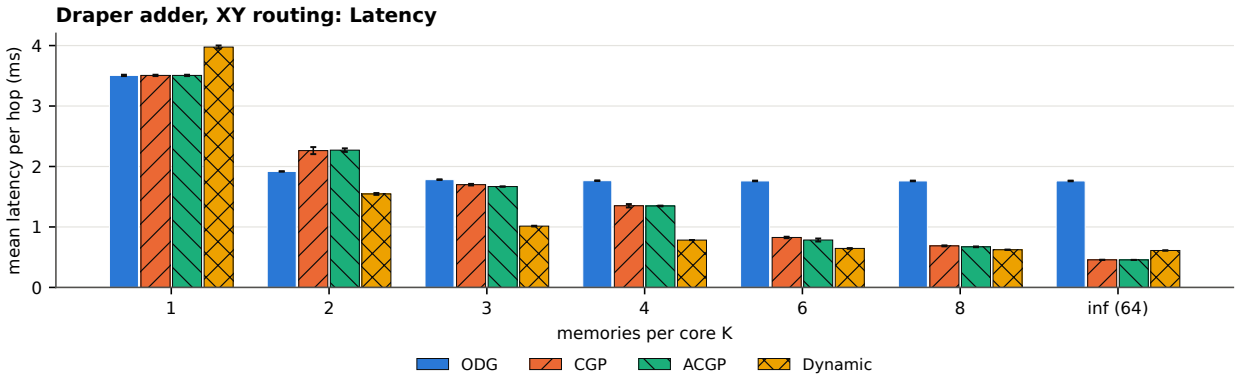


Figure 2.84: Draper adder, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9441	0.9413	0.9400	0.9410	0.9404	0.9405
ACGP	0.9498	0.9441	0.9412	0.9395	0.9410	0.9405	0.9405
Dynamic	0.9480	0.9454	0.9459	0.9465	0.9467	0.9467	0.9467

Table 2.85: Draper adder, XY routing: fidelity (mean delivered EPR fidelity) for every K.

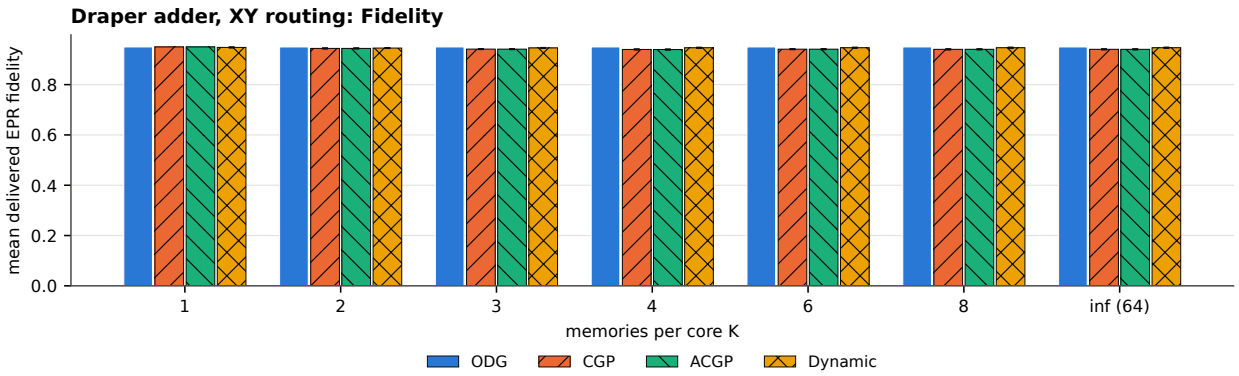


Figure 2.85: Draper adder, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	29.2	49.7	65.5	70.9	81.5	99.2
ACGP	0.0	28.4	51.6	65.8	73.8	81.7	99.5
Dynamic	21.0	58.1	71.3	77.6	79.7	79.2	79.9

Table 2.86: Draper adder, XY routing: coverage (hops with a ready pair (%)) for every K.

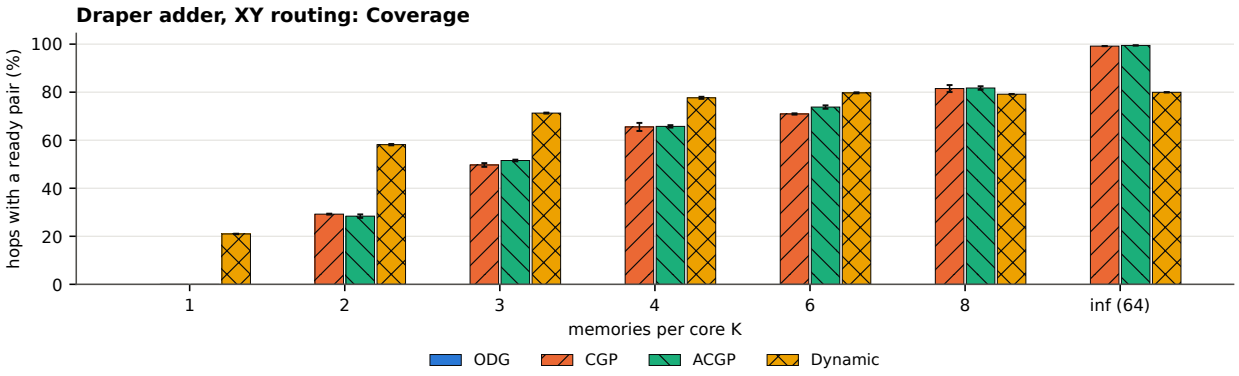


Figure 2.86: Draper adder, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	7.18	7.07	6.00	5.66	6.00	23.84
ACGP	–	7.39	6.42	6.14	5.16	6.03	24.02
Dynamic	41.98	9.78	3.73	3.00	2.68	2.43	2.74

Table 2.87: Draper adder, XY routing: wastage (pairs generated but not used (%)) for every K.

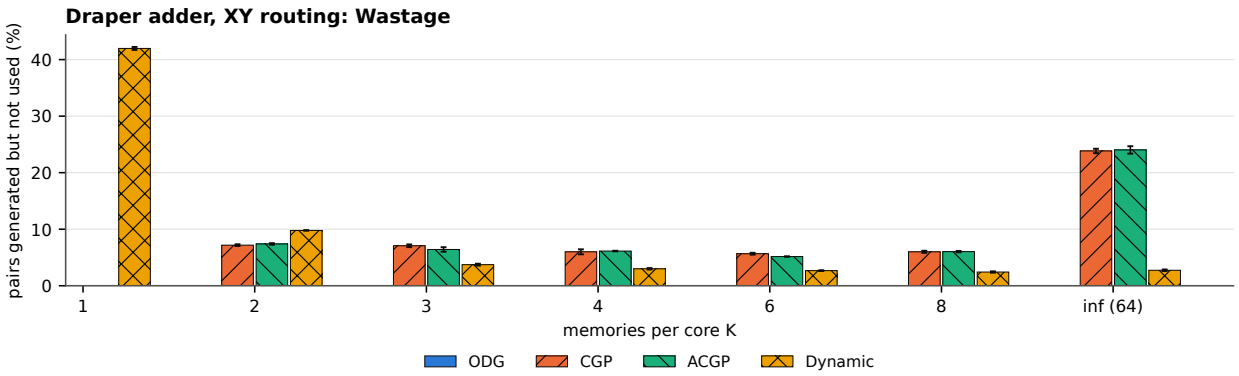


Figure 2.87: Draper adder, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	92.8	92.9	94.0	94.3	94.0	76.2
ACGP	–	92.6	93.6	93.9	94.8	94.0	76.0
Dynamic	58.0	90.2	96.3	97.0	97.3	97.6	97.3

Table 2.88: Draper adder, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

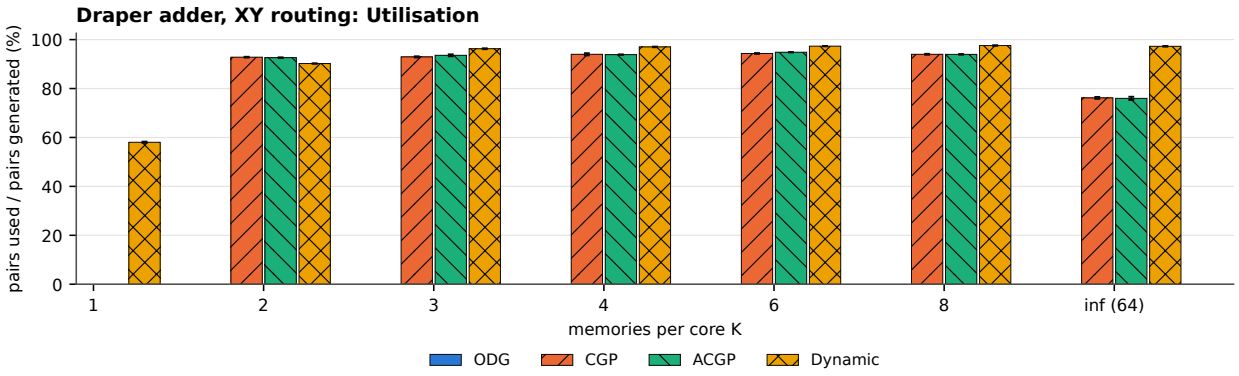


Figure 2.88: Draper adder, XY routing: utilisation for every K.

2.14 Draper adder, history routing

6510 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.14.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.765	4.40	0.9498	0.0	0.0	–	–
CGP	0.689	1.90	0.9410	80.6	67.8	6.24	93.8
ACGP	0.689	1.99	0.9410	81.1	68.1	6.14	93.9
Dynamic (8)	0.628	1.60	0.9467	78.5	64.9	2.58	97.4
Dynamic (∞)	0.611	1.50	0.9466	80.0	66.9	2.45	97.5

Table 2.89: Draper adder, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

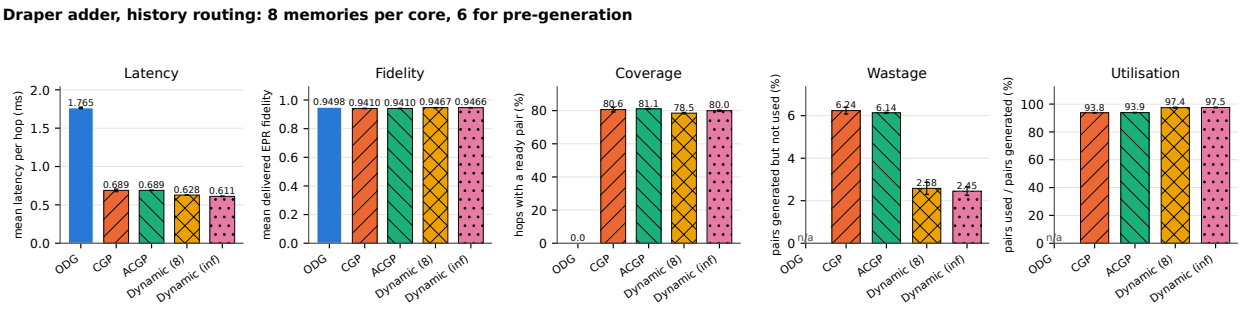


Figure 2.89: Draper adder, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.14.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.768	4.40	0.9498	0.0	0.0	–	–
CGP	1.352	4.45	0.9404	65.0	47.5	6.34	93.7
ACGP	1.292	4.15	0.9409	67.2	51.0	5.98	94.0
Dynamic (4)	0.788	2.40	0.9462	77.5	64.5	2.86	97.1
Dynamic (∞)	0.611	1.50	0.9466	80.0	66.9	2.45	97.5

Table 2.90: Draper adder, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

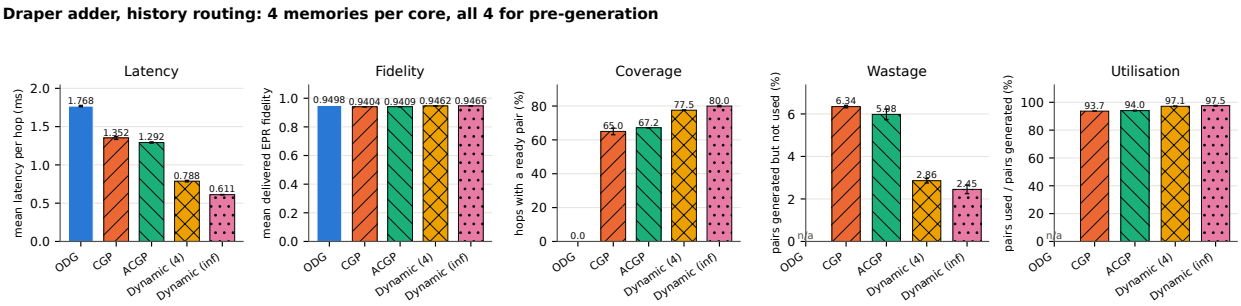


Figure 2.90: Draper adder, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.14.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	3.445	1.929	1.789	1.768	1.765	1.765	1.765
CGP	3.445	2.233	1.675	1.352	0.823	0.689	0.455
ACGP	3.445	2.209	1.664	1.292	0.845	0.689	0.456
Dynamic	3.857	1.505	0.993	0.788	0.650	0.628	0.611

Table 2.91: Draper adder, history routing: latency (mean latency per hop (ms)) for every K.

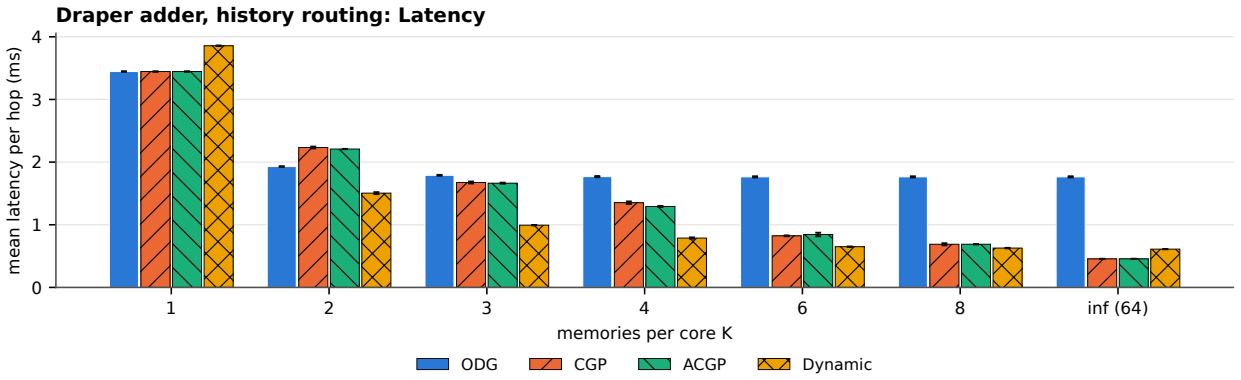


Figure 2.91: Draper adder, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9442	0.9416	0.9404	0.9414	0.9410	0.9407
ACGP	0.9498	0.9445	0.9418	0.9409	0.9410	0.9410	0.9405
Dynamic	0.9478	0.9452	0.9458	0.9462	0.9466	0.9467	0.9466

Table 2.92: Draper adder, history routing: fidelity (mean delivered EPR fidelity) for every K.

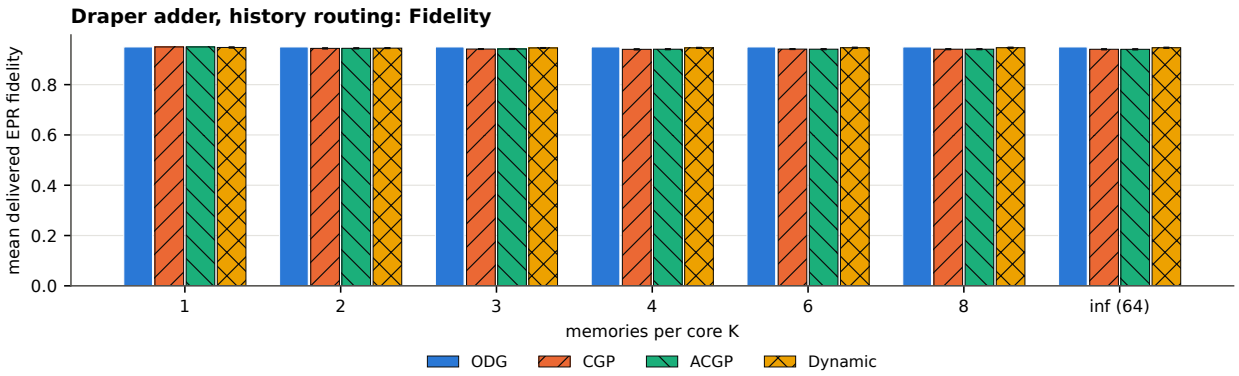


Figure 2.92: Draper adder, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	30.7	50.5	65.0	71.2	80.6	99.4
ACGP	0.0	30.6	52.0	67.2	70.7	81.1	99.3
Dynamic	21.9	61.6	71.9	77.5	79.5	78.5	80.0

Table 2.93: Draper adder, history routing: coverage (hops with a ready pair (%)) for every K.

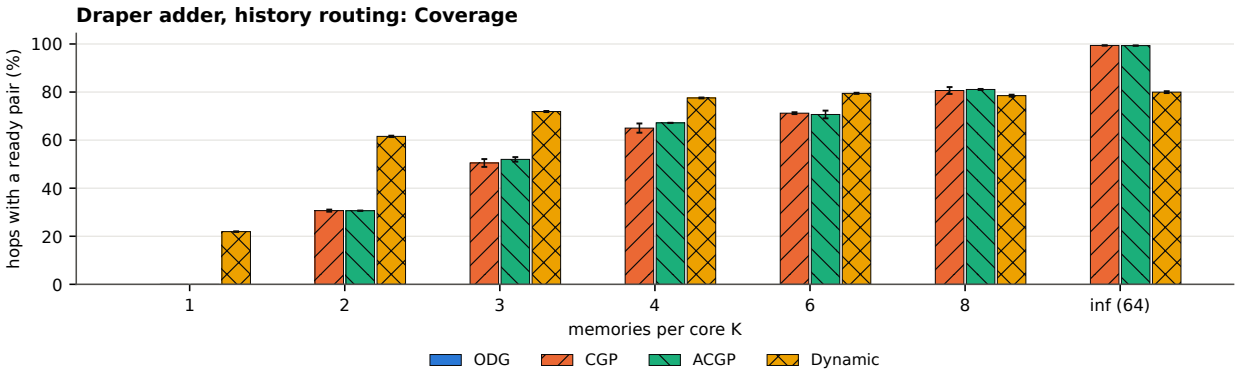


Figure 2.93: Draper adder, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	6.90	6.81	6.34	5.52	6.24	23.86
ACGP	–	6.56	6.56	5.98	5.96	6.14	24.23
Dynamic	42.34	9.89	4.14	2.86	2.60	2.58	2.45

Table 2.94: Draper adder, history routing: wastage (pairs generated but not used (%)) for every K.

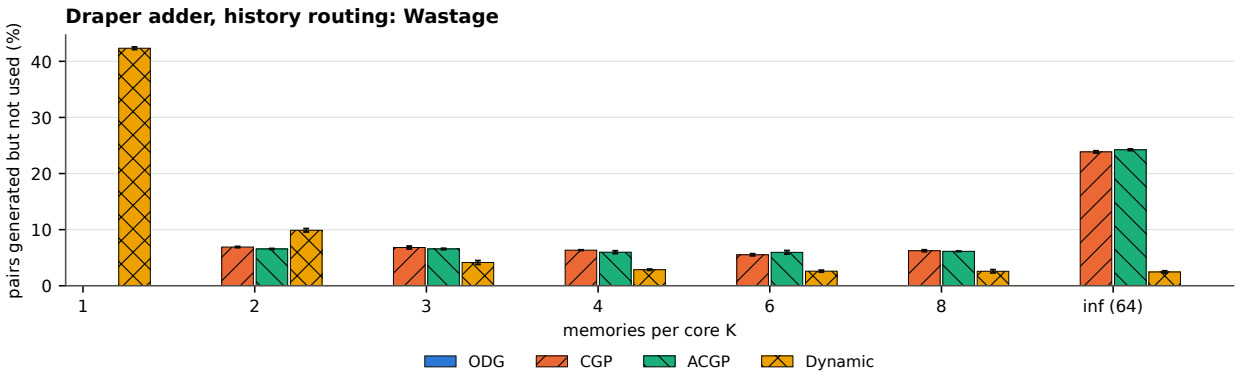


Figure 2.94: Draper adder, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	93.1	93.2	93.7	94.5	93.8	76.1
ACGP	–	93.4	93.4	94.0	94.0	93.9	75.8
Dynamic	57.7	90.1	95.9	97.1	97.4	97.4	97.5

Table 2.95: Draper adder, history routing: utilisation (pairs used / pairs generated (%)) for every K.

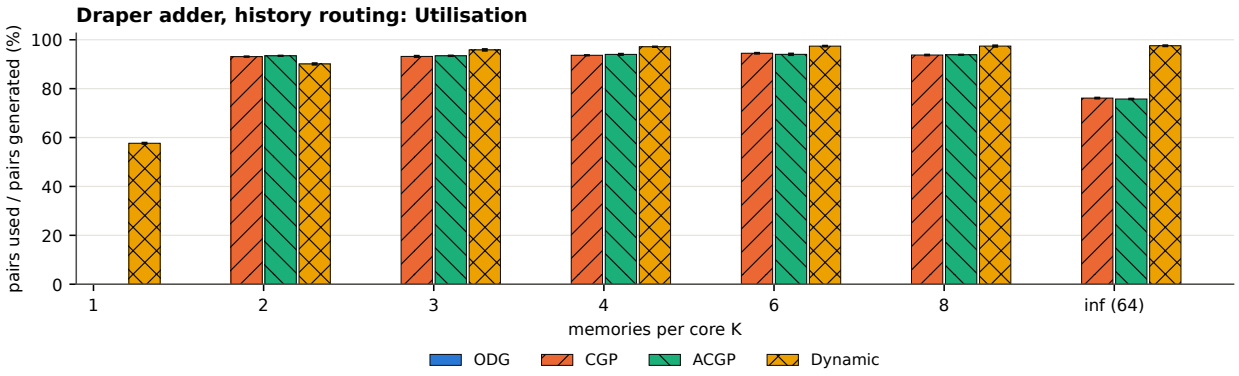


Figure 2.95: Draper adder, history routing: utilisation for every K.

2.15 Graph state, XY routing

136 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.15.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.771	4.45	0.9498	0.0	0.0	–	–
CGP	0.696	2.00	0.9429	79.4	60.3	27.89	72.1
ACGP	0.696	2.00	0.9429	79.4	60.3	27.89	72.1
Dynamic (8)	0.693	2.30	0.9453	74.6	57.8	2.74	97.3
Dynamic (∞)	0.674	1.85	0.9450	77.6	63.8	1.77	98.2

Table 2.96: Graph state, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

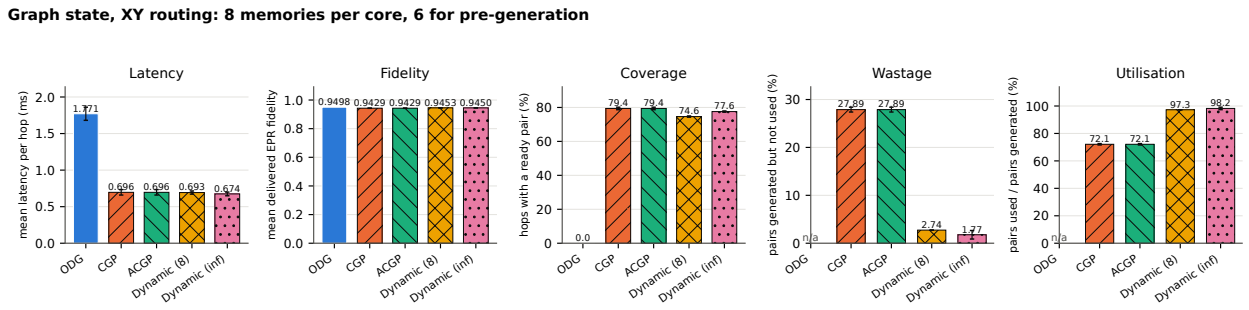


Figure 2.96: Graph state, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.15.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.774	4.45	0.9498	0.0	0.0	–	–
CGP	1.561	5.65	0.9418	61.0	29.3	19.88	80.1
ACGP	1.561	5.65	0.9418	61.0	29.3	19.88	80.1
Dynamic (4)	1.045	3.55	0.9419	70.2	55.2	4.79	95.2
Dynamic (∞)	0.674	1.85	0.9450	77.6	63.8	1.77	98.2

Table 2.97: Graph state, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

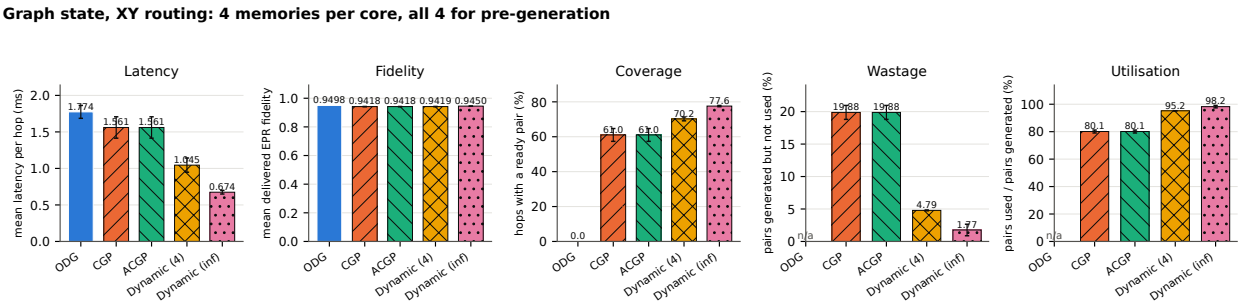


Figure 2.97: Graph state, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.15.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	3.371	2.011	1.807	1.774	1.771	1.771	1.771
CGP	3.371	2.268	1.919	1.561	0.858	0.696	0.556
ACGP	3.371	2.268	1.919	1.561	0.858	0.696	0.556
Dynamic	5.413	1.599	1.204	1.045	0.725	0.693	0.674

Table 2.98: Graph state, XY routing: latency (mean latency per hop (ms)) for every K.

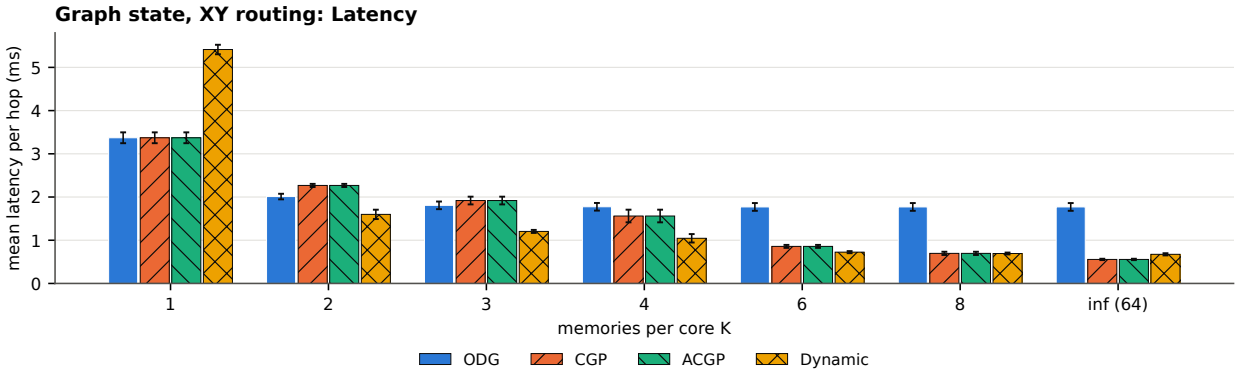


Figure 2.98: Graph state, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9461	0.9421	0.9418	0.9438	0.9429	0.9476
ACGP	0.9498	0.9461	0.9421	0.9418	0.9438	0.9429	0.9476
Dynamic	0.9482	0.9449	0.9427	0.9419	0.9450	0.9453	0.9450

Table 2.99: Graph state, XY routing: fidelity (mean delivered EPR fidelity) for every K.

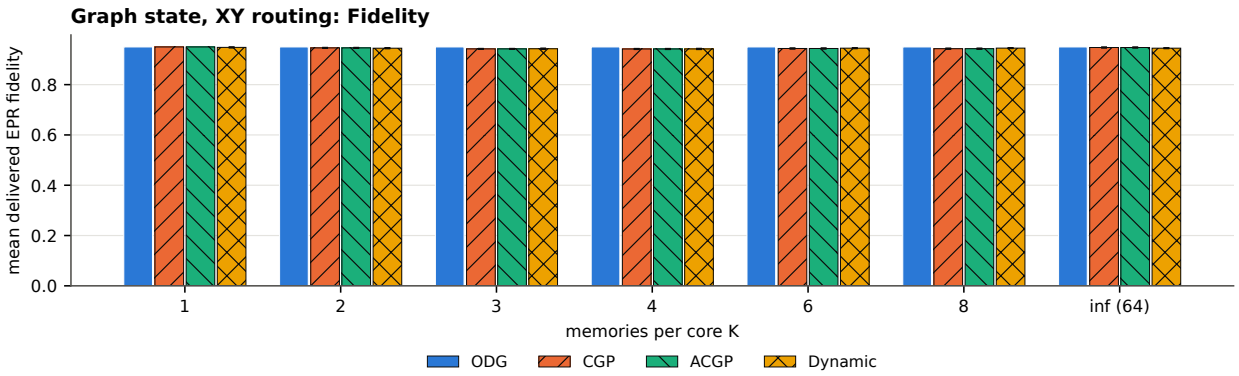


Figure 2.99: Graph state, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	23.5	45.2	61.0	68.0	79.4	89.3
ACGP	0.0	23.5	45.2	61.0	68.0	79.4	89.3
Dynamic	13.2	48.9	61.0	70.2	73.5	74.6	77.6

Table 2.100: Graph state, XY routing: coverage (hops with a ready pair (%)) for every K.

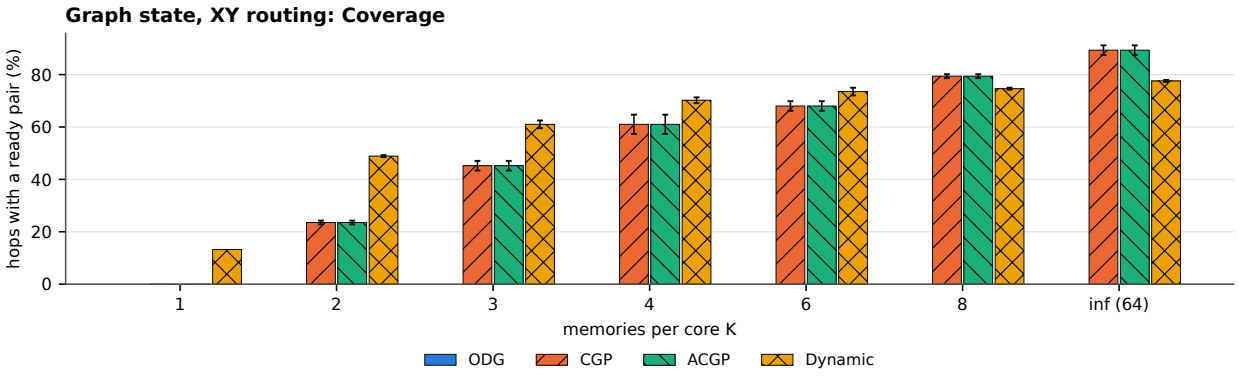


Figure 2.100: Graph state, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	15.85	16.85	19.88	21.96	27.89	72.11
ACGP	–	15.85	16.85	19.88	21.96	27.89	72.11
Dynamic	54.42	11.92	7.16	4.79	4.10	2.74	1.77

Table 2.101: Graph state, XY routing: wastage (pairs generated but not used (%)) for every K.

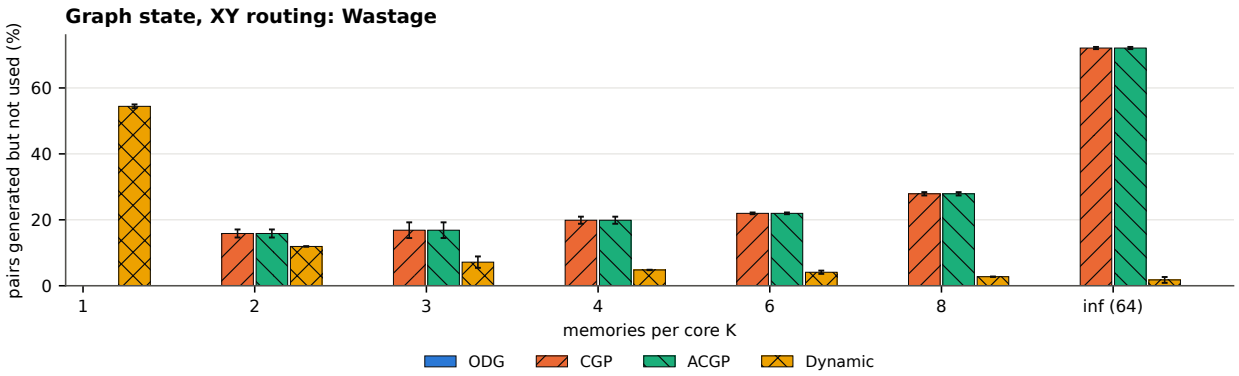


Figure 2.101: Graph state, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	84.1	83.1	80.1	78.0	72.1	27.9
ACGP	–	84.1	83.1	80.1	78.0	72.1	27.9
Dynamic	45.6	88.1	92.8	95.2	95.9	97.3	98.2

Table 2.102: Graph state, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

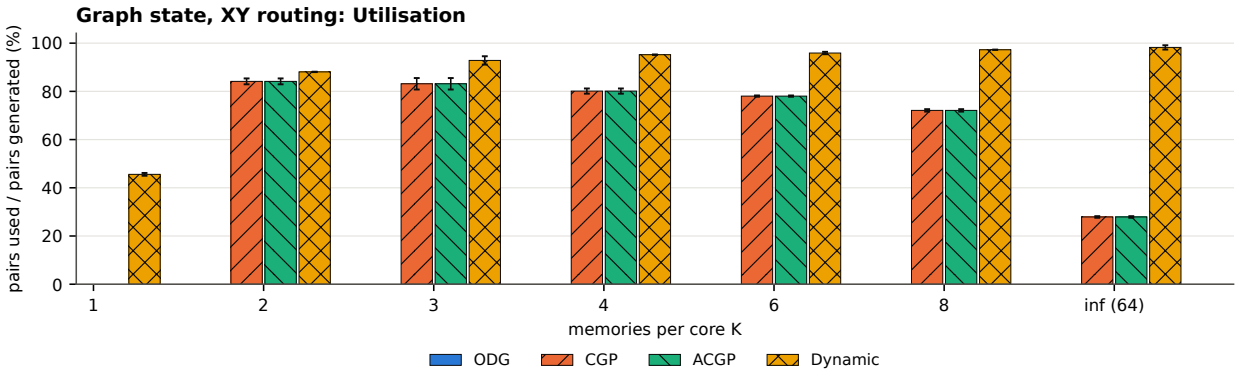


Figure 2.102: Graph state, XY routing: utilisation for every K.

2.16 Graph state, history routing

136 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.16.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.738	3.95	0.9498	0.0	0.0	–	–
CGP	0.799	2.35	0.9421	74.6	51.7	29.72	70.3
ACGP	0.799	2.35	0.9421	74.6	51.7	29.72	70.3
Dynamic (8)	0.716	2.00	0.9449	74.3	54.3	1.34	98.7
Dynamic (∞)	0.681	1.95	0.9448	77.2	60.3	2.14	97.9

Table 2.103: Graph state, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

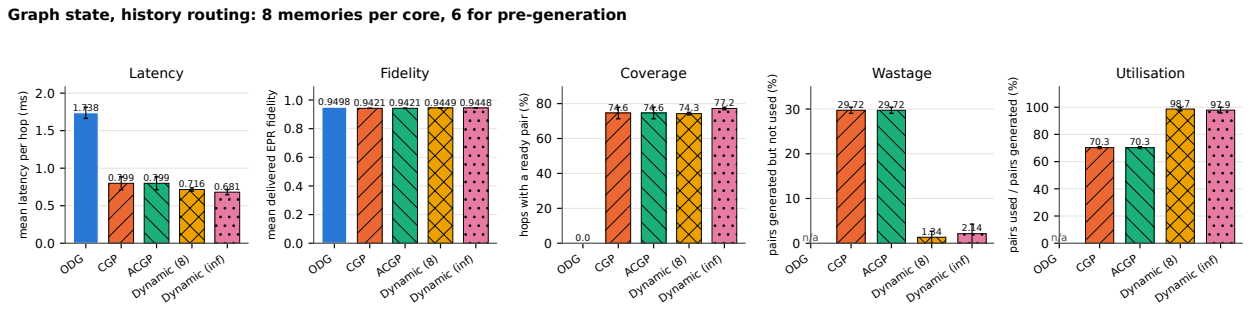


Figure 2.103: Graph state, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.16.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.744	4.20	0.9498	0.0	0.0	–	–
CGP	1.398	5.00	0.9414	61.4	30.2	19.72	80.3
ACGP	1.398	5.00	0.9414	61.4	30.2	19.72	80.3
Dynamic (4)	0.924	2.90	0.9447	68.8	46.6	2.42	97.6
Dynamic (∞)	0.681	1.95	0.9448	77.2	60.3	2.14	97.9

Table 2.104: Graph state, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

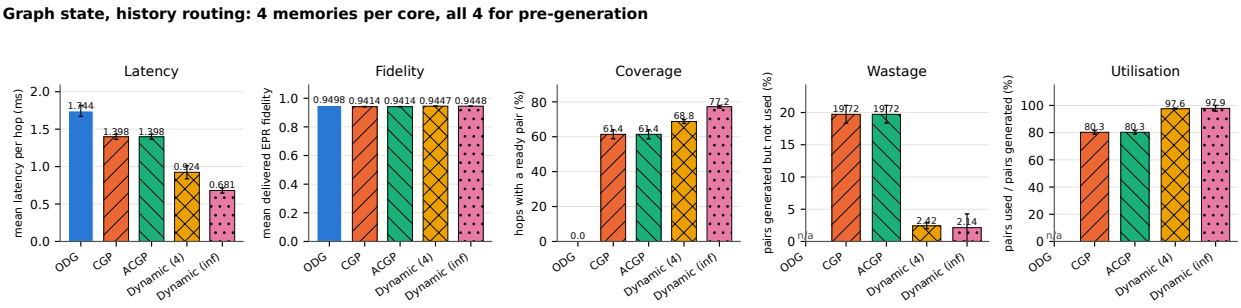


Figure 2.104: Graph state, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.16.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	3.791	1.986	1.774	1.744	1.738	1.738	1.738
CGP	3.791	2.374	1.763	1.398	0.802	0.799	0.555
ACGP	3.791	2.374	1.763	1.398	0.802	0.799	0.555
Dynamic	4.994	1.782	1.176	0.924	0.745	0.716	0.681

Table 2.105: Graph state, history routing: latency (mean latency per hop (ms)) for every K.

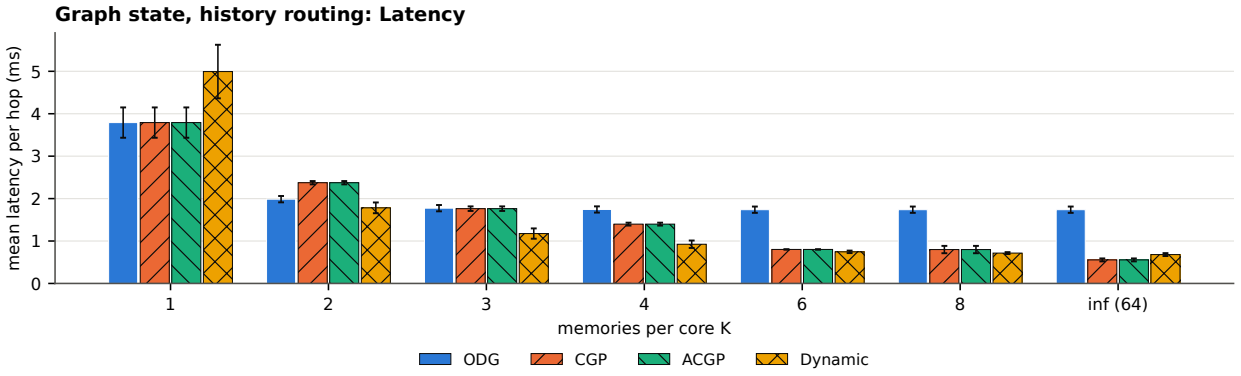


Figure 2.105: Graph state, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9450	0.9424	0.9414	0.9435	0.9421	0.9475
ACGP	0.9498	0.9450	0.9424	0.9414	0.9435	0.9421	0.9475
Dynamic	0.9481	0.9450	0.9450	0.9447	0.9447	0.9449	0.9448

Table 2.106: Graph state, history routing: fidelity (mean delivered EPR fidelity) for every K.

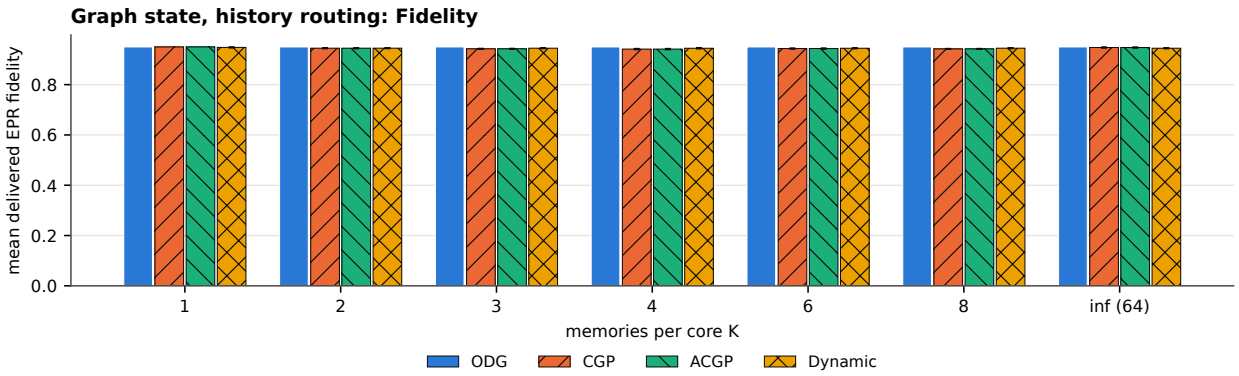


Figure 2.106: Graph state, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	23.2	46.7	61.4	70.6	74.6	90.8
ACGP	0.0	23.2	46.7	61.4	70.6	74.6	90.8
Dynamic	13.2	46.0	59.9	68.8	74.6	74.3	77.2

Table 2.107: Graph state, history routing: coverage (hops with a ready pair (%)) for every K.

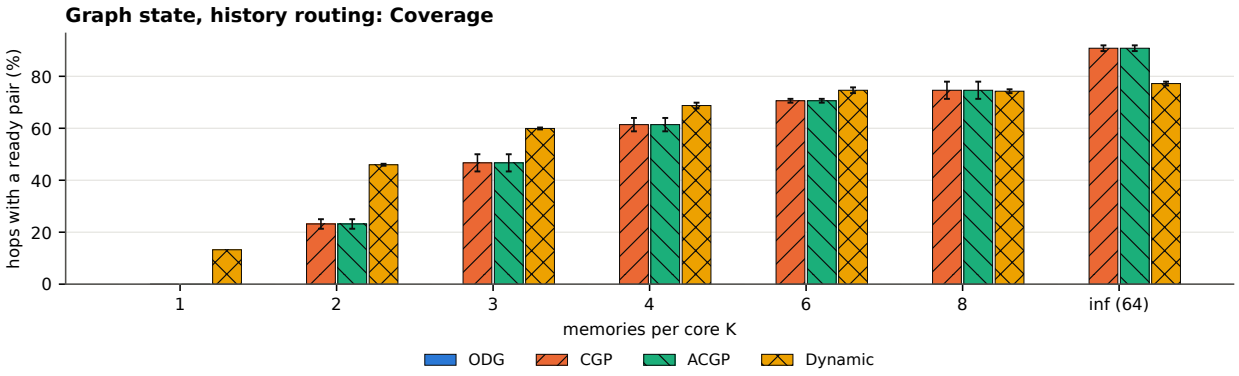


Figure 2.107: Graph state, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	16.53	18.55	19.72	22.28	29.72	72.91
ACGP	–	16.53	18.55	19.72	22.28	29.72	72.91
Dynamic	55.00	11.11	2.83	2.42	0.88	1.34	2.14

Table 2.108: Graph state, history routing: wastage (pairs generated but not used (%)) for every K.

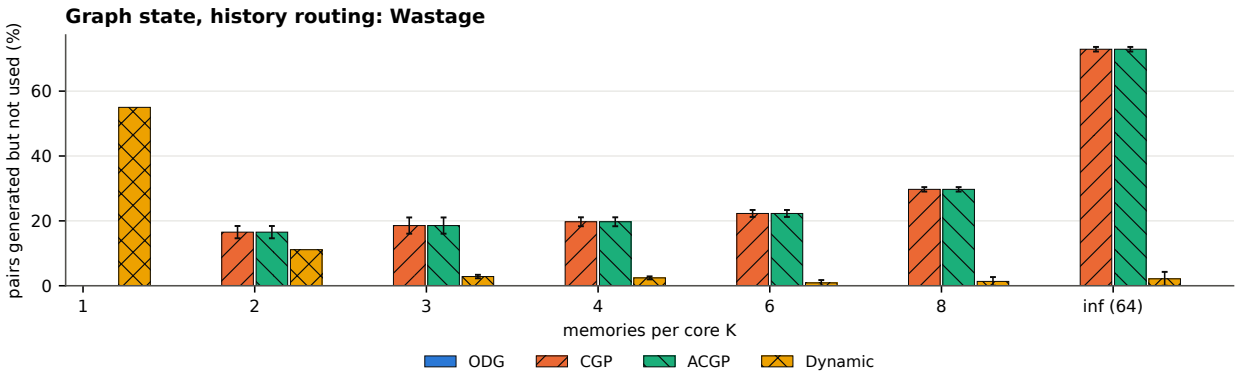


Figure 2.108: Graph state, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	83.5	81.4	80.3	77.7	70.3	27.1
ACGP	–	83.5	81.4	80.3	77.7	70.3	27.1
Dynamic	45.0	88.9	97.2	97.6	99.1	98.7	97.9

Table 2.109: Graph state, history routing: utilisation (pairs used / pairs generated (%)) for every K.

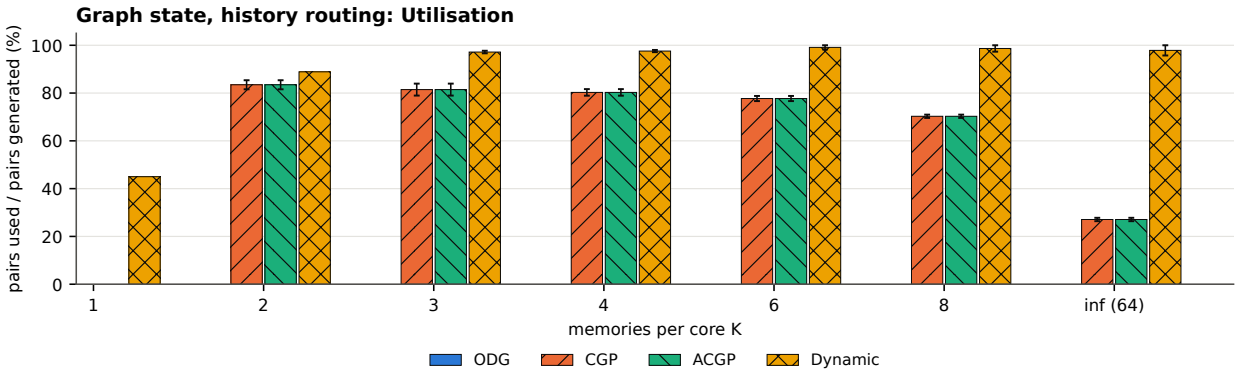


Figure 2.109: Graph state, history routing: utilisation for every K.

2.17 BMW cardinality, XY routing

306 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.17.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.639	3.80	0.9498	0.0	0.0	–	–
CGP	0.741	2.00	0.9243	70.1	49.4	27.87	72.1
ACGP	0.740	2.00	0.9243	70.3	49.7	27.80	72.2
Dynamic (8)	0.705	1.70	0.9476	67.0	45.1	5.94	94.1
Dynamic (∞)	0.705	1.70	0.9476	67.0	45.1	5.94	94.1

Table 2.110: BMW cardinality, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

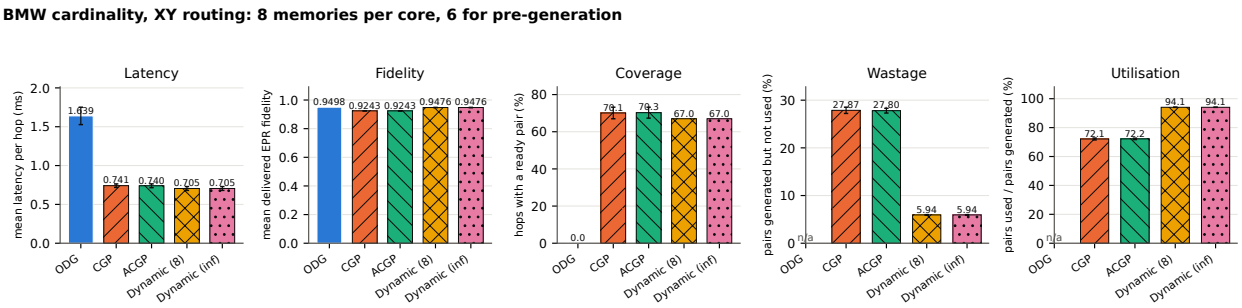


Figure 2.110: BMW cardinality, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.17.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.639	3.80	0.9498	0.0	0.0	–	–
CGP	0.867	2.45	0.9299	56.4	30.3	16.90	83.1
ACGP	0.868	2.45	0.9301	55.7	29.4	16.97	83.0
Dynamic (4)	0.725	1.90	0.9475	65.4	42.9	7.17	92.8
Dynamic (∞)	0.705	1.70	0.9476	67.0	45.1	5.94	94.1

Table 2.111: BMW cardinality, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

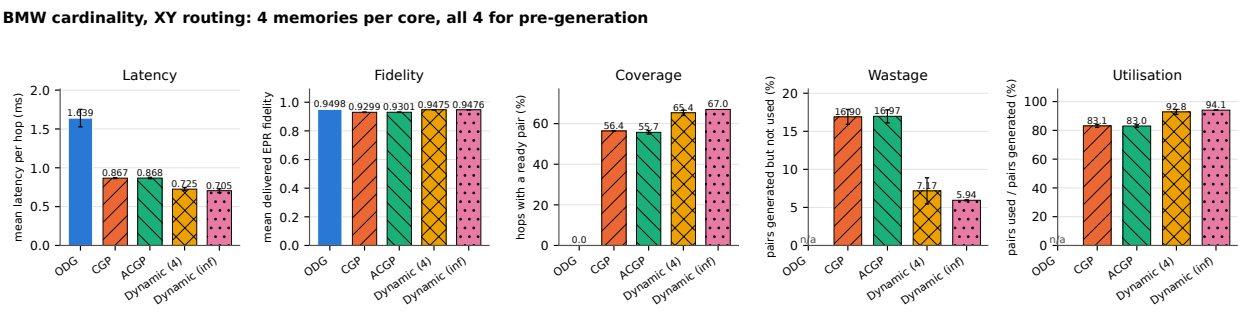


Figure 2.111: BMW cardinality, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.17.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.639	1.639	1.639	1.639	1.639	1.639	1.639
CGP	1.639	1.357	1.022	0.867	0.638	0.741	0.468
ACGP	1.639	1.350	1.022	0.868	0.638	0.740	0.468
Dynamic	2.123	1.042	0.767	0.725	0.705	0.705	0.705

Table 2.112: BMW cardinality, XY routing: latency (mean latency per hop (ms)) for every K.

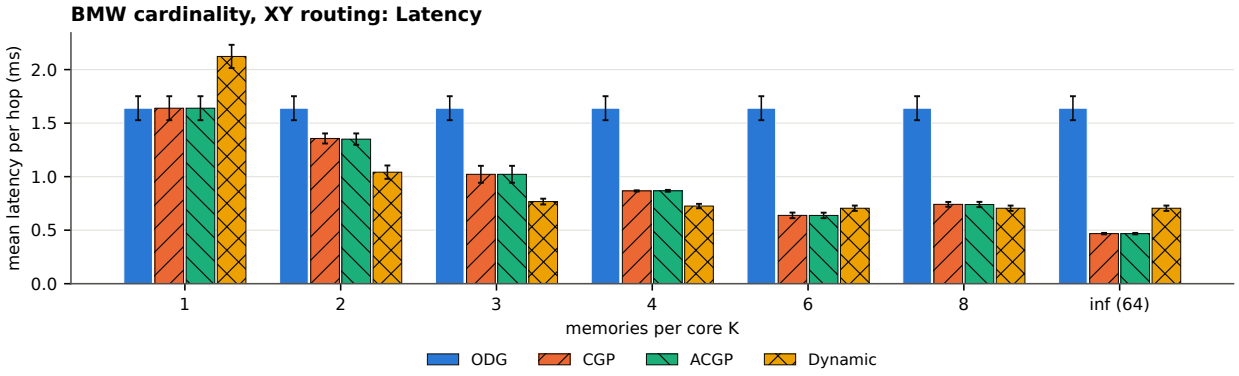


Figure 2.112: BMW cardinality, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9425	0.9375	0.9299	0.9271	0.9243	0.9253
ACGP	0.9498	0.9425	0.9375	0.9301	0.9271	0.9243	0.9253
Dynamic	0.9487	0.9480	0.9477	0.9475	0.9476	0.9476	0.9476

Table 2.113: BMW cardinality, XY routing: fidelity (mean delivered EPR fidelity) for every K.

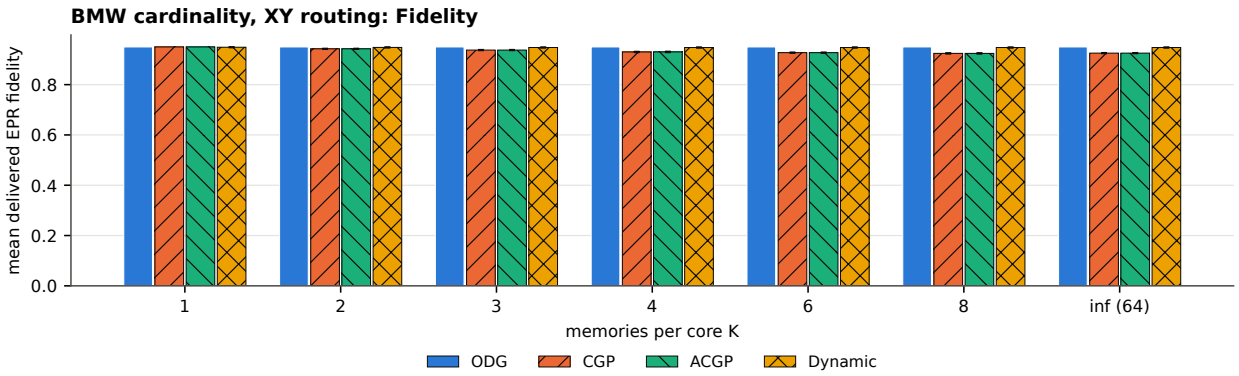


Figure 2.113: BMW cardinality, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	17.8	42.8	56.4	75.7	70.1	99.3
ACGP	0.0	18.1	42.8	55.7	75.7	70.3	99.3
Dynamic	50.7	41.8	60.6	65.4	67.0	67.0	67.0

Table 2.114: BMW cardinality, XY routing: coverage (hops with a ready pair (%)) for every K.

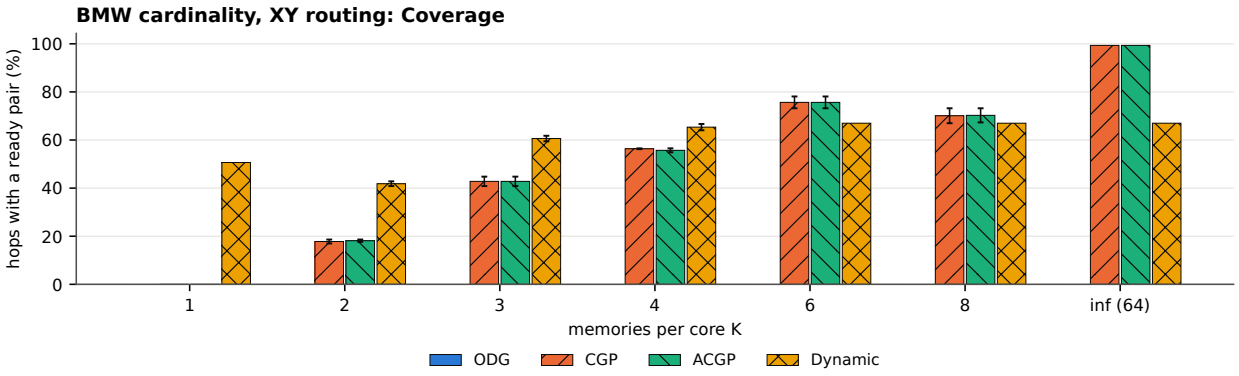


Figure 2.114: BMW cardinality, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	22.06	15.27	16.90	13.37	27.87	54.36
ACGP	–	19.66	15.27	16.97	13.10	27.80	54.36
Dynamic	28.24	23.86	10.47	7.17	5.94	5.94	5.94

Table 2.115: BMW cardinality, XY routing: wastage (pairs generated but not used (%)) for every K.

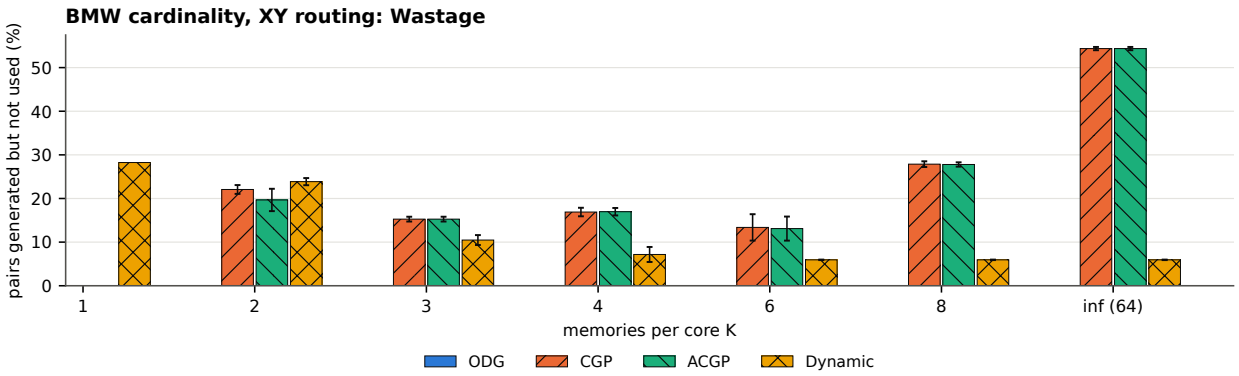


Figure 2.115: BMW cardinality, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	77.9	84.7	83.1	86.6	72.1	45.6
ACGP	–	80.3	84.7	83.0	86.9	72.2	45.6
Dynamic	71.8	76.1	89.5	92.8	94.1	94.1	94.1

Table 2.116: BMW cardinality, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

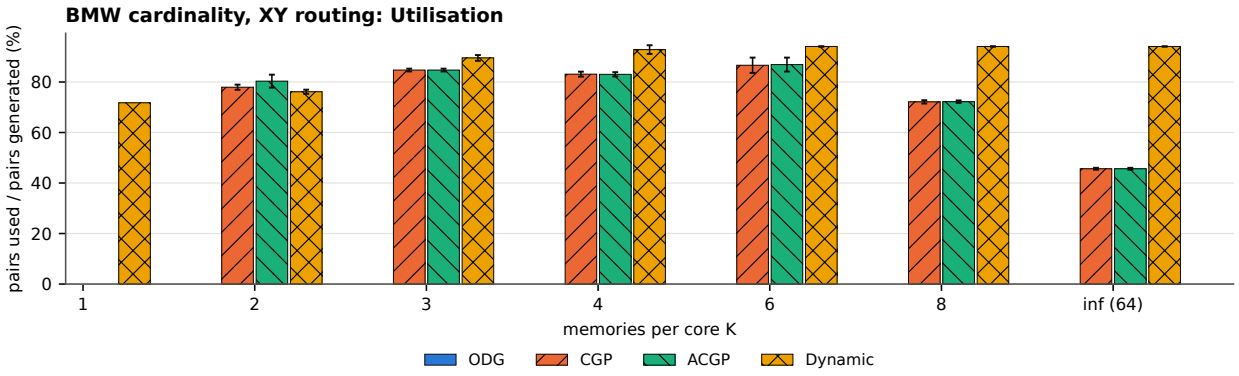


Figure 2.116: BMW cardinality, XY routing: utilisation for every K.

2.18 BMW cardinality, history routing

306 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.18.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.728	4.20	0.9498	0.0	0.0	–	–
CGP	0.767	2.20	0.9262	68.8	50.0	27.90	72.1
ACGP	0.766	2.20	0.9261	69.1	50.3	27.46	72.5
Dynamic (8)	0.724	1.90	0.9476	66.8	46.3	5.84	94.2
Dynamic (∞)	0.724	1.90	0.9476	66.8	46.3	5.84	94.2

Table 2.117: BMW cardinality, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

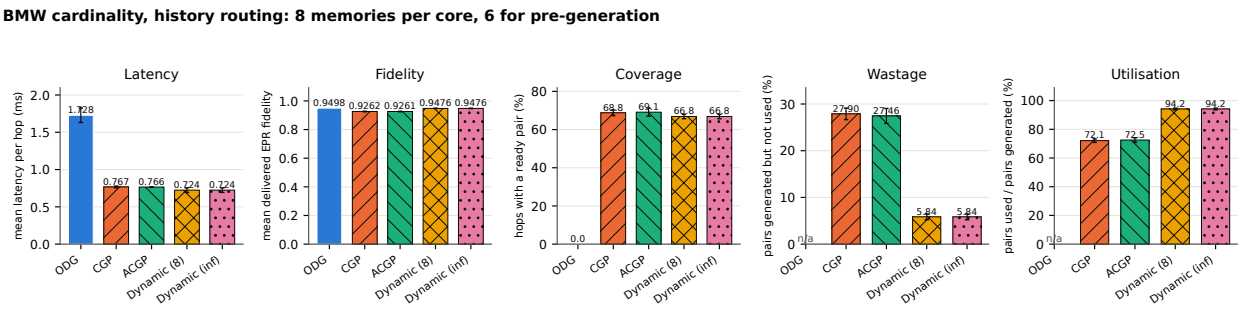


Figure 2.117: BMW cardinality, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.18.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.728	4.20	0.9498	0.0	0.0	–	–
CGP	0.853	2.40	0.9349	58.2	34.0	17.09	82.9
ACGP	0.863	2.60	0.9347	57.5	34.0	17.12	82.9
Dynamic (4)	0.744	1.95	0.9476	63.9	42.0	7.27	92.7
Dynamic (∞)	0.724	1.90	0.9476	66.8	46.3	5.84	94.2

Table 2.118: BMW cardinality, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

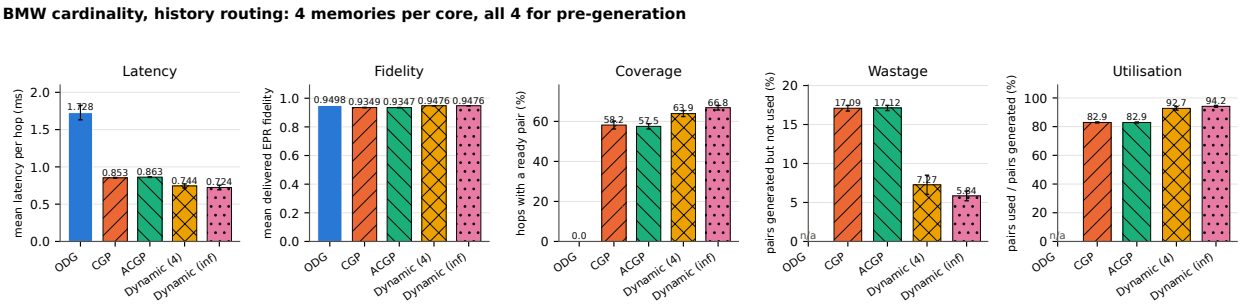


Figure 2.118: BMW cardinality, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.18.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	1.728	1.728	1.728	1.728	1.728	1.728	1.728
CGP	1.728	1.441	1.090	0.853	0.653	0.767	0.468
ACGP	1.728	1.434	1.104	0.863	0.653	0.766	0.468
Dynamic	2.178	1.120	0.815	0.744	0.724	0.724	0.724

Table 2.119: BMW cardinality, history routing: latency (mean latency per hop (ms)) for every K.

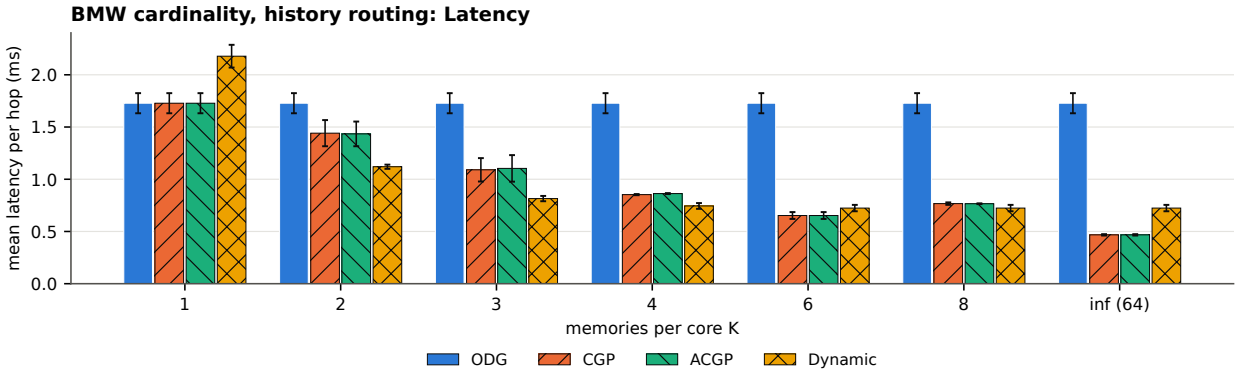


Figure 2.119: BMW cardinality, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9445	0.9372	0.9349	0.9280	0.9262	0.9272
ACGP	0.9498	0.9445	0.9371	0.9347	0.9280	0.9261	0.9272
Dynamic	0.9486	0.9479	0.9478	0.9476	0.9476	0.9476	0.9476

Table 2.120: BMW cardinality, history routing: fidelity (mean delivered EPR fidelity) for every K.

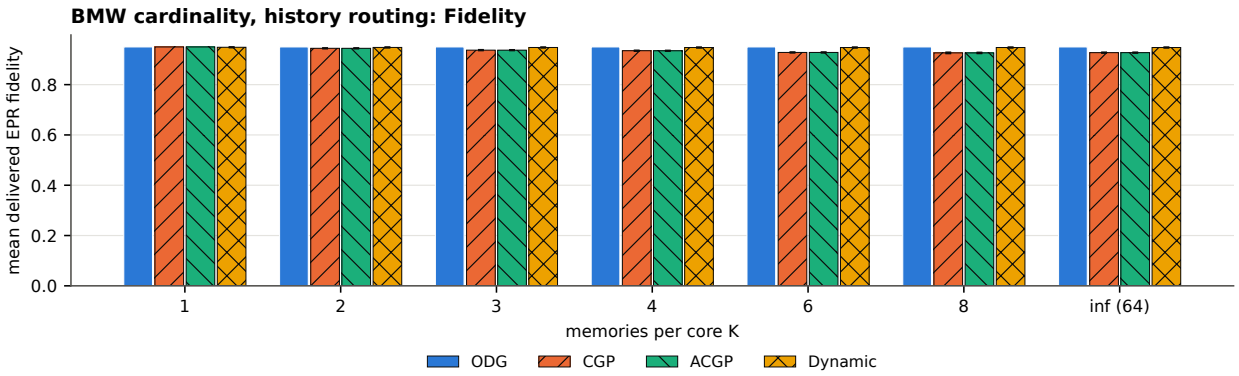


Figure 2.120: BMW cardinality, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	19.1	41.8	58.2	77.1	68.8	99.3
ACGP	0.0	19.4	41.8	57.5	77.1	69.1	99.3
Dynamic	50.7	44.4	61.3	63.9	66.8	66.8	66.8

Table 2.121: BMW cardinality, history routing: coverage (hops with a ready pair (%)) for every K.

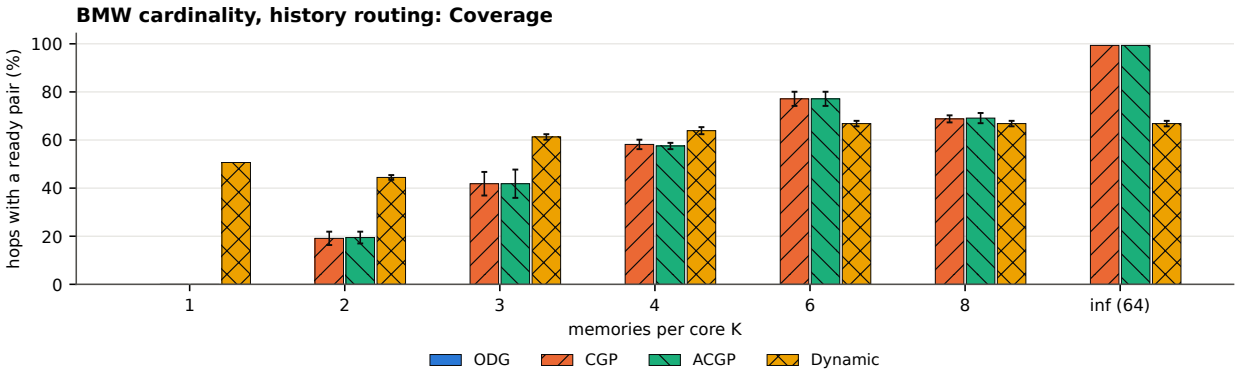


Figure 2.121: BMW cardinality, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	22.54	15.56	17.09	13.60	27.90	54.29
ACGP	–	21.79	15.38	17.12	13.60	27.46	54.29
Dynamic	28.24	21.87	11.86	7.27	5.84	5.84	5.84

Table 2.122: BMW cardinality, history routing: wastage (pairs generated but not used (%)) for every K.

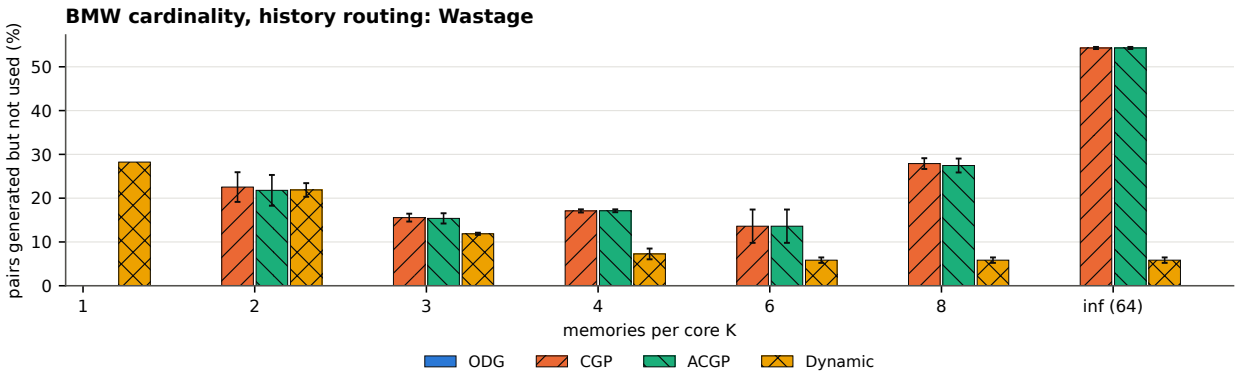


Figure 2.122: BMW cardinality, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	77.5	84.4	82.9	86.4	72.1	45.7
ACGP	–	78.2	84.6	82.9	86.4	72.5	45.7
Dynamic	71.8	78.1	88.1	92.7	94.2	94.2	94.2

Table 2.123: BMW cardinality, history routing: utilisation (pairs used / pairs generated (%)) for every K.

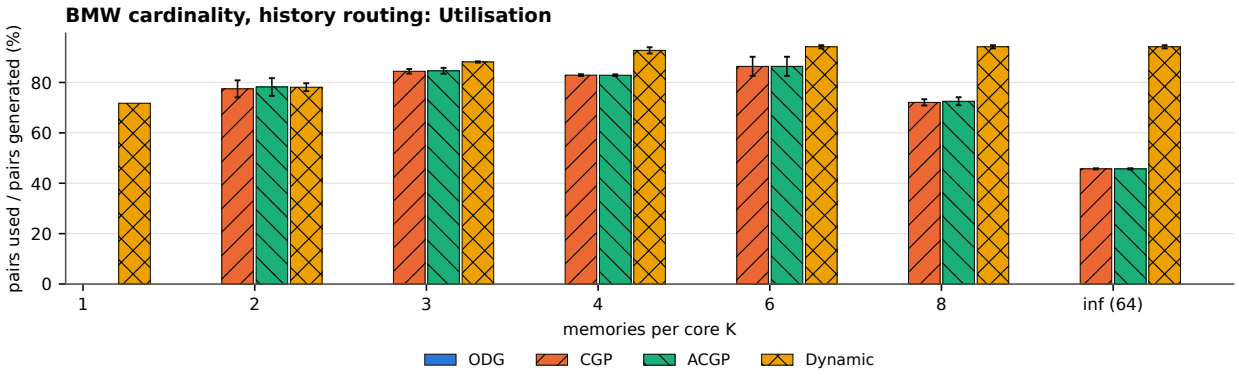


Figure 2.123: BMW cardinality, history routing: utilisation for every K.

2.19 BMW copula, XY routing

9510 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.19.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.872	4.70	0.9498	0.0	0.0	–	–
CGP	1.219	4.00	0.9417	74.7	64.6	4.07	95.9
ACGP	1.212	3.85	0.9418	75.4	65.3	4.20	95.8
Dynamic (8)	0.839	2.85	0.9448	75.0	69.3	1.53	98.5
Dynamic (∞)	0.657	1.80	0.9456	80.9	73.9	1.86	98.1

Table 2.124: BMW copula, XY routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

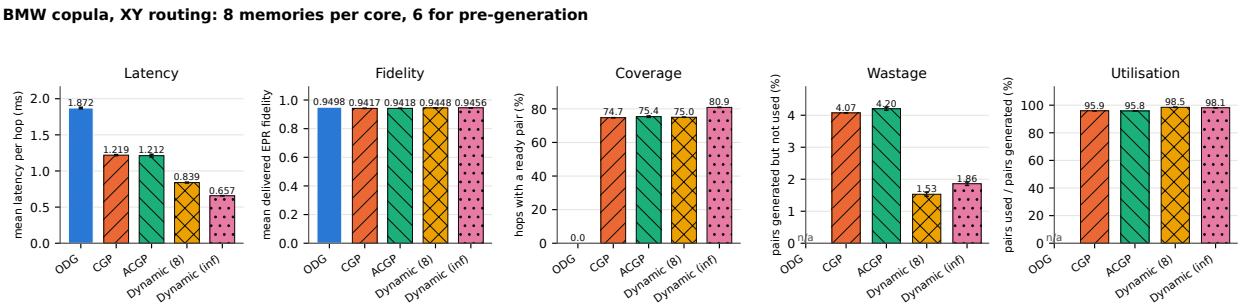


Figure 2.124: BMW copula, XY routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.19.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	2.155	5.75	0.9498	0.0	0.0	–	–
CGP	2.932	10.85	0.9401	63.6	50.9	5.51	94.5
ACGP	2.903	10.45	0.9401	63.2	51.8	5.49	94.5
Dynamic (4)	1.424	5.15	0.9443	63.5	58.4	3.08	96.9
Dynamic (∞)	0.657	1.80	0.9456	80.9	73.9	1.86	98.1

Table 2.125: BMW copula, XY routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

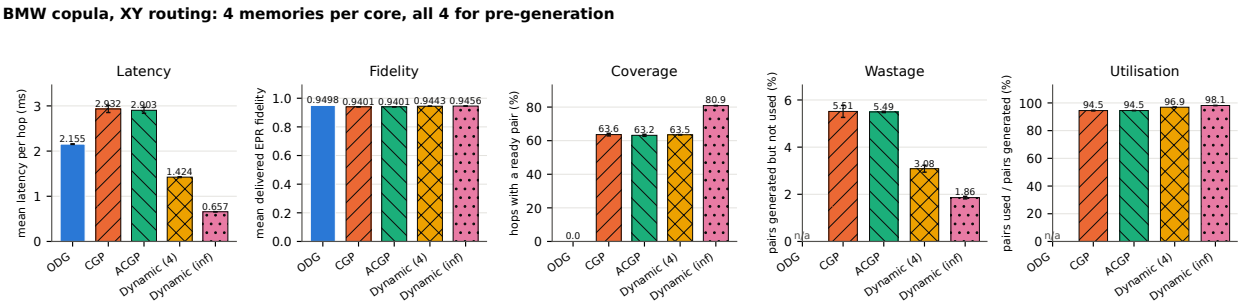


Figure 2.125: BMW copula, XY routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.19.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	8.370	3.504	2.513	2.155	1.930	1.872	1.810
CGP	8.370	5.168	3.726	2.932	1.537	1.219	0.491
ACGP	8.370	5.168	3.697	2.903	1.541	1.212	0.490
Dynamic	9.773	3.354	1.968	1.424	0.985	0.839	0.657

Table 2.126: BMW copula, XY routing: latency (mean latency per hop (ms)) for every K.

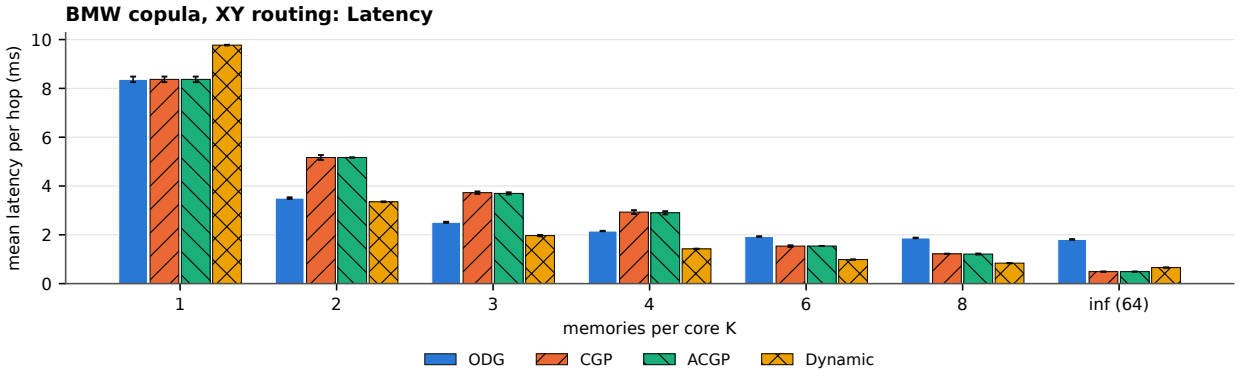


Figure 2.126: BMW copula, XY routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9445	0.9415	0.9401	0.9430	0.9417	0.9419
ACGP	0.9498	0.9444	0.9414	0.9401	0.9421	0.9418	0.9419
Dynamic	0.9481	0.9456	0.9450	0.9443	0.9446	0.9448	0.9456

Table 2.127: BMW copula, XY routing: fidelity (mean delivered EPR fidelity) for every K.

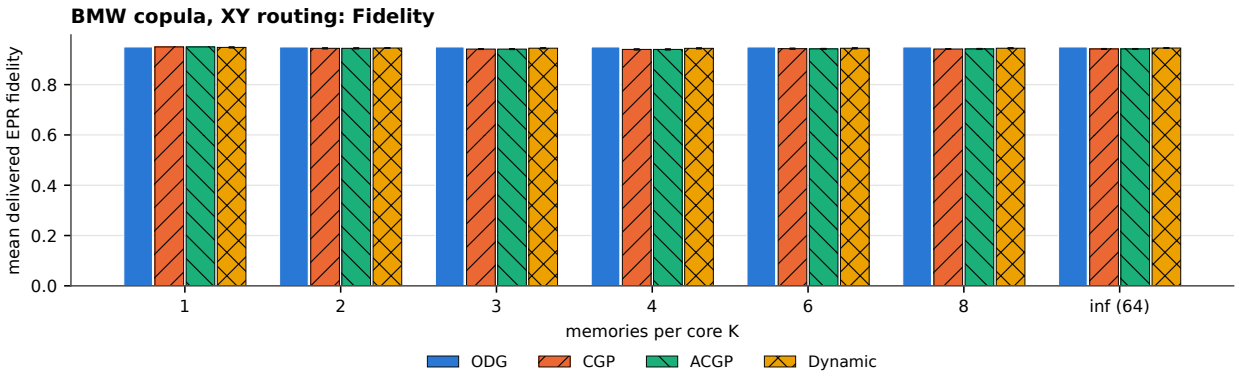


Figure 2.127: BMW copula, XY routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	27.0	49.1	63.6	61.4	74.7	96.9
ACGP	0.0	28.5	50.0	63.2	64.6	75.4	97.0
Dynamic	16.3	40.2	53.2	63.5	74.6	75.0	80.9

Table 2.128: BMW copula, XY routing: coverage (hops with a ready pair (%)) for every K.

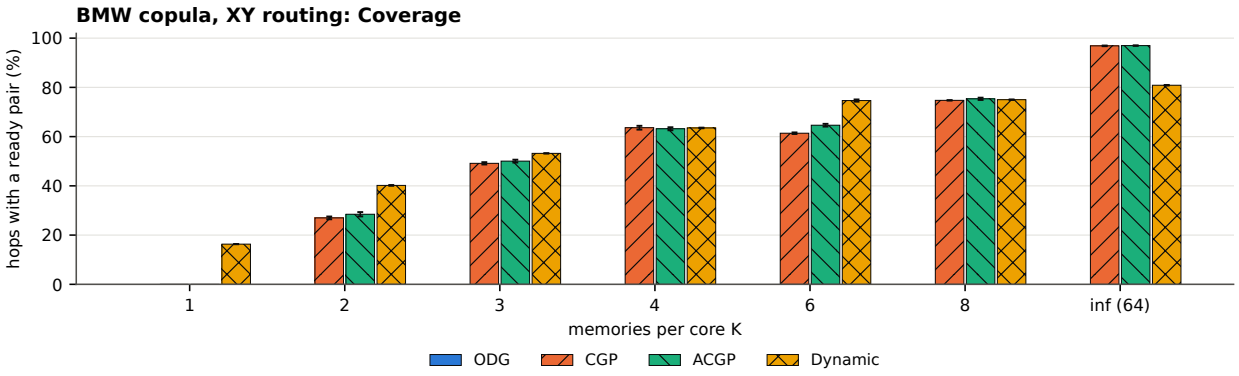


Figure 2.128: BMW copula, XY routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	6.41	5.78	5.51	4.00	4.07	13.06
ACGP	–	6.09	5.64	5.49	4.12	4.20	13.10
Dynamic	29.90	7.35	3.74	3.08	2.27	1.53	1.86

Table 2.129: BMW copula, XY routing: wastage (pairs generated but not used (%)) for every K.

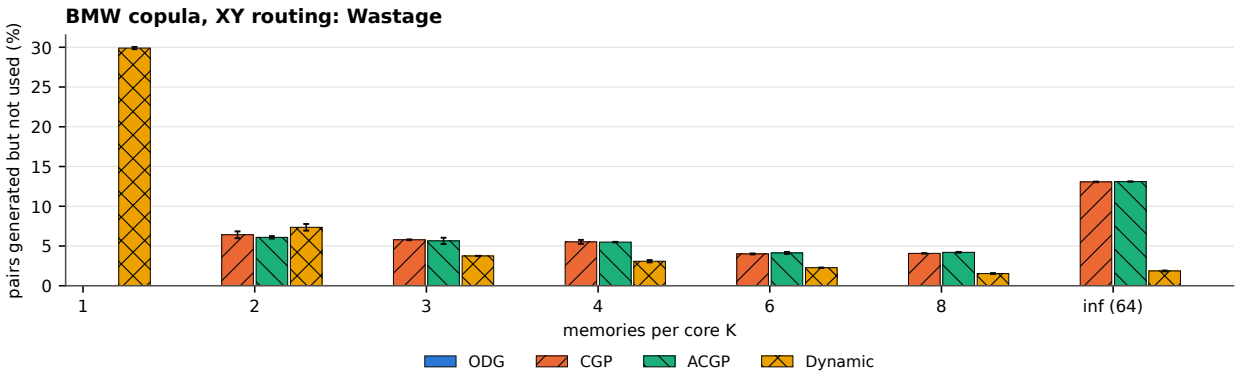


Figure 2.129: BMW copula, XY routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	93.6	94.2	94.5	96.0	95.9	86.9
ACGP	–	93.9	94.4	94.5	95.9	95.8	86.9
Dynamic	70.1	92.6	96.3	96.9	97.7	98.5	98.1

Table 2.130: BMW copula, XY routing: utilisation (pairs used / pairs generated (%)) for every K.

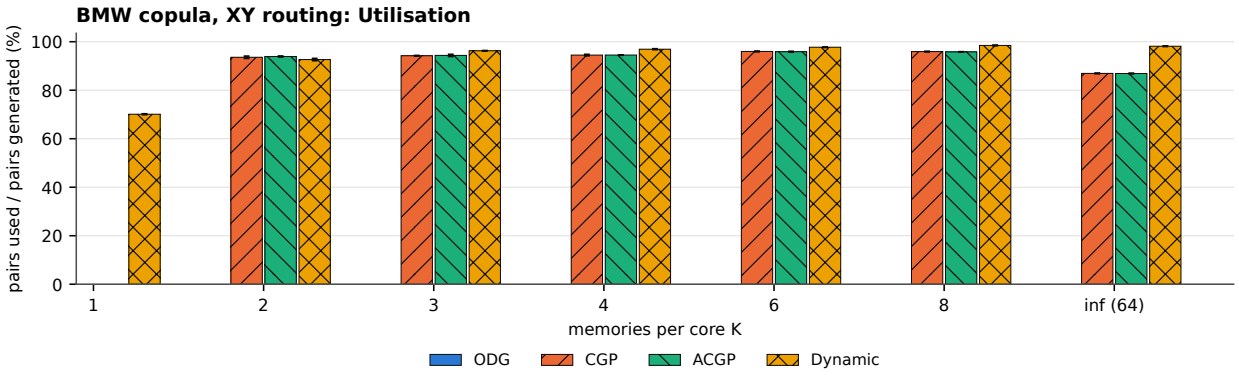


Figure 2.130: BMW copula, XY routing: utilisation for every K.

2.20 BMW copula, history routing

9510 hops; seeds 60, 61; every run completed every hop and passed the physical audit (Checks sheet of the workbook).

2.20.1 Headline: 8 memories per core, 6 for pre-generation

ODG, CGP, ACGP and Dynamic (8) use 8 memories per core, of which 6 may hold pre-generated pairs; Dynamic (∞) uses 64, which no core ever fills.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	1.863	4.70	0.9498	0.0	0.0	–	–
CGP	1.169	3.70	0.9424	74.6	63.5	4.11	95.9
ACGP	1.145	3.65	0.9425	75.9	64.9	4.01	96.0
Dynamic (8)	0.794	2.55	0.9453	76.8	70.1	1.84	98.2
Dynamic (∞)	0.645	1.70	0.9457	81.9	74.9	1.87	98.1

Table 2.131: BMW copula, history routing, 8 memories per core, 6 for pre-generation: mean of the seeds.

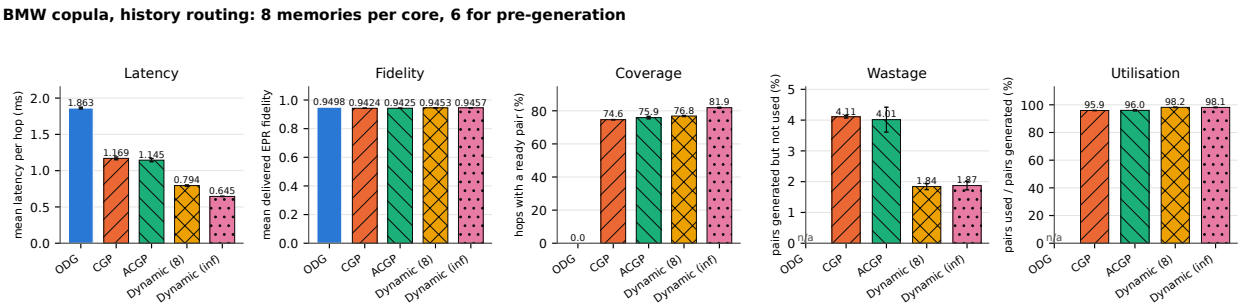


Figure 2.131: BMW copula, history routing, 8 memories per core, 6 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.20.2 Headline: 4 memories per core, all 4 for pre-generation

ODG, CGP, ACGP and Dynamic (4) use 4 memories per core and ODG and Dynamic may use all 4 for pre-generation; CGP and ACGP cache at most 3 (all 4 starves them), the same as in the K sweep; Dynamic (∞) as above.

Configuration	Latency (ms)	p95 (ms)	Fidelity	Hop cov. (%)	Chain full (%)	Wastage (%)	Utilisation (%)
ODG	2.095	5.60	0.9498	0.0	0.0	–	–
CGP	2.718	9.52	0.9411	64.0	51.0	5.09	94.9
ACGP	2.662	9.30	0.9412	63.9	51.2	4.72	95.3
Dynamic (4)	1.325	4.70	0.9446	65.6	59.5	3.34	96.7
Dynamic (∞)	0.645	1.70	0.9457	81.9	74.9	1.87	98.1

Table 2.132: BMW copula, history routing, 4 memories per core, all 4 for pre-generation: mean of the seeds.

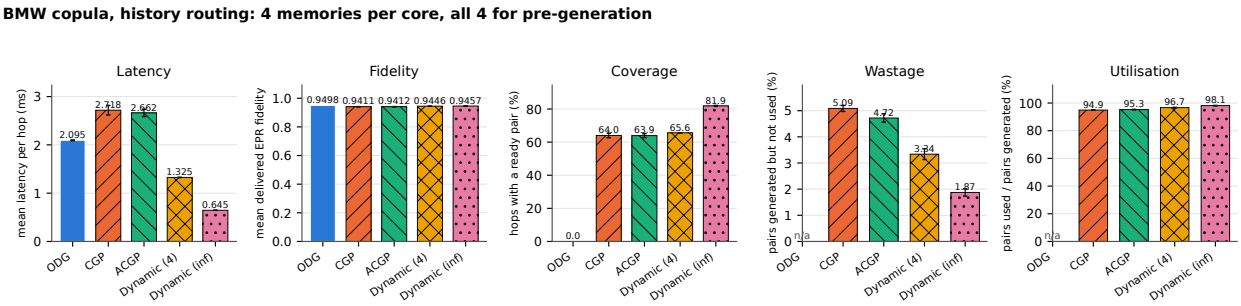


Figure 2.132: BMW copula, history routing, 4 memories per core, all 4 for pre-generation. Bars are the mean of the seeds, whiskers the seed range; fidelity bars start at zero, so read the labels.

2.20.3 Memories per core K

Dynamic may use all K memories, except 6 of 8 at $K = 8$; CGP and ACGP cache $\lfloor 0.75K \rfloor$ pairs (so at $K = 1$ they are ODG). $K = \infty$ is 64 memories.

Latency	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	7.865	3.271	2.392	2.095	1.915	1.863	1.802
CGP	7.865	4.796	3.399	2.718	1.416	1.169	0.488
ACGP	7.865	4.804	3.448	2.662	1.416	1.145	0.488
Dynamic	9.278	3.024	1.805	1.325	0.920	0.794	0.645

Table 2.133: BMW copula, history routing: latency (mean latency per hop (ms)) for every K.

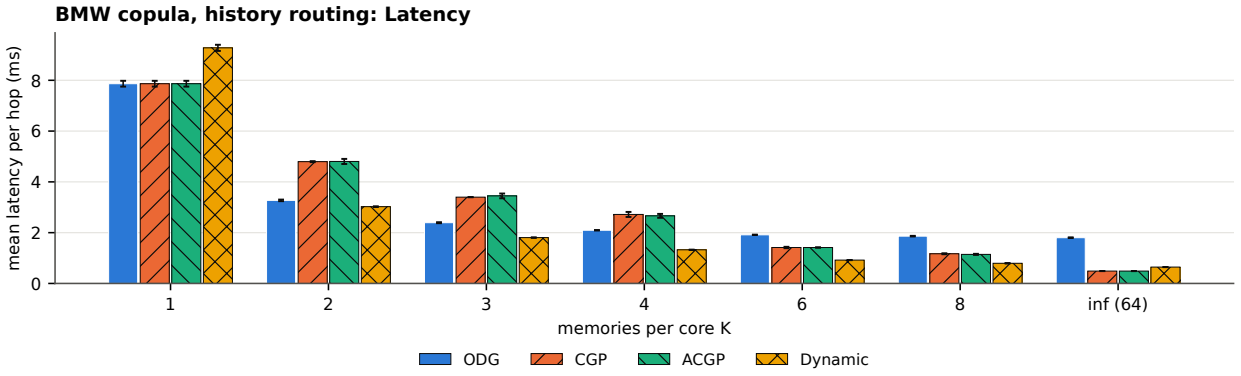


Figure 2.133: BMW copula, history routing: latency for every K.

Fidelity	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498	0.9498
CGP	0.9498	0.9452	0.9428	0.9411	0.9434	0.9424	0.9419
ACGP	0.9498	0.9453	0.9428	0.9412	0.9435	0.9425	0.9418
Dynamic	0.9480	0.9457	0.9451	0.9446	0.9451	0.9453	0.9457

Table 2.134: BMW copula, history routing: fidelity (mean delivered EPR fidelity) for every K.

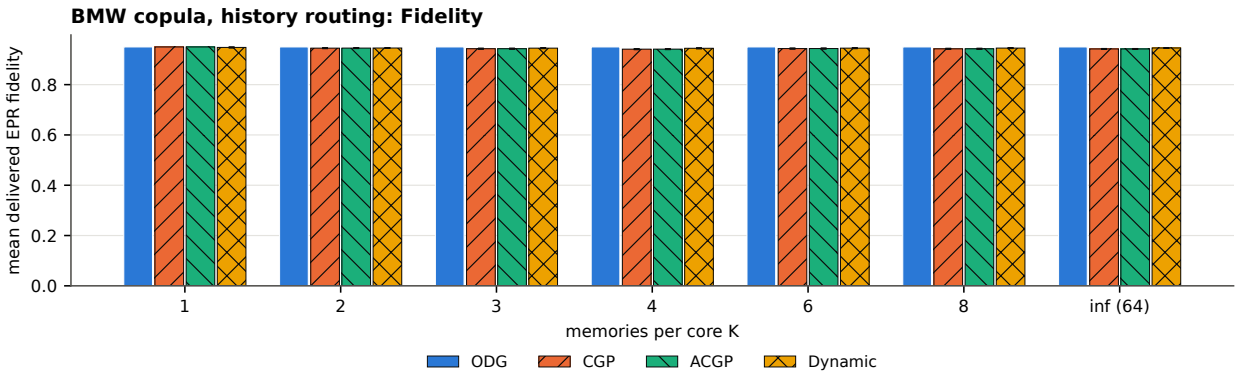


Figure 2.134: BMW copula, history routing: fidelity for every K.

Coverage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CGP	0.0	27.8	49.9	64.0	64.2	74.6	97.1
ACGP	0.0	27.6	51.0	63.9	64.5	75.9	97.2
Dynamic	17.0	41.2	54.4	65.6	76.6	76.8	81.9

Table 2.135: BMW copula, history routing: coverage (hops with a ready pair (%)) for every K.

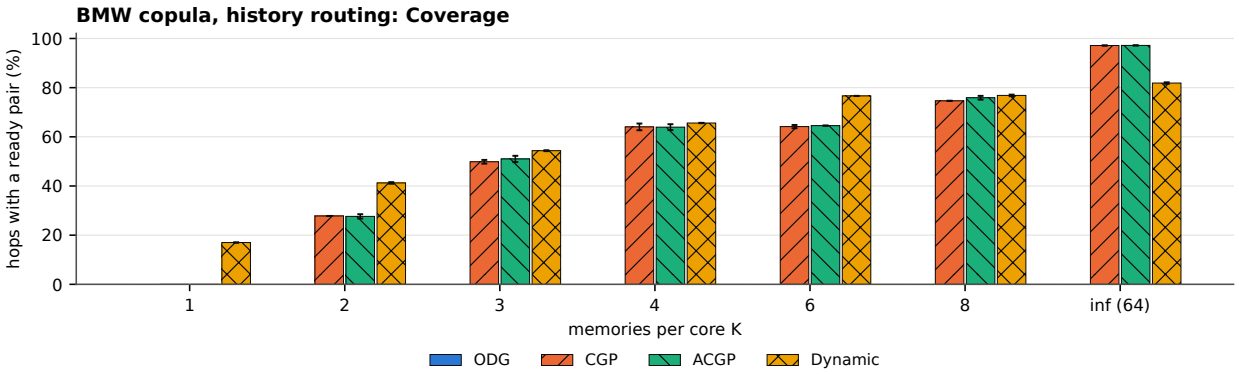


Figure 2.135: BMW copula, history routing: coverage for every K.

Wastage	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	6.03	4.81	5.09	3.46	4.11	12.95
ACGP	–	5.72	4.98	4.72	3.58	4.01	13.02
Dynamic	29.38	8.28	4.41	3.34	2.17	1.84	1.87

Table 2.136: BMW copula, history routing: wastage (pairs generated but not used (%)) for every K.

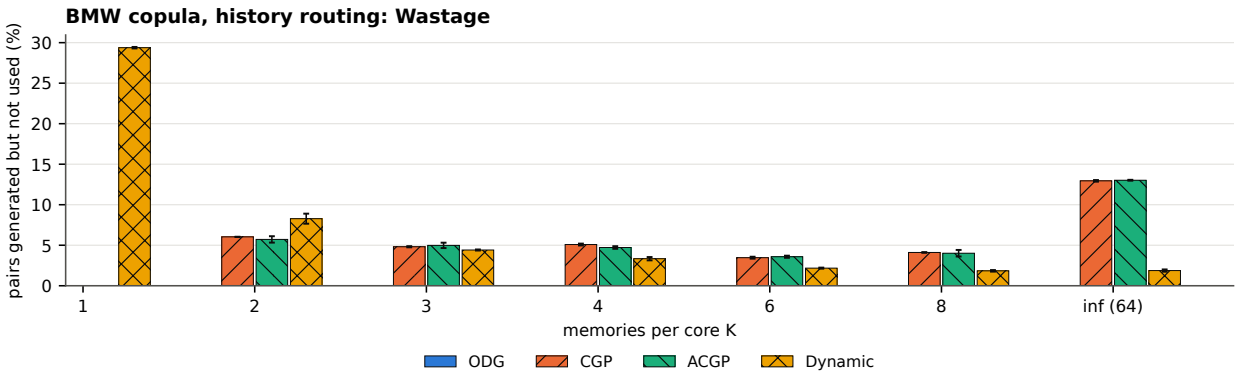


Figure 2.136: BMW copula, history routing: wastage for every K.

Utilisation	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 6$	$K = 8$	$K = \infty$
ODG	–	–	–	–	–	–	–
CGP	–	94.0	95.2	94.9	96.5	95.9	87.1
ACGP	–	94.3	95.0	95.3	96.4	96.0	87.0
Dynamic	70.6	91.7	95.6	96.7	97.8	98.2	98.1

Table 2.137: BMW copula, history routing: utilisation (pairs used / pairs generated (%)) for every K.

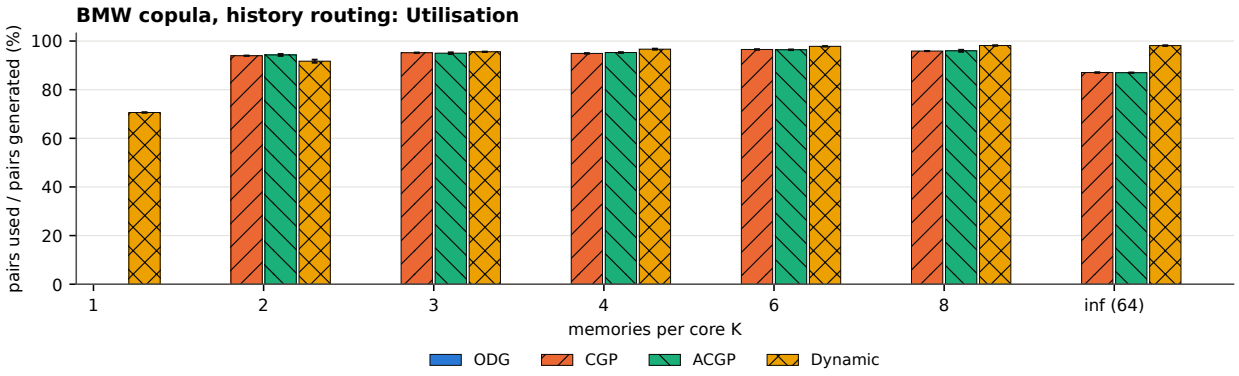


Figure 2.137: BMW copula, history routing: utilisation for every K.